Space Exploration & Aerospace Engineering

Part X

Launch Systems and Infrastructure

Everything in space started on a launch pad. Part X explains launch vehicle engineering and rocket propulsion, surveys the world's launch vehicles including the new generation of reusable rockets, and describes spaceports and the ground systems that support them.



Chapters 34–37 · pages 610–693 of the complete manual

First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

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First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example "Chapter 24") and to the appendices refer to the complete book.



PART X

Launch Systems and Infrastructure

Everything in space started on a launch pad. Part X explains launch vehicle engineering and rocket propulsion, surveys the world's launch vehicles including the new generation of reusable rockets, and describes spaceports and the ground systems that support them.

34 Launch Vehicle Engineering
35 Rocket Propulsion
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Image: NASA (NASA/Joel Kowsky)

34 CHAPTER Launch Vehicle Engineering



Image: NASA/KSC (NASA/Kim Shiflett)

A launch vehicle must accelerate its payload to nearly 8 km/s in under ten minutes while carrying all of its own propellant and surviving aerodynamic loads, noise, vibration and heat — reliably and cheaply. This chapter explains how engineers meet that challenge. It builds the velocity budget and shows how launch latitude changes it, derives the rocket equation and the logic of staging, follows a simulated two-stage kerosene vehicle from the pad to orbit, and describes tanks, pressurisation, guidance and flight termination. It then turns to the environments a payload must survive, the payload penalty of landing a booster, the economics of launch, and what seventy years of launch statistics reveal about reliability.

IN THIS CHAPTER

34.1 Getting to Orbit
34.2 Staging
34.3 The Ascent Trajectory
34.4 Structures and Materials
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34.8 Reliability and Testing

A rocket standing on the pad is mostly propellant. A Falcon 9 weighs about 549 t at lift-off, almost 490 t of it liquid oxygen and kerosene; the satellites on top make up only 3–4 % of the total. Within about eight minutes it must lift its payload above the atmosphere and accelerate it to 7.8 km/s. Launch is the shortest phase of most missions and the most violent: the payload meets its highest loads, loudest noise and strongest vibration in its first minutes of flight.

To keep the numbers concrete, this chapter follows a **reference vehicle**. It is a planar trajectory simulation of a two-stage kerosene rocket with Falcon 9-class masses and engines, flown from Cape Canaveral to a 200 km orbit (Figure 34.5). The engines themselves are the subject of Chapter

35, the world's launch vehicles of Chapter 36, and launch sites and ground systems of Chapter 37; the orbital mechanics used here is developed in Chapter 3.

34.1 Getting to Orbit

34.1.1 The Velocity Budget

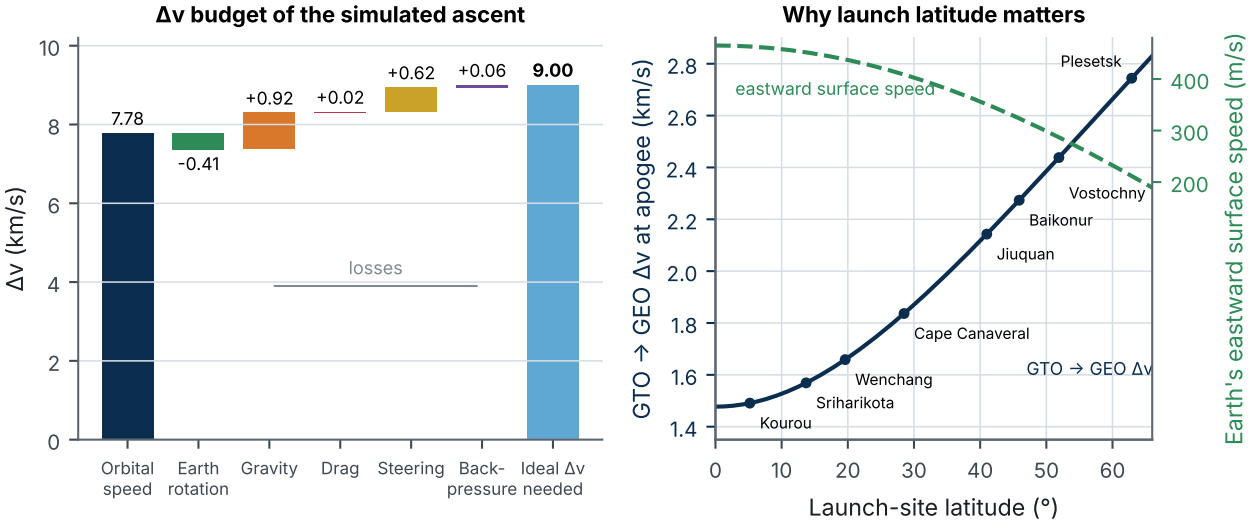
A satellite in a circular orbit 200 km above Earth moves at 7.78 km/s. Each kilogram carries about 30 MJ of kinetic energy and almost 2 MJ of potential energy. The launch vehicle supplies it by throwing propellant backwards, and the measure of what it can do is the **velocity change** Δv it can give itself (Figure 34.1).

THE ROCKET EQUATION

A rocket of mass m ejects propellant at speed v_e relative to itself. When it expels a small mass $-dm$, conservation of momentum gives $m\,dv = -v_e\,dm$. Integrating from the initial mass m_0 to the final mass m_f:

$$\Delta v = v_e \ln \frac{m_0}{m_f} = I_{sp}\, g_0 \ln \frac{m_0}{m_f}$$

where I_{sp} is the specific impulse (Chapter 35) and g_0 = 9.806 65 m/s². The ratio m_0/m_f is the **mass ratio**. The equation, published by Konstantin Tsiolkovsky in 1903, gives the velocity change in empty space, with no gravity or air.



Left: losses integrated along the simulated ascent of the 4-panel ascent figure (ideal Δv = Σ v_e ln(m_0/m_f), vacuum Isp).
Right: GTO perigee 200 km, inclination = site latitude; the apogee burn circularises and removes the inclination. Surface speed = 465 m/s × cos(latitude).

FIGURE 34.1 Left: Δv budget of the simulated reference ascent to a 200 km circular orbit from Cape Canaveral. Earth's rotation reduces the requirement by 0.41 km/s; gravity and steering losses add about 1.5 km/s. Right: how launch-site latitude changes the eastward surface speed and the Δv a satellite needs to turn a geostationary transfer orbit into a geostationary one.

Four **losses** separate this ideal Δv from the final orbital speed:

- **Gravity loss**, $\int g \sin\gamma\, dt$, where γ is the flight-path angle above the horizontal: while the rocket climbs, part of its thrust only holds it up. A rocket that hovered would burn all its propellant and gain nothing.
- **Drag loss**, $\int D/m\, dt$, the momentum given to the air.

- **Steering loss**, $\int (T/m)(1-\cos\alpha)\,dt$, when the thrust T points at an angle α away from the velocity.
- **Back-pressure loss**: a nozzle produces less thrust in the atmosphere than in a vacuum (Chapter 35).

Earth's rotation helps, because the pad already moves east:

$$\Delta v_{ideal} = v_{orbit} - v_{rotation} + \Delta v_{gravity} + \Delta v_{drag} + \Delta v_{steering} + \Delta v_{pressure}$$

Figure 34.1 shows the terms for the reference vehicle, integrated along its simulated trajectory: it needs an ideal Δv of 9.00 km/s. Gravity loss (0.92 km/s) is the largest. Steering loss (0.62 km/s) is almost as large. At ignition the second stage's thrust is less than its weight, so it must point its nose above the flight path to avoid falling back while it gains speed. It trades gravity loss for steering loss. Drag costs only 0.02 km/s, because a large, dense vehicle has so much mass per unit of frontal area; small launchers lose more, often 0.1–0.2 km/s. Without the 0.41 km/s gift of Earth's rotation the reference vehicle would need 9.4 km/s, consistent with the usual rule of thumb of 9.3–9.6 km/s to low Earth orbit (LEO).

34.1.2 Latitude and Inclination

The ground at latitude φ moves east at 465 m/s × cos φ. A rocket launched on azimuth Az (measured from north) into an orbit of inclination i must satisfy sin Az = cos i / cos φ. The useful part of the rotation it inherits works out to 465 m/s × cos i. It depends only on the orbit, not the site. A launch due east gives the lowest inclination reachable directly, i = φ. A launch to the International Space Station (51.6°) gains 289 m/s; a polar orbit gains nothing; a sun-synchronous orbit, inclined about 97–98°, is slightly retrograde and costs about 60 m/s extra.

Reaching an inclination *lower* than the site latitude needs a **dog-leg** (steering sideways during ascent) or a plane change in orbit, both expensive: at orbital speed each degree costs $2v \sin 0.5° \approx$ 136 m/s. Latitude therefore matters most for geostationary missions (Table 34.1). A satellite released into a geostationary transfer orbit (GTO) inclined at the site latitude must remove that inclination at apogee, where it moves slowly and plane changes are cheapest, while circularising. From Kourou (5.2° N) this costs 1.49 km/s; from Cape Canaveral (28.5° N), 1.84 km/s; from Baikonur (45.9° N), 2.27 km/s. For a satellite with a 320 s apogee engine, the 0.35 km/s difference between Kourou and the Cape raises the propellant share of its mass from 38 % to 44 %. Alternatives are a "supersynchronous" transfer, whose higher apogee makes the plane change cheaper, or direct injection into geostationary orbit by the upper stage.

TABLE 34.1 Latitude effects at selected launch sites. GTO → GEO Δv assumes a 200 km perigee and an inclination equal to the site latitude.

Launch site	Latitude	Eastward surface speed	GTO → GEO Δv
Kourou (Guiana Space Centre)	5.2° N	463 m/s	1.49 km/s
Sriharikota (India)	13.7° N	452 m/s	1.57 km/s
Wenchang (China)	19.6° N	438 m/s	1.66 km/s
Cape Canaveral / Kennedy	28.5° N	409 m/s	1.84 km/s
Mahia (New Zealand)	39.3° S	360 m/s	2.10 km/s
Baikonur (Kazakhstan)	45.9° N	324 m/s	2.27 km/s
Plesetsk (Russia)	62.9° N	212 m/s	2.74 km/s

EXAMPLE: THE ΔV BUDGET TO A 500 KM, 53° ORBIT FROM CAPE CANAVERAL

1. Launch azimuth: $\sin Az = \cos 53° / \cos 28.5° = 0.602 / 0.879 = 0.685$, so $Az = 43°$ — the vehicle flies north-east along the US coast.
2. Earth-rotation gain: $465 \times \cos 53° = 280$ m/s, instead of 409 m/s for a due-east launch.
3. Insertion into a 200 km parking orbit (7 784 m/s), then a Hohmann transfer to 500 km: 86 m/s to raise apogee plus 85 m/s to circularise (Chapter 3).
4. Ascent losses from Figure 34.1: 0.92 + 0.62 + 0.02 + 0.06 = 1.62 km/s.
5. Ideal Δv = 7 784 − 280 + 1 620 + 172 ≈ 9 300 m/s. This is 0.30 km/s more than the due-east 200 km orbit — about 2.3 t of droneship payload in the model of Figure 34.14. This is one reason constellations are often deployed low and raise their own orbits with electric propulsion.

34.2 Staging

34.2.1 Why One Stage Is Not Enough

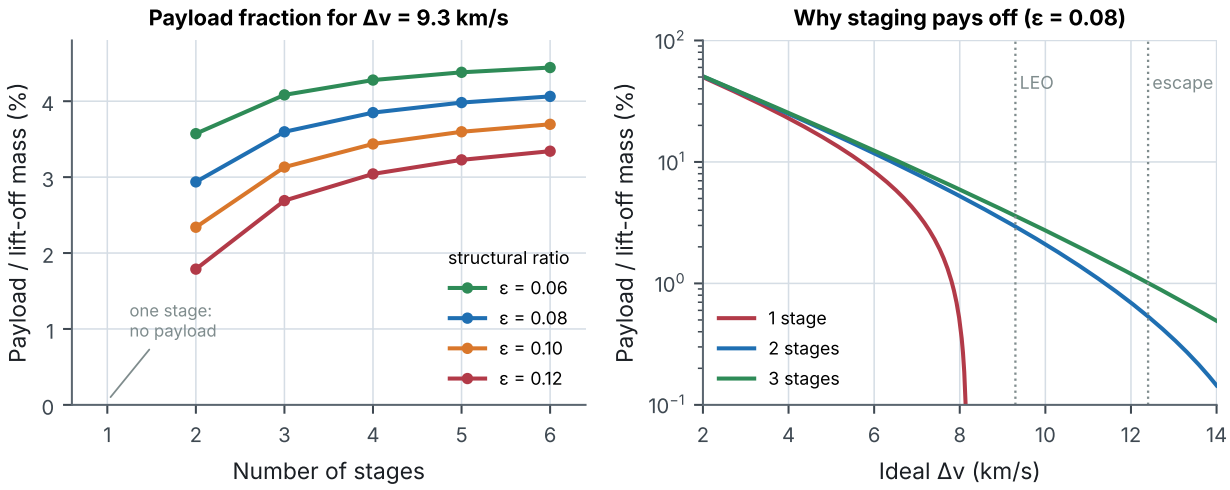
Every stage carries structure — tanks, engines, plumbing, avionics — that becomes dead weight once its propellant is gone. It is described by the **structural ratio** $\varepsilon = m_s/(m_s + m_p)$, the structure's share of a loaded stage; good modern stages achieve $\varepsilon \approx 0.05$–0.10. A single stage with an average specific impulse of 330 s that must deliver 9.3 km/s needs a mass ratio of $e^{9300/3236} = 17.7$: only 5.6 % of the lift-off mass may remain at burnout. If structure alone takes 8 %, the rocket cannot reach orbit even empty. This is the fundamental difficulty of **single-stage-to-orbit** vehicles, studied for decades but never flown to orbit (Chapter 10).

34.2.2 The Rocket Equation for Several Stages

Staging throws structure away as soon as it is empty. Each stage k contributes $\Delta v_k = v_{e,k} \ln(m_{0,k}/m_{f,k})$, where the "payload" of stage k is everything above it, and the contributions add. For n identical stages the optimum is an equal split of Δv, and the overall payload fraction is

$$\frac{m_{payload}}{m_0} = \left(\frac{e^{-\Delta v/(n v_e)} - \varepsilon}{1 - \varepsilon} \right)^n$$

Figure 34.2 evaluates this formula. Going from one stage to two turns an impossible vehicle into one that delivers 3 % of its lift-off mass. A third stage adds about a fifth more. Beyond that the gains are small, while every extra stage adds engines, a separation event, an in-flight ignition and cost. Most modern launchers therefore have two stages. Real stages are not identical. Hydrogen upper stages (Isp near 450 s) take a larger share of the Δv. A reusable first stage is given a *smaller* share, so that it separates slowly and close to the launch site.



Tsiolkovsky equation for identical stages (Isp = 330 s for every stage; structural ratio ε = structure /(structure + propellant)); equal Δv split, which is optimal for identical stages.

FIGURE 34.2 Left: payload fraction for a 9.3 km/s mission as a function of the number of identical stages and their structural ratio. Right: payload fraction versus total Δv for one, two and three stages with ε = 0.08; a single stage runs out of capability just above 8 km/s.

EXAMPLE: TWO STAGES VERSUS ONE

Take Isp = 330 s (v_e = 3 236 m/s), Δv = 9.3 km/s and a lift-off mass of 550 t.

1. One stage, ε = 0.08: the final-to-initial mass ratio is $e^{-9300/3236}$ = 0.0565, less than ε, so no payload is possible. Even with an exceptionally light structure (ε = 0.05) the payload fraction is (0.0565 − 0.05)/0.95 = 0.7 %, or 3.8 t.
2. Two equal stages, ε = 0.08: each provides 4.65 km/s, so $e^{-4650/3236}$ = 0.238. Each stage's payload ratio is (0.238 − 0.08)/0.92 = 0.171, and the overall fraction is 0.171^2 = 2.9 %, or **16 t**.
3. With ε = 0.06 this rises to 3.6 % (19.6 t); with ε = 0.10 it falls to 2.3 % (12.9 t). Each percentage point of structural ratio is worth about 1.7 t of payload, a tenth of the total, which is why engineers fight for every kilogram of tank wall.

34.2.3 Serial, Parallel and Cross-Feed

In **serial** staging the stages sit on top of one another and fire in sequence, as in the Saturn V (Figure 34.3) and Falcon 9. In **parallel** staging, boosters strapped around a core ignite on the ground and drop away first. The R-7 of 1957, ancestor of today's Soyuz, uses four, and Ariane 6 uses two or four solid boosters. The Space Launch System (SLS) uses two, and Falcon Heavy two Falcon 9 first stages. Parallel staging avoids lighting engines in flight, but the core must carry side loads. The Space Shuttle's orbiter engines burned propellant from an external tank that was discarded just short of orbit (Figure 34.4). **Cross-feed** means pumping propellant from the boosters into the core, so that its tanks are still full when they separate. It would give serial-staging performance with parallel hardware. It was proposed for Falcon Heavy but has not flown.

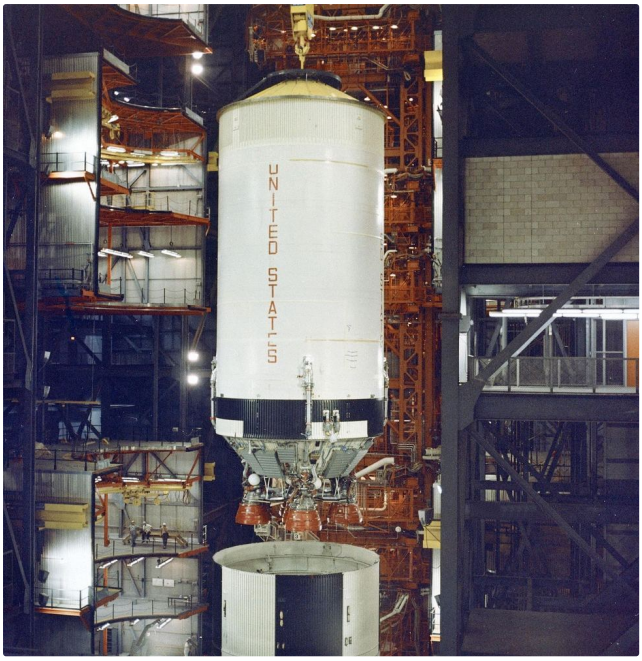


FIGURE 34.3 The second stage (S-II) of the Saturn V for Apollo 6 being lowered onto the first stage in the Vehicle Assembly Building in 1967. Serial staging stacks each stage on top of the previous one. Image: NASA/MSFC.



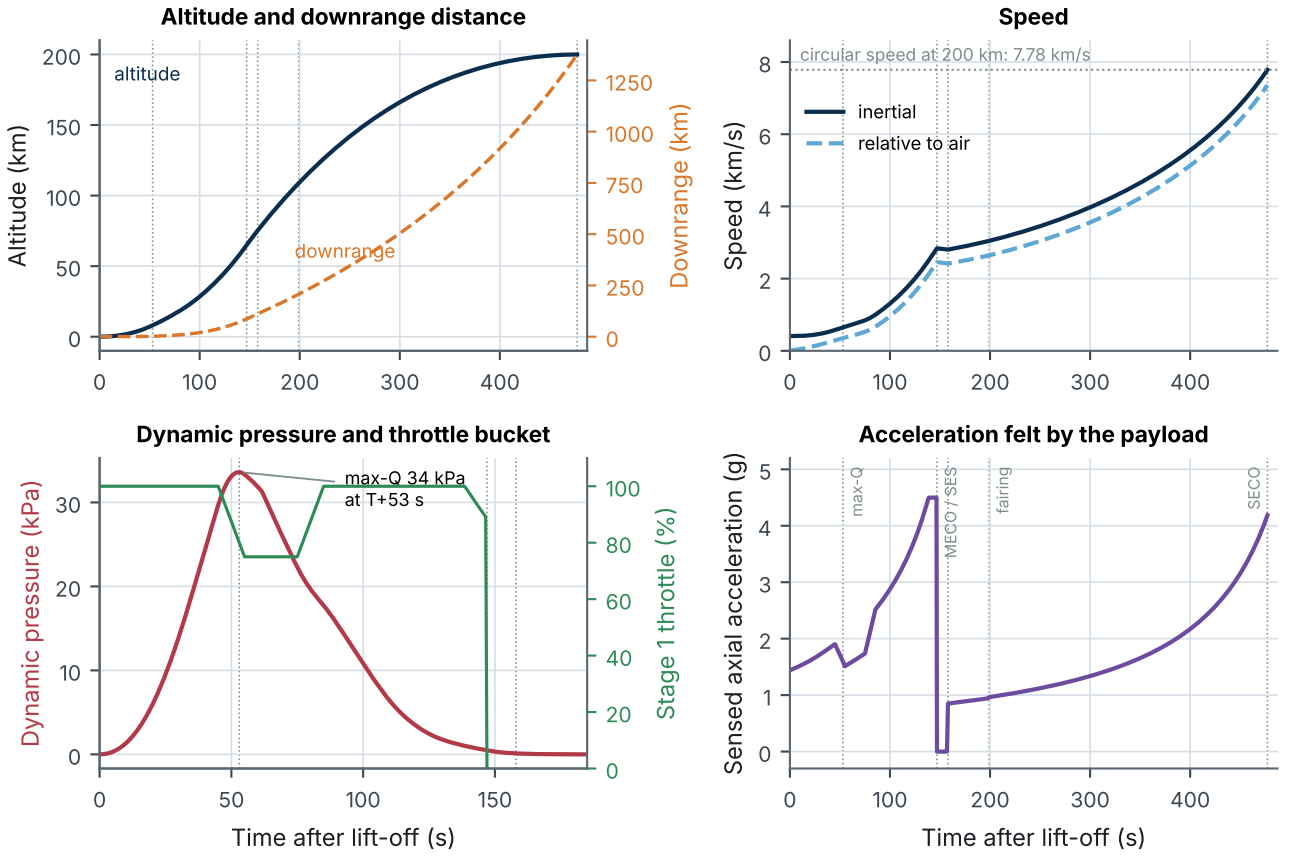
FIGURE 34.4 The Space Shuttle's external tank falling away after separation in March 1989 (STS-29), photographed from Discovery. The tank fed the orbiter's main engines and was discarded just short of orbit. Image: NASA/JSC.

34.3 The Ascent Trajectory

34.3.1 Lift-Off, Pitch-Over and the Gravity Turn

A liquid-fuelled vehicle ignites its engines a few seconds before release and is held down while its computers confirm that every engine is running normally; if not, it shuts down safely on the pad. At release its thrust-to-weight ratio is typically 1.2–1.5 (1.44 for the reference vehicle). It rises vertically to clear the tower, rolls to its launch azimuth, then makes a small **pitch-over** — 4.8° between 8 and 18 s in the simulation. In a **gravity turn** the vehicle keeps its nose along the airflow, at nearly zero **angle of attack** α. Gravity gradually bends the path towards the horizontal. This keeps the aerodynamic side force and the bending of the long, thin structure small. The pitch-over sets the whole first-stage trajectory: too small and the rocket climbs too steeply; too large and it stays too long in thick air. The simulation chooses it to maximise payload.

Dynamic pressure, $q = \frac{1}{2}\rho v^2$, rises as the vehicle accelerates and falls as the air thins; its peak, **max-Q**, is typically 25–40 kPa. In the simulation it reaches 34 kPa at T+53 s, 8 km up, just after Mach 1; the Falcon 9 user's guide lists max-Q at T+67–74 s. The engines are throttled through a **throttle bucket** (75 % in the simulation) to limit it. What matters most for the structure is the product **qα**: a gust or **wind shear** at high dynamic pressure forces a non-zero angle of attack and bends the vehicle. Balloons and wind profilers measure the upper winds before each launch, and the trajectory is "wind-biased" and the loads recomputed to confirm that they stay within limits.



Planar simulation, Falcon 9-class vehicle (stage 1: 25.6 t dry, 395.7 t propellant, 7.6–8.2 MN, Isp 283–312 s; stage 2: 3.9 t dry, 92.7 t, 981 kN, 348 s; fairing 1.9 t), 19.1 t payload to a 200 km circular orbit from 28.5° N with 22 t of first-stage propellant kept for a droneship landing; US 1976 atmosphere, rotating Earth, 4.5 g limit.

FIGURE 34.5 Simulated ascent of the reference vehicle to a 200 km orbit with a droneship booster recovery. Altitude climbs steeply while speed builds slowly; max-Q is limited by throttling the first stage; the sensed acceleration rises as propellant burns off, is capped at 4.5 g, and drops to zero during the coast between staging and second-stage ignition.

As propellant burns off, acceleration grows (Figure 34.5) from 1.4 g towards 4–5 g, where vehicles cap it by throttling or shutting down engines.

34.3.2 Staging, Fairing Jettison and the Upper Stage

At **main-engine cut-off** (MECO), 147 s after lift-off, the reference first stage is 65 km up and moving at 2.5 km/s relative to the air. Pneumatic pushers separate the stages, and the second stage coasts briefly while its propellant settles before igniting. The **payload fairing** is jettisoned once the aerodynamic heating it would let through is negligible. Falcon releases it when the **free-molecular heating** rate, $\tfrac{1}{2}\rho v^3$, falls below 1 135 W/m^2. In the simulation this happens at T+199 s and about 109 km.

The upper stage then flies above the atmosphere under closed-loop guidance. Its thrust-to-weight ratio is only 0.85 at ignition, so it pitches up to hold altitude while it accelerates; by **second-engine cut-off** (SECO) at T+477 s its acceleration has climbed above 4 g. For higher orbits the stage usually stops in a low **parking orbit**, coasts, and restarts to raise the apogee (Table 34.2). Restarting after a long coast requires settling the propellant with small thrusters and keeping engine and propellants within temperature limits. Afterwards a responsible stage performs a **deorbit burn** or vents its propellant and pressurant to “passivate” itself, so that it cannot explode and add to the debris population (Chapter 21).

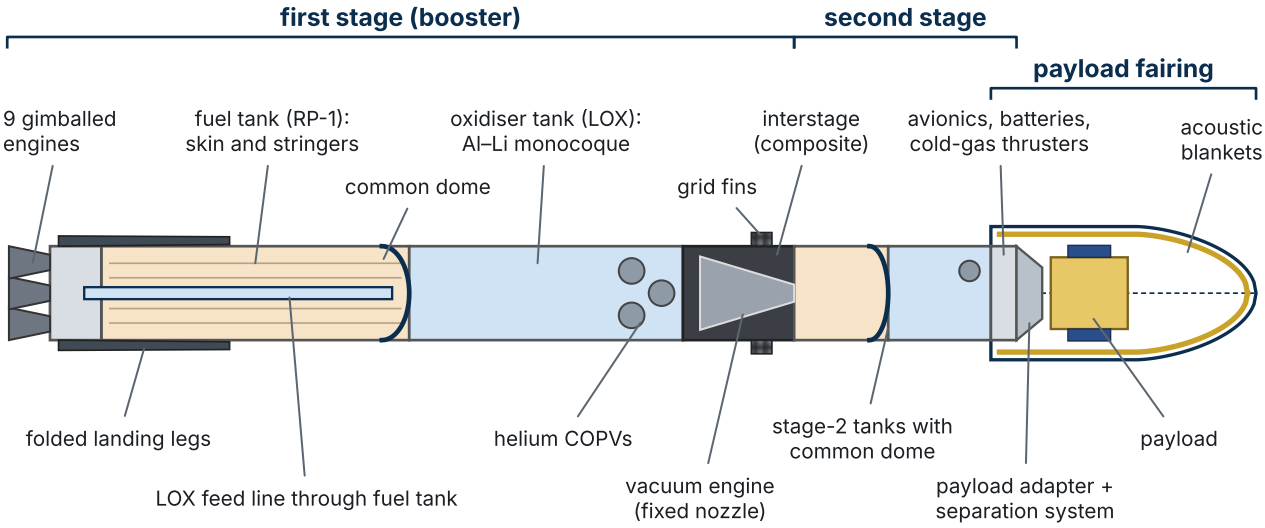
TABLE 34.2 Sample Falcon 9 flight timelines from the Falcon User's Guide (May 2025). Every mission's profile is different.

Event	LEO mission	GTO mission
Lift-off	T+0	T+0
Maximum dynamic pressure	T+67 s	T+74 s
First-stage main-engine cut-off	T+145 s	T+147 s
Stage separation	T+148 s	T+151 s
Second-stage ignition (1)	T+156 s	T+158 s
Fairing separation	T+195 s	T+222 s
Second-stage cut-off (1)	T+514 s	T+484 s
Second-stage ignition (2)	T+3 086 s	T+1 636 s
Second-stage cut-off (2)	T+3 090 s	T+1 696 s
Spacecraft separation	T+3 390 s	T+1 996 s

34.4 Structures and Materials

34.4.1 Tanks, Domes and Stiffeners

A launch vehicle is essentially a stack of thin-walled pressure vessels with engines at one end and a payload at the other (Figure 34.6). A first stage 3.7 m across has tank walls only a few millimetres thick, and structure is typically 5–8 % of its loaded mass.



Schematic of a generic two-stage kerosene vehicle, loosely based on Falcon 9 (diameter exaggerated about ×1.5; length ≈ 70 m).

FIGURE 34.6 Anatomy of a two-stage kerosene launch vehicle. The first stage carries its oxygen tank above its fuel tank, with a common dome between them and a feed line running through the fuel tank; the composite interstage houses the second stage's vacuum engine; the fairing, lined with acoustic blankets, encloses the payload on its adapter.

Aluminium alloys have been the standard since the 1950s (Table 34.3). The Saturn V's first stage, the original Shuttle external tank and the SLS core stage use alloy 2219, which welds well and stays tough at cryogenic temperatures. **Aluminium–lithium** alloys are lighter and stiffer. The Shuttle's Super Lightweight Tank, first flown on STS-91 in June 1998, used alloy 2195, about 30 % stronger and 5 % less dense than 2219. It weighed 3 400 kg less than its predecessor. Falcon 9's

tanks are aluminium–lithium joined by **friction-stir welding**, in which a rotating tool stirs the metal together below its melting point. Al–Li is harder to weld and less tolerant of flaws, and SLS returned to 2219.

Stainless steel, used in the balloon tanks of Atlas and Centaur, has returned in SpaceX's Starship. It is heavier but cheap, easy to weld, stronger when cold, and keeps its strength at several hundred degrees Celsius — an advantage for a stage that must survive re-entry. **Carbon-fibre composites** promise the lightest tanks. Rocket Lab's Electron uses them throughout, and its larger Neutron is built around composite tanks. However, voids found in a composite tank dome helped push Neutron's first launch into 2027. Composite tanks must resist micro-cracking, leakage and, for oxygen, ignition.

TABLE 34.3 Tank and structure materials. Strengths are typical room-temperature tensile values; the metals are stronger at cryogenic temperature, and composite values depend strongly on the lay-up.

Material	Density (g/cm³)	Typical strength (MPa)	Strength / density vs 2219	Examples
Aluminium 2219	2.84	420–460	1.0	Saturn V S-IC, SLS core stage
Aluminium–lithium (e.g. 2195)	2.71	550–600	≈ 1.4	Shuttle SLWT (2195), Falcon 9
Stainless steel (300 series)	7.9	500–600 (more when cold)	≈ 0.5	Atlas, Centaur, Starship
Carbon-fibre/epoxy laminate	1.55–1.6	600–1 000	≈ 3	Electron, Neutron, interstages

A **common bulkhead** — a single dome separating the two propellants — saves the length and mass of a second dome. Falcon 9 uses one, as did both upper stages of the Saturn V, where the S-II's bulkhead had to be insulated to stop −183 °C oxygen freezing against −253 °C hydrogen. Walls in compression must resist **buckling**, so they are stiffened. Some use stringers attached to a thin skin (**skin-and-stringer**, as in Falcon 9's fuel tank). Others use ribs machined out of thick plate in a triangular **isogrid** or rectangular **orthogrid** pattern (Figure 34.7), which removes most of the metal but leaves a light, damage-tolerant structure without fasteners.



FIGURE 34.7 A completed liquid-hydrogen tank barrel for the SLS core stage lifted off the Vertical Weld Center at NASA's Michoud Assembly Facility in February 2025. The orthogrid stiffening machined into the inner surface is visible. Image: NASA/MAF (NASA/Evan Deroche).



FIGURE 34.8 The SLS liquid-hydrogen tank structural test article being placed in a test stand at NASA Marshall in January 2019. In December 2019 this test article was loaded to more than 260 % of its expected flight loads before it buckled and ruptured. Image: NASA/MSFC (Tyler Martin).

34.4.2 Pressurisation

Tanks are pressurised to push propellant into the turbopumps without cavitation and to stiffen the walls, so that internal pressure carries part of the compressive and bending loads. **Balloon tanks** such as those of Atlas had no stiffeners at all and had to stay pressurised even on the ground. The pressurant is usually helium, stored at very high pressure in **composite-overwrapped pressure vessels** (COPVs). These are often submerged in the cold liquid oxygen, where the gas is denser. Falcon heats the helium before it enters the tanks, so that less is needed. Failures of these systems destroyed two Falcon 9s (Table 34.6). **Autogenous** pressurisation avoids helium altogether by feeding warmed propellant gas back into the tanks, as Starship does with methane and oxygen.

34.4.3 Loads, Vibration and POGO

Axial compression peaks at the highest acceleration, just before first-stage cut-off; **bending** peaks around max-qα and in ground winds on the pad; internal pressure sizes the tank walls; and **acoustic** loads shake panels and equipment. A dangerous coupling is **POGO**, named after the pogo stick: a longitudinal vibration changes the pressure at the pump inlets, which changes the thrust, which reinforces the vibration. On Apollo 13 in April 1970, POGO oscillations shut down the centre engine of the Saturn V's second stage about two minutes early. The cure is to detune the feed system with gas-filled **accumulators** in the propellant lines; baffles damp propellant **slosh**.

Structures are designed with **factors of safety**, typically 1.1 on yield and 1.25–1.4 on ultimate strength, and proven by test, sometimes to destruction (Figure 34.8). Wind tunnels still provide the aerodynamic loads. In 2025 NASA's Ames Research Center tested a 1.2 % scale model of the Super Heavy Version 3 booster. The tests measured the forces on it during its return through the atmosphere (Figure 34.10).

34.4.4 Fairings and Adapters

The **payload fairing** is a two-piece shell, usually carbon-fibre composite, lined with acoustic blankets; Falcon's is 5.2 m in diameter and 13.2 m tall. Some fairings are split by pyrotechnic **frangible joints**, which fracture a grooved structural seam (Figure 34.9). Falcon's standard fairing releases mechanical latches and pushes the halves apart pneumatically. This system can be tested before flight and produces little shock or debris. Below the payload, a **payload adapter** carries loads into the upper stage and holds the **separation system**. This is typically a **clamp band** of standard diameter (937, 1 194 or 1 666 mm on Falcon). It is released by cutting its bolts, and springs push the spacecraft away.

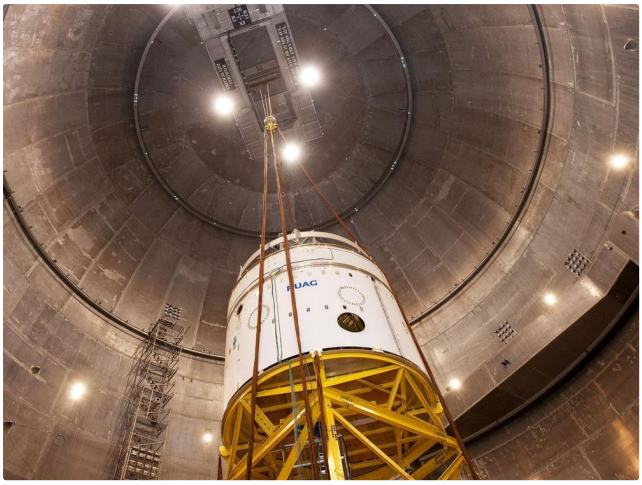


FIGURE 34.9 An Ariane 5 payload fairing in NASA's Space Power Facility at Plum Brook, Ohio, in 2012, before a full-scale test of a new horizontal pyrotechnic separation system that blew the two halves apart. Image: NASA/GRC.

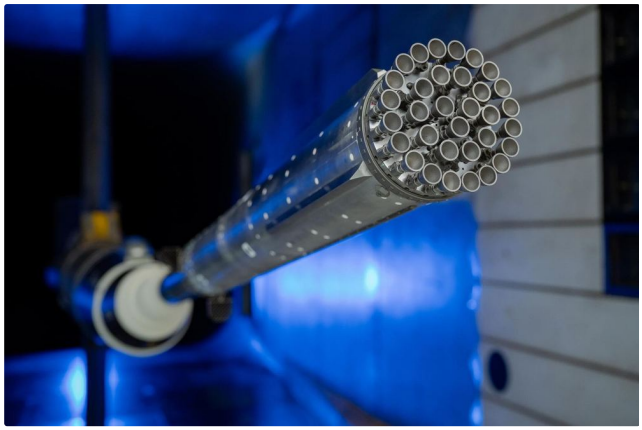


FIGURE 34.10 A 1.2 % scale model of the Super Heavy Version 3 booster in a wind tunnel at NASA's Ames Research Center in 2025, used to measure the aerodynamic forces during re-entry. Image: NASA/MSFC (NASA/Brandon Torres).

34.5 Avionics, Guidance, Navigation and Control

34.5.1 Navigation and Guidance

Navigation rests on an **inertial measurement unit** (IMU): gyroscopes, typically ring-laser or fibre-optic, measure rotation, and accelerometers measure specific force; integrating them gives attitude, velocity and position without outside help. Its errors grow with time, so it is blended with satellite navigation (Chapter 18) in a **Kalman filter**.

Guidance decides where to steer. In the dense atmosphere most vehicles fly a pre-computed, **open-loop** pitch programme that keeps α near zero. Above it, **closed-loop** guidance recomputes, many times a second, the steering needed to reach the target orbit from the current state. The Saturn V's Iterative Guidance Mode and the Shuttle's **Powered Explicit Guidance** both rest on one result. In uniform gravity, the fuel-optimal steering law is the **linear-tangent law**, in which the tangent of the pitch angle varies linearly with time. The reference simulation uses this law for its second stage, solving for the two coefficients that place it on a 200 km circular orbit.

34.5.2 Control and Redundancy

Liquid engines are mounted on **gimbals** and swivelled by a few degrees for **thrust vector control** (TVC). Actuators are hydraulic or, increasingly, electromechanical. Falcon's first-stage actuators use high-pressure kerosene from the engine as hydraulic fluid, removing a separate hydraulic system. The solid first stage of Vega-C and Starship's Raptors use electric actuators. Single-engine stages control roll with small thrusters, such as Falcon's nitrogen cold-gas jets. Most rockets are aerodynamically unstable — their centre of pressure lies ahead of the centre of gravity — so the control loop must actively hold them straight without reacting to structural vibration or propellant slosh. Returning boosters steer with **grid fins**, lattice-like surfaces that work at hypersonic speed and fold flat during ascent.

Avionics are **fault tolerant**: Falcon uses a three-string architecture that votes between triplicated computers and sensors. Several engines give propulsion redundancy — Falcon 9 can shut down a misbehaving first-stage engine and burn the others longer (Chapter 35). A study by The Aerospace Corporation is cited in the Falcon user's guide. It found that 91 % of launch failures in the two decades before 2001 were caused by engines, avionics or stage separation. These are the areas where redundancy and simplicity pay off most.

34.5.3 Range Safety and Flight Termination

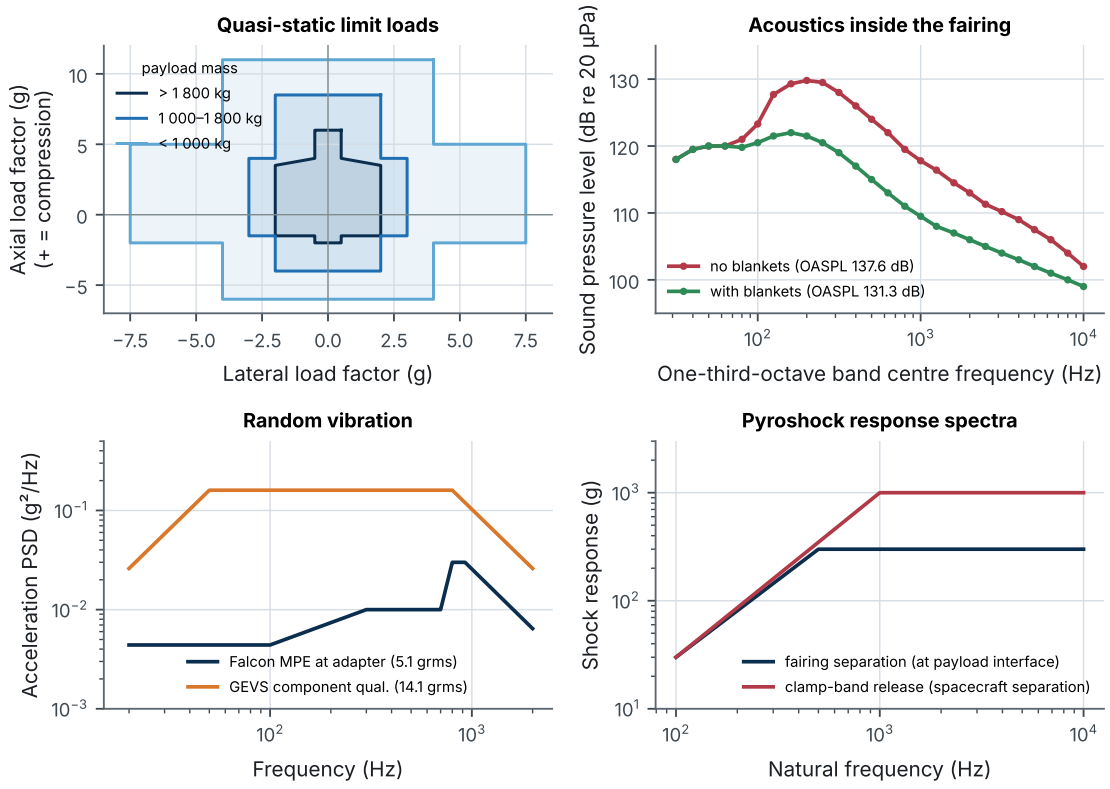
A rocket that strays threatens people on the ground. Ranges compute its **instantaneous impact point** — where it would land if thrust stopped — in real time. Every vehicle carries a **flight termination system**. Its charges cut open the tanks to disperse the propellant before it can reach populated areas. For decades a range safety officer watched radar and telemetry and sent the destruct command by radio. An **autonomous flight safety system** runs the same rules on board, using the vehicle's own GNSS and inertial navigation; its first operational use was on Falcon 9's CRS-10 launch on 19 February 2017. It removes tracking radars and ground crews and allows launches in quick succession. In the United States the Federal Aviation Administration licenses commercial launches under 14 CFR Part 450, which caps the collective risk to the public at an expected 1×10^{-4} casualties per launch.

34.6 The Payload's Ride

34.6.1 Environments

The **launch vehicle user's guide** defines the environments a satellite must survive and the interfaces it must meet. Figure 34.11 and Table 34.4 summarise those of Falcon 9, taken from SpaceX's guide.

- **Quasi-static loads** combine steady acceleration and low-frequency vibration into load factors: up to +6 g axially and ±2 g sideways for satellites heavier than 1 800 kg, more for lighter ones.
- **Sine vibration** (5–100 Hz) comes from engine transients and structural modes; to avoid coupling, Falcon asks for payload fundamental frequencies above 25 Hz axially and 10 Hz laterally.
- **Acoustic** noise inside the fairing, from engine exhaust at lift-off and turbulent airflow in transonic flight, reaches 131 dB overall with acoustic blankets and 138 dB without. Solar arrays and antennas suffer most.
- **Random vibration** through the adapter (5.1 grms) matters mainly for small components. Fairing and spacecraft separation cause **shock** of up to 1 000 g at kilohertz frequencies. This can break crystals, relays and solder joints near the source.
- The payload must also tolerate heating, rapid depressurisation and contamination; it is processed in clean rooms (ISO class 8 on Falcon) and encapsulated before transport to the pad (Figure 34.12).



Data: SpaceX Falcon User's Guide (May 2025) — limit loads, P95/50 acoustic MPE for the standard fairing (Eastern Range, 60 % fill), random-vibration MPE at the top of the payload adapter, shock response spectra; GEVS = NASA GSFC-STD-7000B generalised qualification level for components up to 22.7 kg. Qualification adds 25 % (loads) or 3 dB (acoustics, vibration).

FIGURE 34.11 Payload environments of Falcon 9, from the Falcon User's Guide (May 2025): quasi-static limit-load envelopes for three payload-mass classes; acoustic spectra inside the fairing with and without blankets; random vibration at the payload adapter compared with NASA's generalised component test level; and shock response spectra for fairing and spacecraft separation.

TABLE 34.4 Falcon 9 payload environments and the corresponding qualification tests (Falcon User's Guide, May 2025). MPE = maximum predicted environment.

Environment	Falcon 9 maximum predicted value	Typical verification test
Axial quasi-static load (> 1 800 kg)	+6.0 g compression, −2.0 g tension	Static load or sine burst, 1.25 × limit
Lateral quasi-static load (> 1 800 kg)	±2.0 g	As above
Sine vibration	0.5–1.0 g, 5–100 Hz	Sine sweep, 1.25 × limit
Acoustic (with blankets)	131.4 dB overall	Reverberant chamber, MPE + 3 dB
Random vibration at adapter	5.13 grms, 20–2 000 Hz	Shaker, MPE + 3 dB
Separation shock (clamp band)	up to 1 000 g, 1–10 kHz	Separation or pyroshock test
Fairing jettison	heating < 1 135 W/m²	Thermal analysis

Satellites are verified against these environments with margin. A **qualification** test, on a unit that will not fly, exceeds the predicted loads by 25 % or the acoustic and vibration levels by 3 dB. An **acceptance** test applies the predicted levels to flight hardware to reveal workmanship defects. Many programmes use **protoflight** testing, at qualification levels for a shorter time on the flight unit (Chapter 4). Late in development the launch provider runs a **coupled loads analysis**, combining finite-element models of rocket and satellite to predict the loads at every interface.

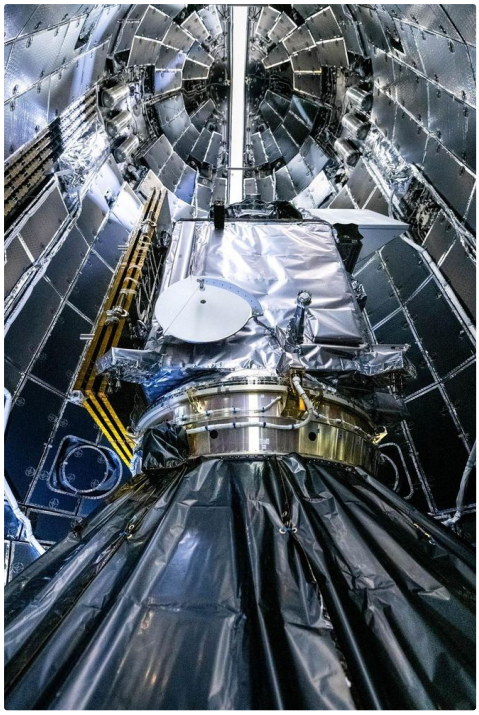


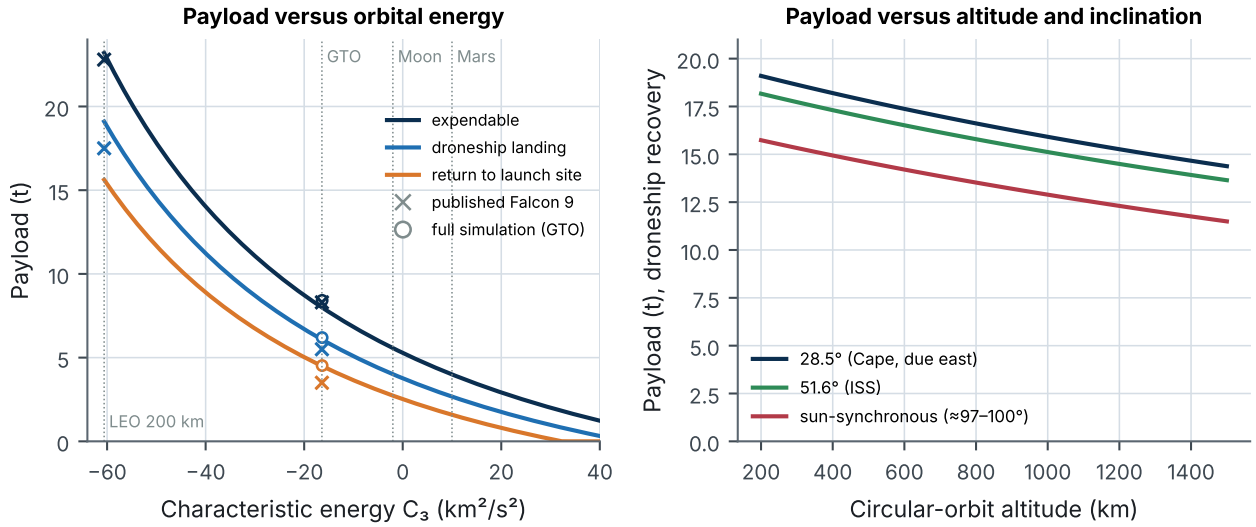
FIGURE 34.12 NASA's PACE Earth-observation satellite being encapsulated in a Falcon 9 fairing in January 2024. The work is done in a clean room to protect the instruments from contamination. Image: NASA/KSC (NASA Goddard Katie Mellos).



FIGURE 34.13 The Orion capsule that flew around the Moon on Artemis I, now an environmental test article, during vibro-acoustic testing at NASA's Armstrong Test Facility, Ohio, in 2024. Image: NASA/GRC (NASA/Sara Lowthian-Hanna).

34.6.2 Reading a Performance Curve

The other key part of a user's guide is the **performance curve** — payload against target orbit. Figure 34.14 shows the reference vehicle's, from a rocket-equation model calibrated to the trajectory simulation. For orbits reaching far from Earth the natural measure is the **characteristic energy** $C_3 = v_\infty^2$ (Chapter 24). It is negative for bound orbits (−60.6 km²/s² for a 200 km circular orbit, −16.4 km²/s² for GTO) and zero for escape. It is about 10 km²/s² for a typical trip to Mars. For low orbits, payload falls gently with altitude and with inclination. The model reproduces Falcon 9's published expendable capabilities well: 23.2 t to a 200 km orbit (published 22.8 t) and 8.4 t to GTO (8.3 t).



Rocket-equation model of the simulated vehicle; LEO requirement calibrated to the trajectory simulation (expendable 23.2 t, droneship 19.1 t, RTLS 15.6 t to 200 km); higher orbits add impulsive Δv from a 200 km parking orbit; inclination changes the Earth-rotation gain (465 m/s × cos i), no dog-legs; booster reserves held at their LEO values.

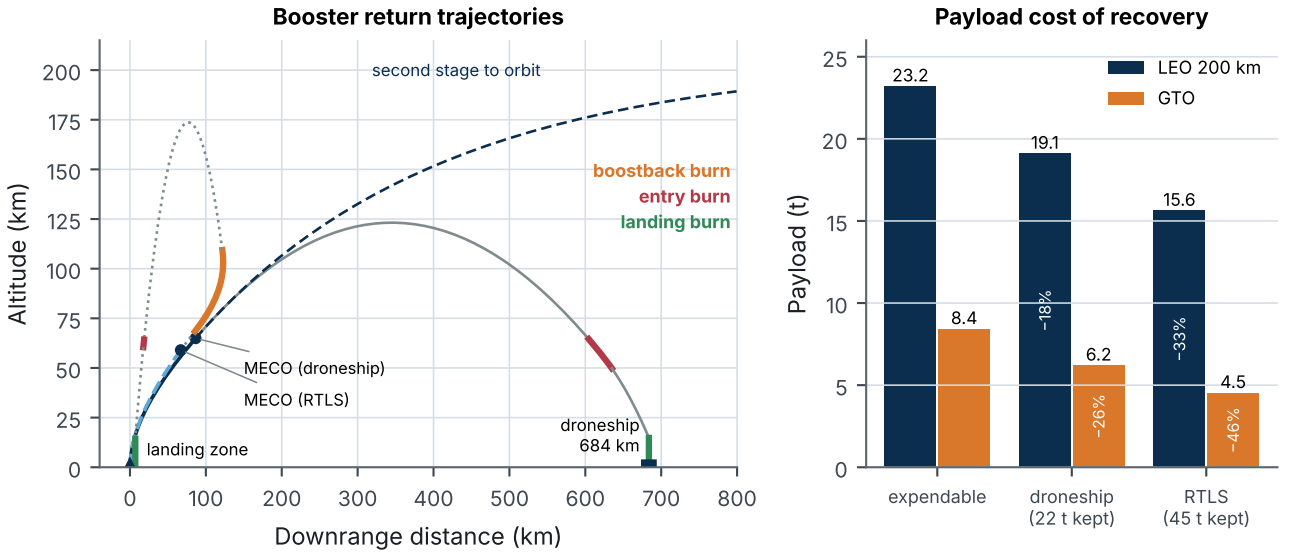
FIGURE 34.14 Left: payload of the reference vehicle versus orbital energy for three booster-recovery modes, compared with published Falcon 9 values (crosses) and the full trajectory simulation (circles). Right: payload to circular orbits of different altitude and inclination with droneship recovery.

34.7 Reusability

34.7.1 Landing a Booster

For most of the space age every stage was thrown away. The Space Shuttle (Chapter 10) recovered its orbiter and boosters but needed months of refurbishment. The modern approach, pioneered by SpaceX's first Falcon 9 landing in December 2015 and first re-flight in 2017, flies the booster back under its own power to a vertical landing on legs. It must keep propellant for up to three burns. A **boostback** burn reverses its course, and an **entry burn** slows it before the dense atmosphere, limiting heating and loads. A **landing burn** is timed so that velocity reaches zero just as the legs touch down, because even at minimum throttle one engine lifts more than the empty stage weighs. In a **return to launch site** (RTLS) the booster cancels its downrange velocity and flies back to a landing zone near the pad. In a **droneship** landing it continues on its ballistic arc to an autonomous barge hundreds of kilometres downrange. This usually needs no boostback burn.

Figure 34.15 shows both for the reference vehicle. Every kilogram kept in the booster is a kilogram not used to accelerate the upper stage, so the first stage cuts off earlier and slower. In the simulation a droneship landing keeps 22 t of propellant and costs 18 % of the LEO payload; RTLS keeps 45 t and costs 33 %. GTO missions need a faster staging speed and so more braking: the penalties grow to 26 % and 46 %. Falcon 9's published figures show larger penalties (22.8 and 17.5 t to LEO; 8.3, 5.5 and 3.5 t to GTO), partly because the model ignores the legs, grid fins and performance reserves.



Same model as the ascent figure. Booster: 25.6 t dry, 3 engines for boostback (15° above horizontal) and entry burns (below 65 km, to 1.3 km/s), one engine at 75 % for landing; kept propellant = burns + 1.5 t residuals, iterated to consistency; payload re-optimised for each case. GTO = 200 × 35 786 km.

FIGURE 34.15 Left: simulated booster return trajectories for a droneship landing about 680 km downrange and for a return to the launch site, with the boostback (orange), entry (red) and landing (green) burns highlighted. Right: payload delivered to LEO and GTO in each recovery mode.

EXAMPLE: THE PROPELLANT BILL FOR LANDING

The reference booster has a dry mass of 25.6 t, plus 1.5 t of unusable residual propellant. Its exhaust speed is 3.06 km/s in vacuum and 2.78 km/s at sea level. Work backwards from touchdown with $m_{before} = m_{after}\, e^{\Delta v / v_e}$:

1. **Droneship.** Landing burn, 0.44 km/s at sea level: 27.1 t × $e^{0.44/2.78}$ = 31.8 t. Entry burn, 1.23 km/s: 31.8 × $e^{1.23/3.06}$ = 47.5 t. Propellant kept at MECO: 47.5 − 25.6 = **21.9 t** (simulation: 22.0 t).
2. **RTLS.** At MECO the stage is moving downrange at 1.87 km/s. Landing burn 0.38 km/s: 27.1 × $e^{0.38/2.78}$ = 31.1 t; entry burn 0.21 km/s: 33.3 t; boostback burn 2.32 km/s: 33.3 × $e^{2.32/3.06}$ = 71.0 t. Propellant kept: **45.4 t**, more than twice as much.
3. Payload: in the simulation each tonne kept in the booster costs about 0.17–0.19 t of LEO payload, roughly one tonne for every five to six held back. This is because the upper stage still provides most of the Δv. RTLS suits light payloads, where saving the time and cost of a droneship voyage matters more than the lost mass.

34.7.2 Records, Catches and Refurbishment

Falcon has turned landing into routine. SpaceX made its 600th Falcon booster landing on 19 April 2026, and the fleet leader, B1067, flew for the 37th time on 25 August 2026. By February 2025 SpaceX had also re-flown fairing halves, recovered from the sea, on 307 missions. Between flights engines, heat shielding, grid fins and legs are inspected and a few parts replaced; Falcon Heavy's side boosters routinely land side by side (Figure 34.16).

Others are following. Blue Origin's New Glenn landed its first booster in November 2025 and re-flew it in April 2026. China recovered its first orbital booster on 10 July 2026. On its maiden flight, the first stage of the new **Long March 10B** was caught at sea by hooks that snagged cables tensioned over a net on the ship *Linghangzhe*. In August 2026 LandSpace's stainless-steel **Zhuque-3** landed on legs at a desert site in Gansu province, although a fire broke out after touchdown. SpaceX's Super Heavy booster is caught by the launch tower's mechanical "chopstick" arms

instead of landing on legs. The first catch came on Starship Flight 5 on 13 October 2024 and was repeated on Flights 7 and 8 in 2025. A caught booster re-flew on Flight 9. On 28 September 2026 Starship reached orbit for the first time on Flight 14; catching the ship, needed to make the whole vehicle reusable, was planned for a later flight.

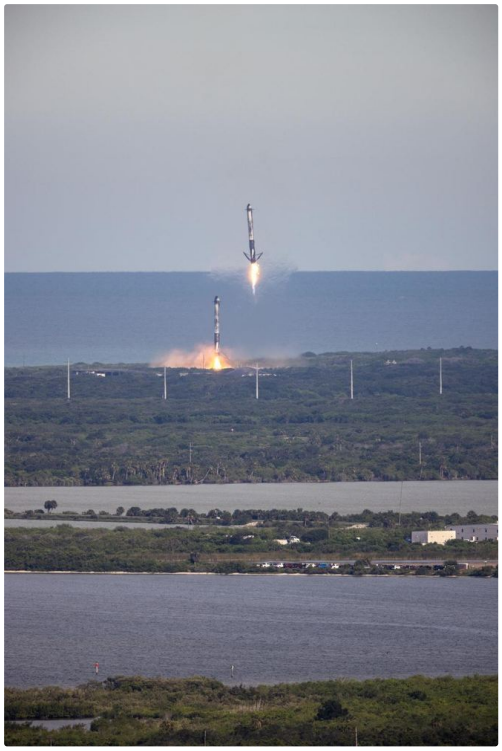


FIGURE 34.16 Falcon Heavy's two side boosters returning to Landing Zones 1 and 2 at Cape Canaveral after launching the GOES-U weather satellite in June 2024. Image: NASA/KSC (NASA/Amber Jean Notvest).

34.7.3 The Economics of Launch

Reuse matters because hardware dominates launch cost: flying a first stage ten or thirty times spreads the cost of most of a vehicle's engines and structure. It pays only at a high flight rate, because factories, recovery ships and launch teams are fixed costs. Falcon 9 flew 165 times in 2025, half of all orbital launch attempts worldwide. Starlink satellites made up 3 169 of the 4 511 payloads launched that year — an in-house customer that keeps the fleet busy.

Table 34.5 compares launch prices per kilogram; list prices are not costs, and rockets rarely fly full. A 2018 NASA analysis found that vehicles introduced between 1970 and 2000 averaged about $18 500 per kilogram to LEO (2018 dollars), the Space Shuttle $61 700, and Falcon 9 at its advertised price $2 700 — less than a twentieth of the Shuttle's figure. Prices have since risen with inflation and demand: in 2026 SpaceX listed a dedicated Falcon 9 at $74 million and rideshare at $350 000 for 50 kg to sun-synchronous orbit. Cheaper access has transformed demand: the mass launched to orbit grew from about 400 t in 2015 to about 3 500 t in 2025.

TABLE 34.5 Launch prices per kilogram to low Earth orbit. Historical values from Jones (NASA, 2018); 2026 values from SpaceX's published prices in current dollars; per-kilogram values assume a full vehicle.

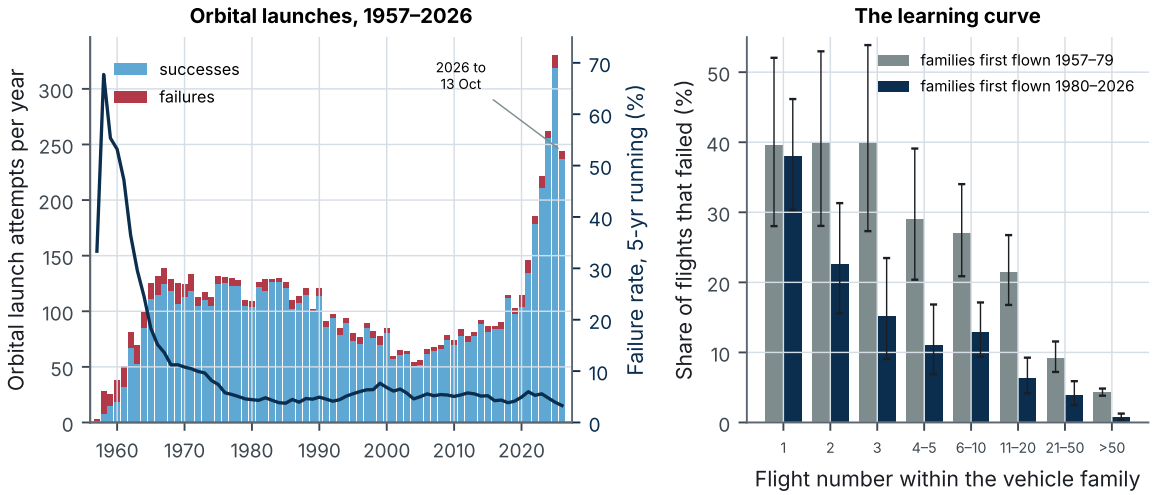
Vehicle	First flight	Price per launch	Payload to LEO	Price per kg
Vanguard	1957	—	very small	≈ $895 000 (2018 $)
Saturn V	1967	$728 M (2018 $)	140 t	$5 200 (2018 $)
Space Shuttle	1981	$1 697 M (2018 $)	27.5 t	$61 700 (2018 $)

Vehicle	First flight	Price per launch	Payload to LEO	Price per kg
Ariane 5 G	1996	—	—	$13 100 (2018 $)
Falcon 9 (2018 list)	2010	$62 M	22.8 t	$2 700
Falcon Heavy (2018 list)	2018	$90 M	63.8 t	$1 400
Falcon 9 (2026 list, booster landed)	2010	$74 M	≈ 17.5 t	≈ $4 200
Falcon 9 rideshare (2026)	—	$350 000 per 50 kg	—	$7 000

34.8 Reliability and Testing

34.8.1 How Often Do Rockets Fail?

Of 330 orbital launch attempts in 2025, 11 failed — 3.3 %. Figure 34.17 puts this in context using Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT). In the late 1950s a third or more of all launches failed. The five-year average fell below 10 % in the early 1970s. It has stayed at roughly 3–8 % since the mid-1970s, even as the launch rate tripled after 2020. The right-hand panel shows that **reliability is learned**. Across the 100 vehicle families first flown since 1980, 38 % of maiden flights failed, 23 % of second flights and 15 % of third flights. From the 21st to the 50th flight the rate was 4 %, and beyond the 50th, 0.7 %. New vehicles meet combinations of conditions that no ground test fully reproduces. Mature vehicles are very reliable: by October 2026 Falcon 9 had made 695 orbital attempts with 2 failures (99.7 %), and Soyuz more than 1 180 with 97.4 % success (Appendix E).



Data: GCAT (J. McDowell, launch list to 13 Oct 2026); partial failures counted as failures. Right: 143 vehicle families; error bars are 90 % Jeffreys intervals.

FIGURE 34.17 Left: orbital launch attempts per year since 1957, split into successes and failures, with the five-year running failure rate. Right: the share of flights that failed as a function of flight number within each vehicle family, for families first flown before and after 1980.

A few flights say little about a new vehicle. With k failures in n flights a reasonable estimate of failure probability is $(k + ½)/(n + 1)$, the mean of the Jeffreys distribution used for the error bars of Figure 34.17. A rocket with ten successes in ten flights thus has an estimated failure probability of about 5 %. The upper end of its 90 % interval lies near 17 %. Only a hundred clean flights bring it below 2 %.

34.8.2 Why Rockets Fail

Propulsion causes the largest share of failures; Chapter 35 describes the engine failures that shaped design practice. Table 34.6 lists failures elsewhere in the vehicle — software, sensors, structures, separation and weather — and their lessons.

TABLE 34.6 Launch failures outside the main engines and their lessons.

Event	What failed	Lesson
Atlas-Centaur AC-67, Mar 1987	Launched into cloud; lightning altered guidance-computer memory	Stricter weather launch-commit criteria
Ariane 5 flight 501, Jun 1996	Reused Ariane 4 inertial-reference software overflowed	Re-validate reused software for the new flight
Taurus XL, Feb 2009 and Mar 2011	Fairing did not separate; traced to falsified tests of aluminium parts	Verify the supply chain
Proton-M, Jul 2013	Angular-rate sensors installed upside down	Error-proof mounting; inspection
Falcon 9 CRS-7, Jun 2015	Strut holding a helium COPV in the upper-stage oxygen tank broke	Test critical small parts individually
Falcon 9 AMOS-6, Sep 2016	Oxygen trapped in a COPV overwrap ignited during fuelling on the pad	Understand cryogenic loading; change procedures
H3 flight 8, Dec 2025	Bonded composite payload adapter, delaminated in manufacture, failed under fairing-separation shock; satellite lost	Shock exposes hidden flaws; bolted adapter adopted

34.8.3 Investigations and Testing

After a failure the operator leads a **mishap investigation** — in the United States under FAA oversight — and the vehicle is grounded until the cause is understood and the fix accepted. The evidence is mainly telemetry, analysed with fault trees that work through every possible cause. Falcon 9's Starlink 9-3 failure of 11 July 2024 shows the process at its fastest. An oxygen leak was traced to a pressure-sensor line cracked by vibration fatigue and a loose clamp; it chilled the engine's ignition system and caused a hard start at relight (Chapter 35). SpaceX removed the line, inspected clamps across the fleet, and the FAA cleared the vehicle on 25 July. Investigations take longer when the cause is less obvious. On New Glenn's third flight, in April 2026, one upper-stage engine fell short of full thrust, stranding its satellite in too low an orbit. Five weeks later the FAA accepted Blue Origin's finding that a cryogenic leak had frozen a hydraulic line. Japan's H3 needed six months: its December 2025 failure was traced to a manufacturing flaw in its bonded carbon-fibre payload adapter that grew under the shock of fairing separation. With a bolted adapter, H3 flew successfully again in June and August 2026.

The guiding rule of testing is **test like you fly**. Components are qualified on test stands, stages are fired whole (Chapter 35) and complete vehicles perform a **static fire** on the pad. Structures are tested to failure to learn their true margins. Two philosophies coexist. The traditional approach spends years on analysis and ground testing to make the first flight as safe as possible. The **iterative**, hardware-rich approach builds cheaper test articles, flies early and expects to lose some — SpaceX's approach with Starship, which lost several upper stages in 2025 before reaching orbit in 2026. Both converge on the lesson of Figure 34.17: reliability is demonstrated by flying.

AI IN LAUNCH-VEHICLE ENGINEERING

Convex optimisation already flies: the G-FOLD algorithm, developed at NASA's Jet Propulsion Laboratory, computes fuel-optimal landing trajectories in milliseconds with guaranteed convergence, and SpaceX has described landing Falcon 9 boosters with automatically generated convex-optimisation code. Machine learning is being added around such algorithms: anomaly detectors scan thousands of telemetry channels for patterns that precede failures, and learned surrogates of loads and engine behaviour speed up design optimisation. Flight-critical software still rests on verifiable algorithms (Chapters 52 and 53).

KEY TAKEAWAYS

- Reaching a 200 km orbit takes about 9.0–9.6 km/s of ideal Δv. This is 7.8 km/s of orbital speed plus about 1.5–2 km/s of gravity, steering, drag and back-pressure losses. Earth's rotation subtracts up to 0.46 km/s (465 m/s × cos *i*).
- Launch latitude matters most for geostationary missions: removing a 28.5° inclination at apogee costs 0.35 km/s more than removing 5°.
- With realistic structures one stage cannot reach orbit with useful payload; two stages capture most of the benefit of staging.
- A gravity turn keeps the angle of attack near zero; engines throttle through max-Q (typically 25–40 kPa) and acceleration is capped at about 4–5 g.
- Tanks are thin pressurised shells of aluminium–lithium, stainless steel or composite; payloads must survive 130–140 dB of noise and loads of several g.
- Landing a booster costs payload — about a fifth for a droneship landing and a third for a return to the launch site — but enables dozens of re-flights.
- About 40 % of maiden flights fail; reliability grows with flight experience to 99 % or better for mature vehicles.

REVIEW QUESTIONS

1. List the four losses in a launch Δv budget and explain why steering loss can rival gravity loss for a vehicle with a low-thrust upper stage.
2. Show that the Earth-rotation gain for a launch into an orbit of inclination *i* is 465 m/s × cos *i*, whatever the site latitude. What is it for a 97.5° sun-synchronous orbit?
3. A 4 000 kg geostationary satellite with a 320 s apogee engine is launched from Kourou or from Cape Canaveral. Estimate the difference in the propellant it needs.
4. For stages with Isp 340 s and ε = 0.07, compute the payload fraction of one, two and three identical stages for Δv = 9.4 km/s.
5. Why do launch vehicles fly a gravity turn? What is max-qα, and how is it managed on launch day?
6. Explain the purpose of a throttle bucket and why acceleration rises during each stage's burn.
7. Compare aluminium–lithium, stainless steel and composite tanks. Why might a reusable stage use steel?
8. What is POGO, and how is it suppressed?
9. A satellite's lowest lateral natural frequency is 8 Hz. Why might the launch provider object, and what analysis would settle the question?
10. Estimate the propellant a 25 t booster must keep for a 3 km/s return-to-launch-site profile with Isp 300 s, and explain why the payload penalty is much smaller than the propellant kept.

35 CHAPTER Rocket Propulsion



Image: NASA/SSC (Danny Nowlin)

A rocket engine is a machine for throwing mass backwards as fast as possible. This chapter explains how engines and motors produce thrust and how performance is measured — specific impulse, characteristic velocity and thrust coefficient — and how a converging–diverging nozzle turns hot gas into a supersonic jet whose performance depends on altitude. It compares liquid, solid and hybrid propellants, explains the cycles that feed liquid engines, from pressure-fed thrusters to full-flow staged combustion and battery-driven pumps, and describes turbopumps, injectors, combustion instability, cooling, throttling and 3-D printing. It covers solid-motor ballistics, hybrid physics and engine testing, using the engines of 2026 as examples.

IN THIS CHAPTER

35.1 How a Rocket Makes Thrust
35.2 Nozzle Flow
35.3 Propellants
35.4 Liquid-Propellant Engines
35.5 Engines of 2026
35.6 Solid Rocket Motors
35.7 Hybrid Rockets
35.8 Testing and Qualification
35.9 Beyond Chemical Propulsion

Every launch vehicle and almost every spacecraft moves by one principle. It carries its own propellant and releases the propellant's chemical energy as heat. It then expels the hot products backwards through a nozzle at several kilometres per second. The reaction to that stream of mass is **thrust**. Engines burning liquid propellants are called **rocket engines**; those with propellant cast as a solid block are **rocket motors**.

A Super Heavy booster lights 33 Raptor engines that together produce about 80 MN — more than eight thousand tonnes-force. By contrast, a Rutherford engine on Rocket Lab's Electron weighs 35 kg and produces 25 kN. Both obey the same few equations. Chapter 34 shows how engines are combined into stages and derives the rocket equation, Chapter 36 surveys the world's launch vehicles, and Chapter 6 covers spacecraft thrusters and electric propulsion.

35.1 How a Rocket Makes Thrust

35.1.1 The Thrust Equation

Draw an imaginary boundary around an engine. Propellant enters at almost zero velocity and leaves through the nozzle exit at velocity v_e with mass flow rate $\dot{m}$. Newton's second law says the force on the engine equals the rate at which momentum leaves the boundary, plus any unbalanced pressure force across the exit plane:

$$F = \dot{m}\, v_e + (p_e - p_a)\, A_e$$

The first term is the **momentum thrust**. The second, the **pressure thrust**, is positive if the gas at the exit plane (pressure p_e, area A_e) is at a higher pressure than the surrounding air (p_a). It is negative if the pressure is lower. In vacuum $p_a = 0$, so every engine produces more thrust in space than at sea level — 8 % more for the Merlin 1D, 23 % for the RS-25. Both terms together define the **effective exhaust velocity** $c = F/\dot{m}$, which therefore changes with altitude.

35.1.2 Specific Impulse

Dividing the effective exhaust velocity by standard gravity g_0 = 9.806 65 m/s² gives the **specific impulse** in seconds:

$$I_{sp} = \frac{F}{\dot{m}\, g_0} = \frac{c}{g_0}$$

Specific impulse is the number of seconds that one kilogram of propellant can produce one kilogram-force of thrust — the "fuel economy" of a rocket. The rocket equation, $\Delta v = I_{sp}\, g_0 \ln(m_0/m_f)$ (Chapter 34), shows that the velocity a stage can deliver grows in direct proportion to it. Typical vacuum values are 250–290 s for solid motors, 310–340 s for kerosene, about 340–380 s for methane and 420–465 s for hydrogen engines. An engine's **thrust-to-weight ratio** measures how compact it is: about 35–60 for hydrogen engines and 70–185 for kerosene and methane engines.

35.1.3 Characteristic Velocity and Thrust Coefficient

Two further parameters separate the work of the combustion chamber from that of the nozzle. When the flow is choked at the nozzle throat (explained below), the mass flow is fixed by the chamber pressure p_c and the throat area A_t:

$$\dot{m} = \frac{p_c A_t}{c^*}, \qquad F = C_F\, p_c A_t, \qquad c = c^* C_F$$

The **characteristic velocity** c^* ("c-star") depends only on the combustion gas — its temperature T_c, mean molar mass $\mathcal{M}$ and ratio of specific heats γ. It is 1 500–1 600 m/s for solid propellants, about 1 800 m/s for kerosene or methane with oxygen and 2 300–2 400 m/s for hydrogen–oxygen.

The ratio of measured to theoretical c^*, the **c* efficiency** (typically 95–99 %), shows how well the injector mixes and burns the propellants. The **thrust coefficient** C_F, typically 1.3–2.0, measures how much the nozzle amplifies the force that the chamber pressure would exert on the throat alone; it depends only on γ, the **expansion ratio** ε = A_e/A_t and the ambient pressure.

35.2 Nozzle Flow

35.2.1 The Ideal Rocket

The nozzle turns heat into directed motion. For the **ideal rocket** — steady, one-dimensional flow of a perfect gas with constant γ, no friction and no heat loss — the energy equation gives the exit velocity:

$$v_e = \sqrt{\frac{2\gamma}{\gamma-1}\,\frac{R_u T_c}{\mathcal{M}}\left[1-\left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right]}$$

where R_u = 8 314 J/(kmol·K). Exhaust velocity therefore rises with $\sqrt{T_c/\mathcal{M}}$: light exhaust matters as much as a hot flame. Hydrogen–oxygen burns slightly cooler than kerosene–oxygen, but its exhaust (mean molar mass about 13.5 kg/kmol) is much lighter than kerosene's (about 23.5). With γ = 1.2 and a pressure ratio of 1 000 the formula gives 4 200 m/s for hydrogen and 3 270 m/s for kerosene. Because the bracket approaches 1 slowly, higher pressure ratios add little velocity.

35.2.2 Area, Mach Number and the Throat

Subsonic gas accelerates in a narrowing duct, and supersonic gas in a widening one. So a rocket nozzle converges to a **throat**, where the flow reaches exactly Mach 1, and then diverges — the **de Laval nozzle**. For γ = 1.2 the throat pressure is 56 % and the throat temperature 91 % of the chamber values, so the throat carries the highest heat load. Downstream the Mach number climbs to 3–5 (Figure 35.1), following the **area–Mach relation**:

$$\frac{A}{A_t} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

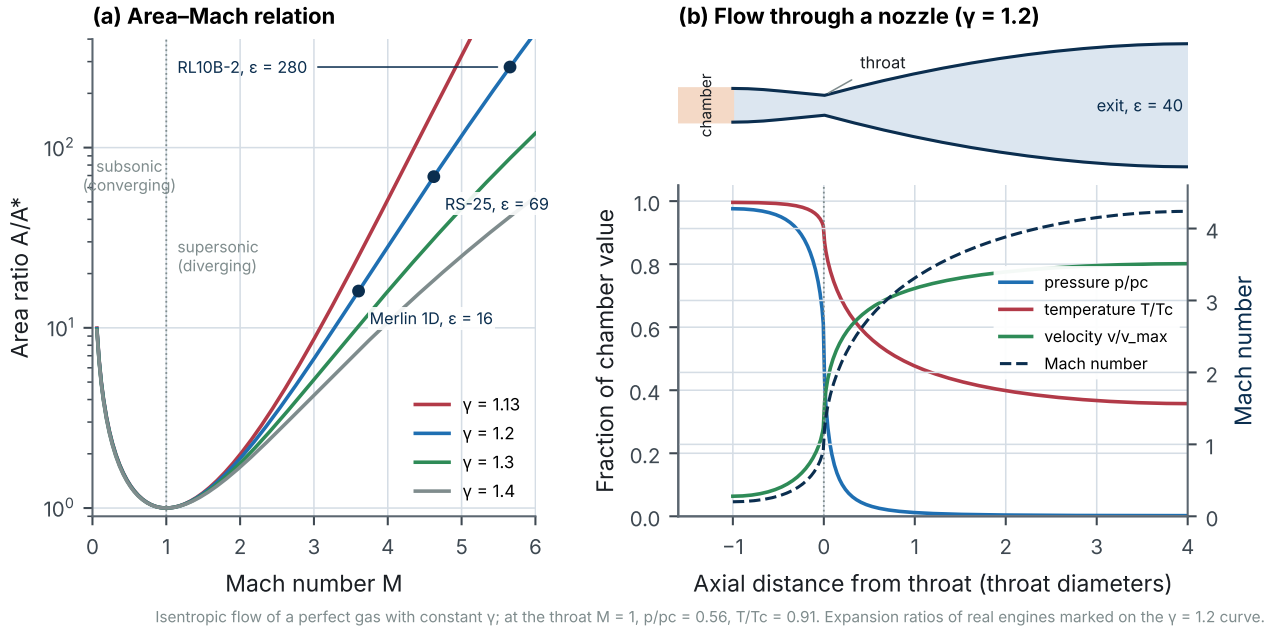


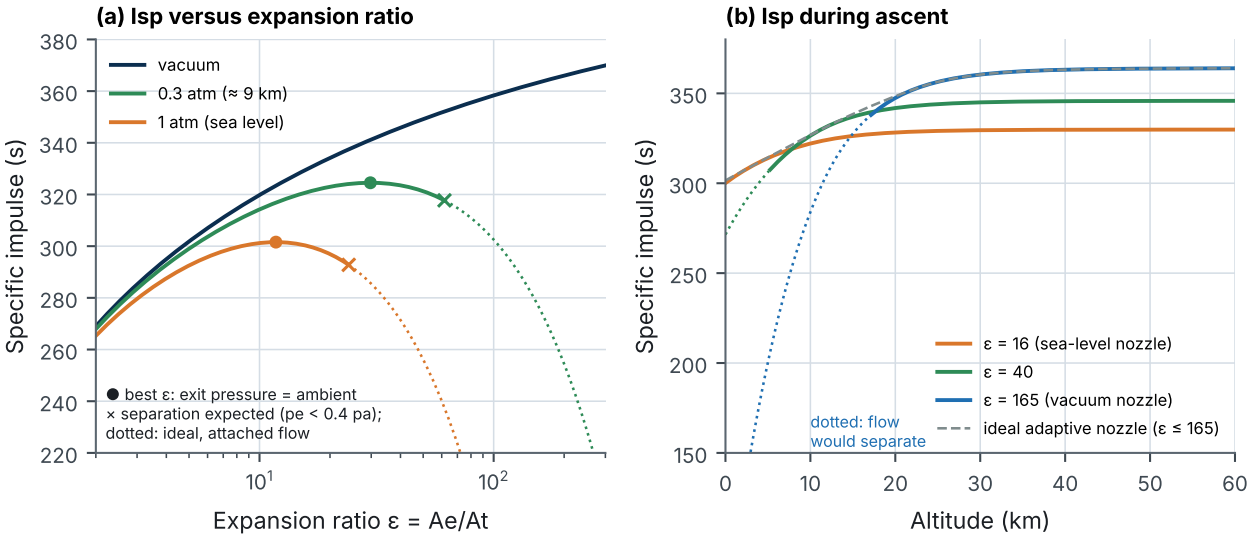
FIGURE 35.1 The de Laval nozzle. (a) Area ratio needed to reach each Mach number for several values of γ, with the expansion ratios of three real engines. (b) Pressure, temperature, velocity and Mach number along a nozzle with ε = 40: pressure falls by almost half before the throat, and most of the velocity is gained within a few throat diameters downstream of it.

Once the throat is **choked**, no disturbance can travel upstream, so the mass flow depends only on chamber conditions and throat area; designers size an engine by choosing p_c and A_t first. A simple 15° conical nozzle loses 1.7 % of thrust because the exhaust leaves at an angle (divergence factor $(1+\cos\alpha)/2$ = 0.983). Most engines use a contoured **bell nozzle**, which gives the same performance at about 80 % of the cone's length.

35.2.3 Expansion Ratio and Altitude

A nozzle gives its best thrust when exit pressure equals ambient pressure. If $p_e > p_a$ it is **under-expanded** and the plume balloons outwards after the exit. If $p_e < p_a$ it is **over-expanded**: the air squeezes the jet into the diamond-shaped shock pattern seen in sea-level engine tests, and the pressure term subtracts thrust. Below roughly 40 % of ambient pressure the flow **separates** from the wall inside the nozzle, and asymmetric separation produces side loads that can break it. A first-stage nozzle is therefore a compromise: it must work at sea level but spends most of its burn at altitude (Figure 35.2).

With the same chamber, a nozzle of ε = 16 gives 300 s at sea level and 330 s in vacuum. A vacuum nozzle of ε = 165, like the Merlin Vacuum's, gives 364 s in space but would separate violently at sea level. Vehicles solve the problem with stages: first-stage engines use moderate expansion ratios (Merlin 1D 16, Raptor 34, RD-180 37), upper-stage engines large ones (Vinci 240, RL10B-2 280). The RL10B-2 and Vinci also have **extendible nozzles** that slide out after staging. There are also **dual-bell** and **aerospike** nozzles, which adapt to the surrounding pressure. They have been tested many times, but none has yet powered an orbital launch.



Ideal rocket, γ = 1.2, chamber pressure 10 MPa, c* = 1 800 m/s (typical of LOX/RP-1 or LOX/CH_4); US Standard Atmosphere; Summerfield separation criterion pe ≈ 0.4 pa.

FIGURE 35.2 Why nozzles are sized for altitude (chamber pressure 10 MPa). (a) Specific impulse versus expansion ratio for three ambient pressures; the best ratio grows from about 12 at sea level to infinity in vacuum. (b) Isp during ascent for a sea-level nozzle (ε = 16), an intermediate one and a vacuum nozzle (ε = 165); the dashed curve is an ideal nozzle that adapts to the surroundings.



FIGURE 35.3 The 27 Merlin engines of a Falcon Heavy at lift-off with NASA's Nancy Grace Roman Space Telescope on 30 August 2026. The sea-level engines are slightly over-expanded at lift-off. Image: NASA/KSC (SpaceX).

Real engines fall short of the ideal through incomplete mixing, boundary-layer losses, exhaust divergence and chemistry that lags the falling temperature. Published vacuum Isp can be compared with ideal values computed for each engine's own conditions. This gives about 92 % for the Merlin 1D, 94 % for the RD-180, 95 % for Raptor, 97 % for the RL10B-2 and 98 % for the RS-25. Solid motors lose a little more because alumina droplets lag behind the gas.

EXAMPLE: SPECIFIC IMPULSE OF THE RS-25 FROM C* AND THE THRUST COEFFICIENT

The RS-25 burns hydrogen and oxygen at a mixture ratio of 6.0, with p_c = 20.6 MPa and ε = 69. Its rated vacuum thrust is 2 279 kN.

1. **Characteristic velocity.** An equilibrium calculation gives a theoretical c^* of 2 322 m/s; with a c* efficiency of 99 %, c^* = 2 299 m/s.
2. **Exit pressure.** With γ = 1.2 the area–Mach relation gives M_e = 4.6 for ε = 69 and p_e/p_c = 0.001 05, so p_e = 21.6 kPa.
3. **Thrust coefficient.** In vacuum C_F = 1.855 (momentum) + 0.072 (pressure term $p_e\varepsilon/p_c$) = 1.927. At sea level subtract $p_a\varepsilon/p_c$ = 0.339: C_F = 1.587.
4. **Specific impulse.** $I_{sp} = c^*C_F/g_0$ = 2 299 × 1.927 / 9.807 = **452 s** in vacuum (published: 452.3 s) and 372 s at sea level (published: 366 s).
5. **Throat size.** $A_t = F/(C_F p_c)$ = 2 279 000 / (1.927 × 20.6 × 10^6) = 0.0574 m^2, a throat 27 cm across. The mass flow $p_c A_t/c^*$ is **514 kg/s** — a tonne of propellant every two seconds.

EXAMPLE: SIZING A FIRST-STAGE NOZZLE

A kerosene engine must give 850 kN at sea level with p_c = 10 MPa, γ = 1.2 and c^* = 1 746 m/s (97 % of 1 800 m/s). To keep the nozzle short and avoid separation, choose an exit pressure of 70 kPa.

1. **Expansion ratio.** p_c/p_e = 143; the isentropic relations give M_e = 3.6 and **ε = 15.6**.
2. **Thrust coefficient.** Momentum term 1.685; pressure term (70 − 101.3) kPa × 15.6 / 10 MPa = −0.049; so C_F = 1.636 at sea level. Since p_e/p_a = 0.69 > 0.4, the flow stays attached.
3. **Throat and exit.** $A_t = F/(C_F p_c)$ = 0.0520 m^2 (diameter 25.7 cm); $A_e = \varepsilon A_t$ = 0.810 m^2 (diameter **1.02 m**).
4. **Performance.** I_{sp} = 291 s at sea level and 320 s in vacuum, where thrust rises to 932 kN; the mass flow is 298 kg/s. The Merlin 1D (ε = 16, 845/914 kN, 282/311 s) is close, with slightly lower real efficiency.

35.3 Propellants

35.3.1 Choosing a Combination

A good combination releases much energy per kilogram, produces light exhaust, is dense, storable, safe and cheap, and can cool the chamber. None is best at everything (Table 35.1, Figure 35.4).

Liquid hydrogen with liquid oxygen (LOX/LH_2) gives the highest Isp of any practical chemical combination. But hydrogen is a cryogenic liquid at 20 K with a density of only 71 kg/m^3. Its tanks are huge and must be insulated against boil-off, and the hydrogen can embrittle metals. Hydrogen powers upper stages and core stages that burn from the ground up (RS-25, Vulcain 2.1, LE-9, YF-77). Engines run fuel-rich, at O/F 5–6 instead of the stoichiometric 7.9, because lighter exhaust and smaller hydrogen tanks more than repay the lost energy.

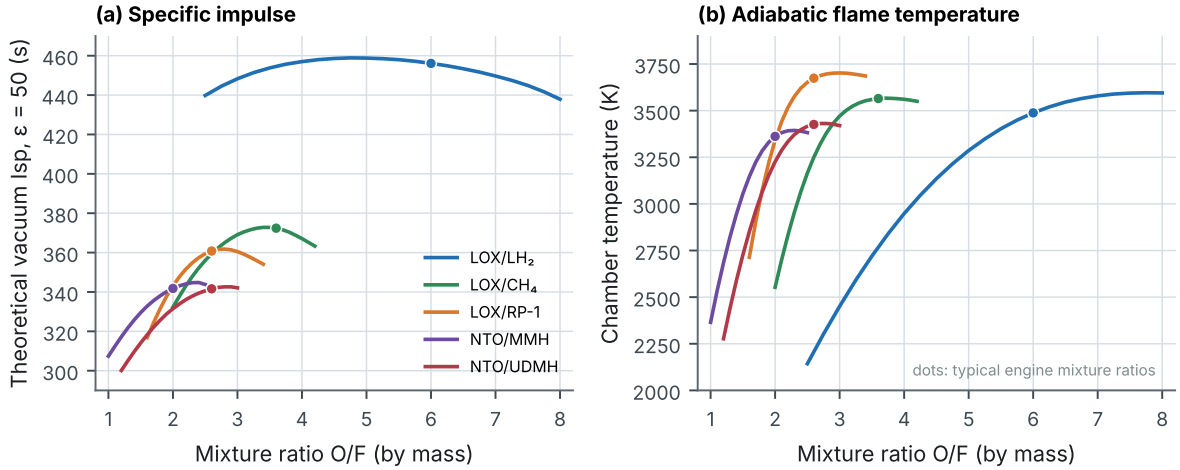
Kerosene with oxygen (LOX/RP-1, a highly refined kerosene) is dense and cheap, and the fuel is stored at room temperature. Its drawbacks are lower Isp and **coking** — carbon deposits in cooling channels and soot on turbines — which complicates reuse. SpaceX chills Falcon 9's oxygen well below its boiling point and its kerosene below room temperature to pack more propellant into

the tanks. **Methane with oxygen** ("methalox") sits between the two: better Isp than kerosene, far denser than hydrogen and little coking. Its boiling point (112 K) is close to that of oxygen (90 K). So the tanks can share a bulkhead, and each propellant can pressurise its own tank with its own vapour. Methane can also be made on Mars. China's Zhuque-2 was the first methane-fuelled rocket to reach orbit, on 12 July 2023, followed by, among others, Vulcan (January 2024), New Glenn (January 2025), China's Zhuque-3 (December 2025) and, in September 2026, Starship.

Storable hypergolic propellants — nitrogen tetroxide (NTO) with monomethylhydrazine (MMH), unsymmetrical dimethylhydrazine (UDMH) or UH25 (UDMH with 25 % hydrazine hydrate) — ignite on contact and stay liquid at room temperature for years. They power spacecraft (Chapter 6), some upper stages and India's Vikas engine, but are highly toxic.

TABLE 35.1 Propellant performance. Liquid values are theoretical (shifting equilibrium, chamber pressure 6.9 MPa, sea-level Isp with optimum expansion, vacuum Isp with ε = 50), computed for this book; solid values are delivered figures of the Shuttle booster and GEM 63XL.

Propellants	O/F	T_c (K)	c^* (m/s)	I_{sp} SL / vac (s)	Bulk density (kg/m³)	Storage
LOX/LH_2	6.0	3 490	2 305	382 / 456	361	Both cryogenic (90 K, 20 K)
LOX/CH_4	3.6	3 565	1 823	307 / 372	833	Both cryogenic (90 K, 112 K)
LOX/RP-1	2.6	3 674	1 793	300 / 361	1 025	Fuel storable
NTO/MMH	2.0	3 362	1 755	288 / 342	1 186	Storable, hypergolic
NTO/UDMH	2.6	3 426	1 726	286 / 342	1 174	Storable, hypergolic
APCP solid	—	—	≈ 1 550	242 / 268–280	≈ 1 800	Cast; storable for years

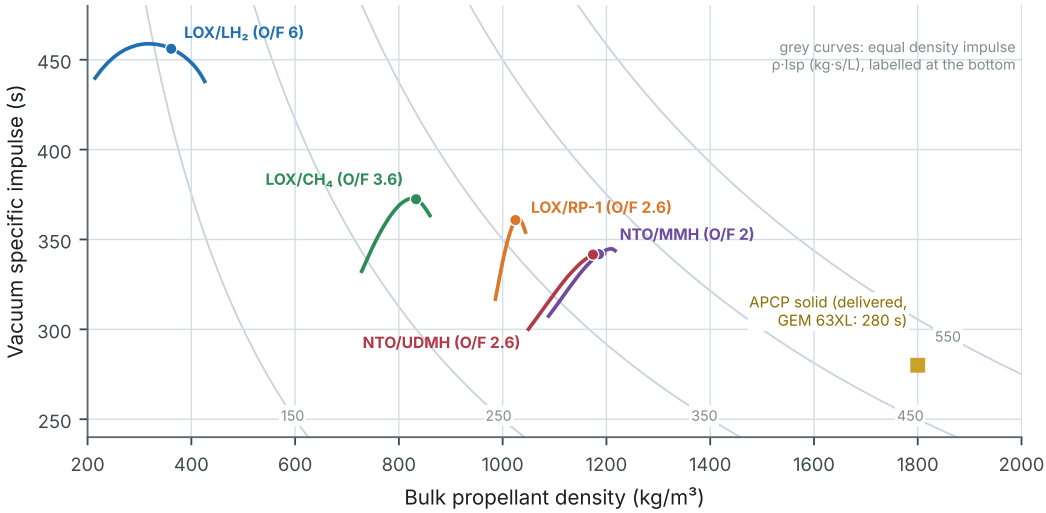


Shifting equilibrium, chamber pressure 6.9 MPa (1 000 psia), computed with an equilibrium code and NASA thermodynamic data; liquid propellants at their normal storage temperatures.

FIGURE 35.4 Theoretical performance of liquid bipropellants versus mixture ratio. (a) Vacuum Isp peaks on the fuel-rich side of stoichiometric because lighter exhaust compensates for less heat. (b) Chamber temperature peaks near stoichiometric; running fuel-rich also keeps walls and turbines cooler. Dots mark mixture ratios typical of real engines.

35.3.2 Density versus Specific Impulse

A stage's tanks, structure and drag grow with the *volume* of its propellant, not its mass. For a fixed tank volume the total impulse is proportional to the **density impulse** ρI_{sp}, where ρ is the bulk density of the combination. By this measure kerosene (370 kg·s/L) beats hydrogen (165 kg·s/L) by more than a factor of two, and solids (about 500 kg·s/L) do better still (Figure 35.5). This is why most first stages and strap-on boosters burn kerosene, methane or solid propellant, while hydrogen is favoured for upper stages, where every second of Isp is multiplied through the rocket equation.



Lines: theoretical Isp (ε = 50, 6.9 MPa, shifting equilibrium) as the mixture ratio varies; dots: typical mixture ratios. Bulk density ρ = (1 + O/F)/(O/F/ρo + 1/ρf).

FIGURE 35.5 The density–impulse trade. Each line shows how theoretical vacuum Isp and bulk density change as the mixture ratio varies; dots mark mixture ratios typical of real engines, and grey curves join combinations with equal density impulse.

35.4 Liquid-Propellant Engines

35.4.1 Feeding the Chamber

Propellants must reach the injector at a pressure 20–30 % above chamber pressure, so that the injector's pressure drop isolates the feed system from chamber fluctuations. The simplest method is to pressurise the tanks with helium (**pressure-fed**). But tank wall thickness grows with pressure ($t = pr/\sigma$ for a cylinder of radius r and allowable stress σ). A 3 m diameter aluminium-alloy tank needs walls 1.5 mm thick at 0.3 MPa but 15 mm at 3 MPa. Pressure-fed engines are therefore limited to chamber pressures of about 0.7–2 MPa and to spacecraft and small stages. Large stages keep their tanks at a few bar and raise the pressure with **turbopumps** — pumps driven by a turbine — and the way the turbine is powered defines the **engine cycle** (Figure 35.6, Table 35.2).

35.4.2 Engine Cycles

A **gas generator** burns a few per cent of the propellants very fuel-rich (below about 1 000 K) to drive the turbine and dumps the exhaust overboard: simple, but it costs several seconds of Isp. In an **expander** cycle the fuel, usually hydrogen, absorbs heat in the cooling jacket and then drives the turbine before being injected. Nothing burns in the turbine, so failures tend to be benign and

restarts easy. But the heat available limits the size of closed expanders such as the RL10 (flying since 1963) and Vinci. The **expander-bleed** variant dumps its turbine flow and allows larger engines such as Japan's 1.47 MN LE-9.

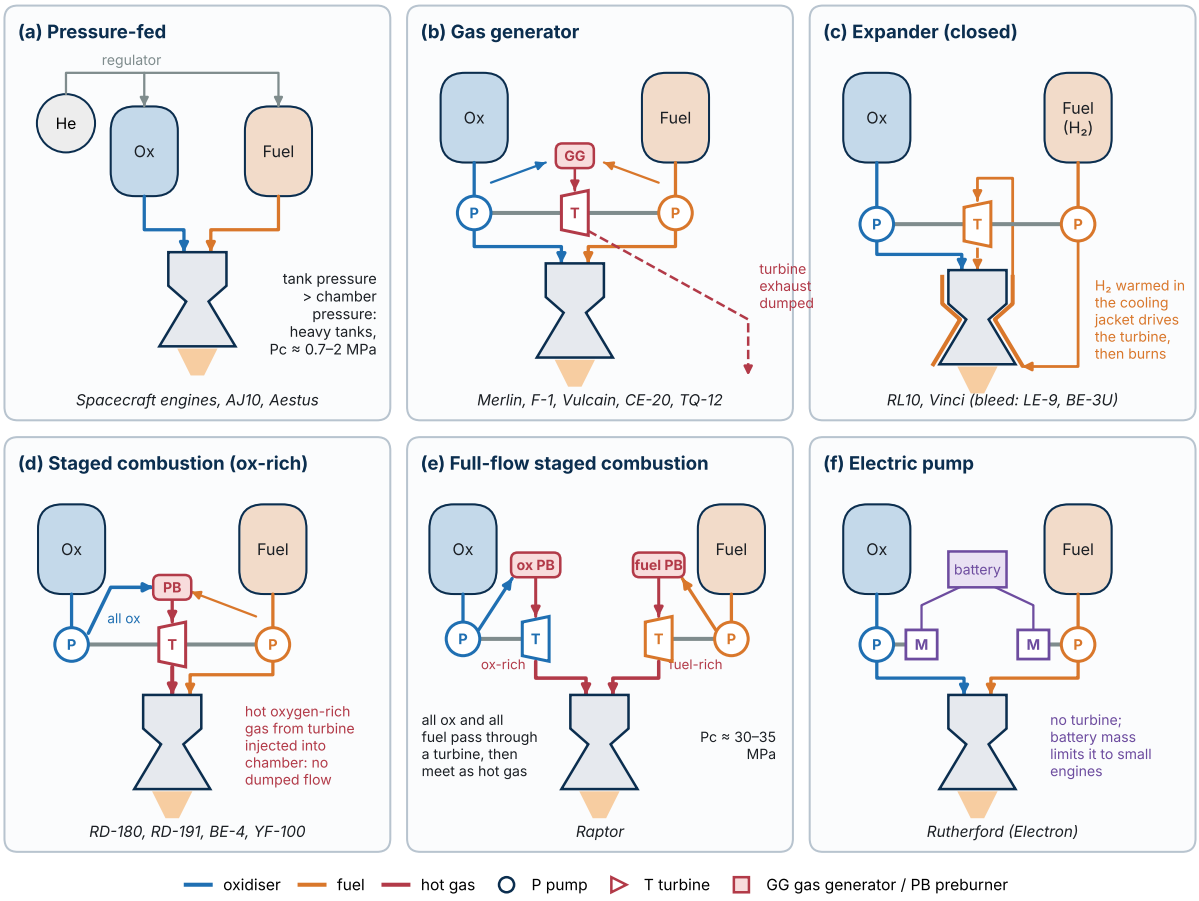


FIGURE 35.6 The main liquid-engine cycles. Blue lines carry oxidiser, orange fuel, red hot gas. Gas-generator engines dump their turbine exhaust; expander engines drive the turbine with fuel heated in the cooling jacket; staged-combustion engines send all turbine exhaust into the main chamber; electric-pump engines use battery-powered motors.

In **staged combustion**, a **preburner** burns all of one propellant with a little of the other, and the hot gas drives the turbine and then flows into the main chamber, so nothing is wasted. The Soviet Union pioneered the **oxidiser-rich** version — the RD-253 (1965), NK-33 and the RD-170 family, including the RD-180 and RD-191. It needs special alloys because hot oxygen attacks most metals. The YF-100, BE-4 and Archimedes use it too. The RS-25 uses the **fuel-rich** version, with one preburner for each turbopump. **Full-flow staged combustion** (FFSC) passes *all* of both propellants through turbines. A fuel-rich preburner drives the fuel pump and an oxygen-rich one the oxygen pump. This gives cooler turbines and no need for a seal between fuel-rich gas and oxygen on one shaft. After Soviet and US ground demonstrators, SpaceX's Raptor, first flown in 2019, became the only FFSC engine to fly; Raptor 3 runs at about 33 MPa, the highest chamber pressure of any flying engine.

Rocket Lab's Rutherford replaces the turbine with two brushless electric motors of about 37 kW each, spinning near 40 000 rpm on lithium-polymer batteries. The pumps are simple and easy to control, but battery mass grows with power and burn time, so the **electric-pump** cycle suits only small engines; Electron's second stage even jettisons spent batteries in flight.

TABLE 35.2 Liquid-engine cycles compared.

Cycle	What drives the turbine	Typical p_c (MPa)	Strengths	Weaknesses	Examples
Pressure-fed	No turbine: tank pressure	0.7–2	Simplest; easy restarts	Heavy tanks; low Isp	Spacecraft engines, AJ10, Aestus
Gas generator	Small fuel-rich combustor; exhaust dumped	5–12	Simple, easy to develop	Wastes a few per cent of flow	Merlin, F-1, Vulcain 2.1, CE-20
Expander (closed / bleed)	Fuel heated in the cooling jacket	4–6 / ≈ 10	Benign failures; restartable	Heat pickup limits size	RL10, Vinci; LE-9, BE-3U
Staged combustion	Preburner gas, then main chamber	10–27	High Isp and pressure	Complex; hot oxygen (ox-rich)	RD-180, BE-4, YF-100; RS-25
Full-flow staged comb.	Two preburners; all flow through turbines	30–33	Cooler turbines; no interpropellant seal	Most complex	Raptor
Electric pump	Battery and electric motors	Low (not published)	Simple; precise control	Battery mass	Rutherford

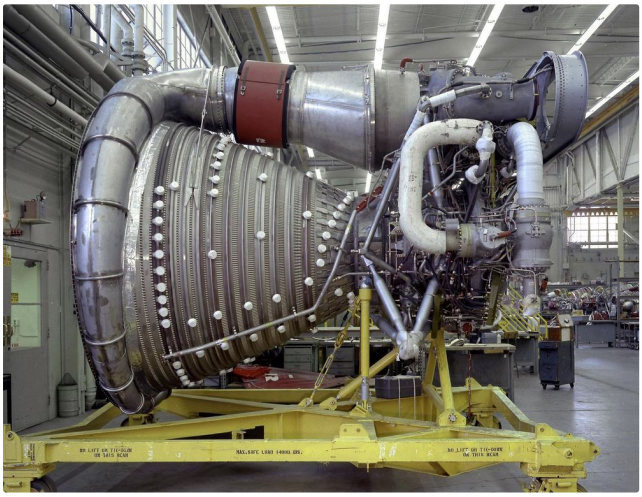


FIGURE 35.7 A complete F-1 engine, 5.8 m tall. Five of these gas-generator engines powered each Saturn V first stage; the turbopump and turbine exhaust manifold are visible on the side of the tube-wall chamber. Image: NASA/MSFC.

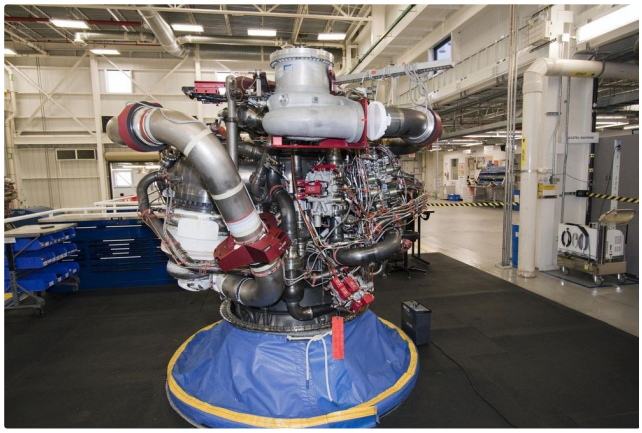


FIGURE 35.8 Space Shuttle main engine No. 1 being fitted with a new turbopump at the Kennedy Space Center in June 2010. The staged-combustion powerhead sits above the nozzle, which is covered in blue. Image: NASA/KSC (NASA_Jack Pfaller).

35.4.3 Turbopumps

Pump power is $P = \dot{m}\,\Delta p/(\rho\,\eta)$, where Δp is the pressure rise and η the efficiency, so light hydrogen is expensive to pump. Raising the RS-25's 73 kg/s of hydrogen (71 kg/m^3) by about 40 MPa at 75 % efficiency needs about 55 MW. This is close to the 53 MW (71 140 hp) rating of its high-pressure fuel turbopump, which spins at about 35 000 rpm (Figure 35.8). At the pump inlet the danger is **cavitation**: if the local pressure falls below the propellant's vapour pressure, bubbles form and collapse, destroying performance and eroding blades. The margin is the **net positive suction head** (NPSH), the excess of inlet pressure over vapour pressure expressed as a height of liquid. Designers provide it with tank pressure, subcooled propellants, **inducers** (helical impellers that tolerate some cavitation) and, on the RS-25, low-pressure boost pumps.

35.4.4 Injectors and Combustion Instability

The **injector** atomises and mixes the propellants through hundreds or thousands of holes. Kerosene engines often use **impinging** jets. Hydrogen engines use **coaxial** elements, in which fast hydrogen gas shears a central stream of liquid oxygen (the RS-25 main injector has 600). Russian engines favour **swirl** elements, and Merlin, like the Apollo lunar descent engine, uses a single central **pintle** that throttles well. A good injector releases 96–99 % of the theoretical energy.

The chamber is also an organ pipe. If heat release couples with a pressure wave, the wave grows until it destroys the engine. Low-frequency instability ("chugging") couples with the feed system — or with the whole vehicle's structure, when it is called **pogo** — while high-frequency acoustic instability ("screech") is the most destructive. In a chamber about 1 m across, where sound travels at about 1 100 m/s, the first transverse (tangential) mode lies near 650 Hz. Cures include **baffles** that divide the injector face, acoustic cavities that absorb the dangerous frequencies, and changes to the injection pattern.

> **TAMING THE F-1**
>
> The F-1, which produced 6.8 MN at sea level, remains the most powerful single-chamber liquid engine ever flown. In June 1962 one destroyed itself on the test stand through high-frequency instability. Rocketdyne and NASA spent more than three years and about 2 000 full-scale tests on injector and baffle variants, setting off small charges inside the running chamber to trigger instability. The final design, with copper baffles and a modified injection pattern, damped such disturbances within about 45 ms. No F-1 failed in flight: 65 engines powered 13 Saturn V launches.

35.4.5 Cooling

Near the throat of a high-pressure engine the heat flux exceeds 100 MW/m^2, about a thousand times that of a kitchen hot-plate, and the gas is hotter than any structural metal can survive. In **regenerative cooling** the fuel flows through channels in the wall before injection, returning the heat to the chamber. Modern chambers use copper alloys with milled channels, while the RS-25 nozzle is made of 1 080 brazed tubes (Figure 35.9). **Film cooling** protects the wall with a layer of fuel or, on Vulcain 2, re-injected turbine exhaust, while **ablative** walls char and erode slowly. Nozzle extensions can also be **radiation-cooled**: made of niobium alloy or carbon–carbon (Merlin Vacuum, RL10B-2), they glow and radiate their heat to space. Finally, **dump cooling** passes a little fuel through the wall of a nozzle extension and out of its exit.

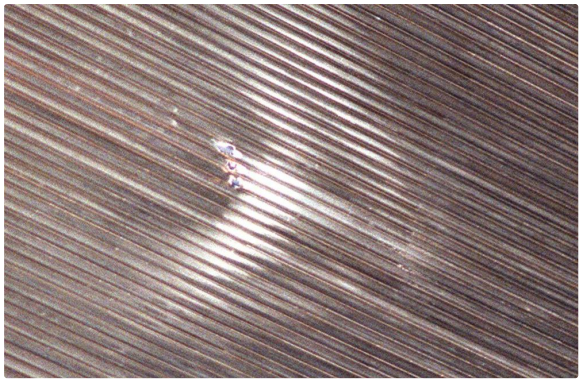


FIGURE 35.9 Coolant tubes inside main engine No. 3 of Space Shuttle Columbia after STS-93 in July 1999. A pin ejected from the injector at ignition punctured three tubes; the resulting hydrogen leak made the engines shut down 0.16 s early, leaving Columbia about 5 m/s short. Image: NASA/KSC.

35.4.6 Ignition, Throttling and Restart

Merlin is ignited by **TEA–TEB**, a mixture of triethylaluminium and triethylborane that bursts into green flame on contact with oxygen; each start consumes a dose, so the supply carried limits restarts. The RS-25, RL10 and Raptor use electrically ignited **torch igniters**, and hypergolic engines need no igniter at all. Restarting in space also needs settled propellant (Chapter 6), chill-down of pumps and lines and careful sequencing; if too much propellant accumulates before ignition, a **hard start** can damage the engine. **Throttling** limits loads near maximum dynamic pressure and makes landing possible; it requires enough injector pressure drop at low thrust to avoid instability. The RS-25 throttles between 67 % and 109 % of rated thrust, the BE-4 between 40 and 100 % and the RD-191 between 27 and 105 %.

35.4.7 Additive Manufacturing

Rocket engines suit **additive manufacturing** (metal 3-D printing): they are built in small numbers from costly alloys with complex internal passages. Laser powder-bed fusion now produces injectors, chambers with integral cooling channels (including NASA's GRCop copper alloys), manifolds and pump housings in one piece instead of dozens of brazed parts. Rutherford is mostly printed. Raptor still uses printed parts, and Raptor 3 moved much of its plumbing into the engine's main structure. Up to half of Prometheus by mass is printed, part of its aim to cost a tenth as much as Vulcain 2. Porosity, rough passages and uncertain fatigue life mean printed parts still need extensive testing.



FIGURE 35.10 Engineers at NASA's Marshall Space Flight Center prepare a 3-D printed turbopump for a test in 2016. At full power it pumped 2.3 m^3 of liquid methane per minute, enough for an engine of about 156 kN. Image: NASA/MSFC (EMMETT GIVEN).

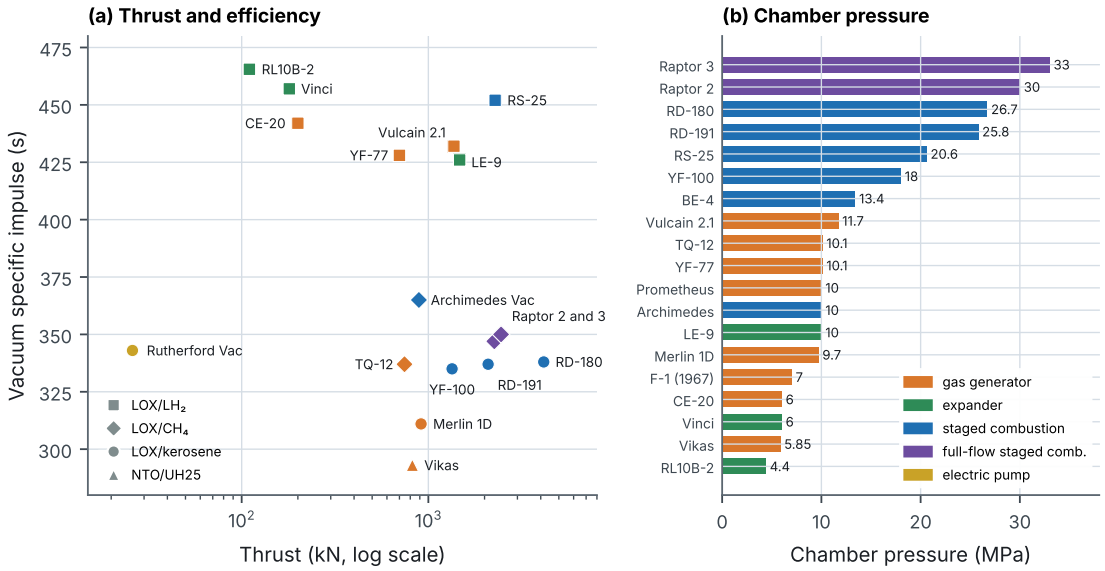
35.5 Engines of 2026

Table 35.3 and Figure 35.11 compare the main engines of 2026. Chamber pressures have risen from about 7 MPa for the F-1 to 33 MPa for Raptor 3, making engines smaller for their thrust. Methane has become the fuel of new reusable boosters in the United States, China and Europe. The newest kerosene and methane engines have thrust-to-weight ratios of 140–185, against about 54 for the RS-25, which shows the weight penalty of pumping hydrogen.

Several engines changed status in 2025–26. Raptor 3 first flew on Starship Flight 12 on 22 May 2026. Blue Origin is raising the BE-4 to about 2.85 MN with subcooled propellants, but a New Glenn booster exploded in a pre-launch hot fire on 28 May 2026 (Table 35.5). India hot-fired the CE-20 at an uprated 22 tonnes-force in 2026, and Prometheus awaits hop tests on Europe's Themis demonstrator. Rocket Lab moved the first launch of Neutron, with its Archimedes engines, to 2027 after problems with a composite tank.

TABLE 35.3 Launch-vehicle engines in 2026. GG = gas generator, SC = staged combustion, FFSC = full-flow staged combustion. Raptor Isp: vacuum value of the sea-level engine; Rutherford and Archimedes vacuum figures: vacuum versions; arrows: uprating announced in 2025. T/W from sea-level thrust where available, otherwise vacuum thrust; "—" not published. CN China, EU Europe, IN India, JP Japan, NZ New Zealand, RU Russia. Sources: manufacturers and agencies, checked October 2026.

Engine	Cycle	Propellants	Thrust SL / vac (kN)	I_{sp} vac (s)	p_c (MPa)	T/W
Merlin 1D (US)	GG	LOX/RP-1	845 / 914	311	9.7	≈ 180
Raptor 2 (US)	FFSC	LOX/CH_4	2 256 / —	≈ 347	30	141
Raptor 3 (US)	FFSC	LOX/CH_4	2 452 / —	≈ 350	33	164
RS-25 (US)	SC fuel-rich	LOX/LH_2	1 859 / 2 279	452	20.6	54
BE-4 (US)	SC ox-rich	LOX/CH_4	2 450 → 2 850	—	13.4	—
BE-3U (US)	Exp. bleed	LOX/LH_2	— / 711 → 890	—	—	—
RL10B-2 (US)	Expander	LOX/LH_2	— / 110	465.5	4.4	≈ 37
RD-180 (RU)	SC ox-rich	LOX/RP-1	3 830 / 4 150	338	26.7	71
RD-191 (RU)	SC ox-rich	LOX/RP-1	1 920 / 2 090	337	25.8	85
Vulcain 2.1 (EU)	GG	LOX/LH_2	— / ≈ 1 370	≈ 432	11.7	—
Vinci (EU)	Expander	LOX/LH_2	— / 180	457	6.0	≈ 33
Prometheus (EU)	GG	LOX/CH_4	≈ 980	—	10	—
YF-100 (CN)	SC ox-rich	LOX/RP-1	1 200 / 1 340	335	18	—
YF-77 (CN)	GG	LOX/LH_2	518 / 700	428	10.1	—
TQ-12 (CN)	GG	LOX/CH_4	658 / 745	337	10.1	—
LE-9 (JP)	Exp. bleed	LOX/LH_2	— / 1 471	426	10	≈ 62
Rutherford (NZ)	Electric pump	LOX/RP-1	25 / 26	343	—	≈ 73
Archimedes (US)	SC ox-rich	LOX/CH_4	730 / 890	≈ 365	10	—
Vikas (IN)	GG	NTO/UH25	737 / 821	293	5.85	—
CE-20 (IN)	GG	LOX/LH_2	— / ≈ 200	442	6.0	≈ 35



Thrust: vacuum where published (sea level for Raptor). Raptor Isp: vacuum value of the sea-level engine. Colours give the cycle in both panels. Sources: manufacturers and agencies (2024–26).

FIGURE 35.11 Launch-vehicle engines of 2026. (a) Thrust and vacuum specific impulse; marker shape gives the propellant, colour the cycle. (b) Chamber pressure: most staged-combustion engines run two to three times higher than gas-generator engines, although the methane-fuelled BE-4 and Archimedes use moderate pressures.

35.6 Solid Rocket Motors

35.6.1 Anatomy

A solid motor is a pressure vessel filled with propellant. The **case** — steel or, increasingly, carbon-fibre composite — is lined with rubbery insulation, and the propellant **grain** is cast inside it around a shaped mandrel that leaves an internal channel. An **igniter** fires hot gas down the channel, and the burning surface advances until it reaches the insulation. The ablative **nozzle**, with a graphite or carbon–carbon throat, is often partly submerged in the case and mounted on a flexible bearing for steering.

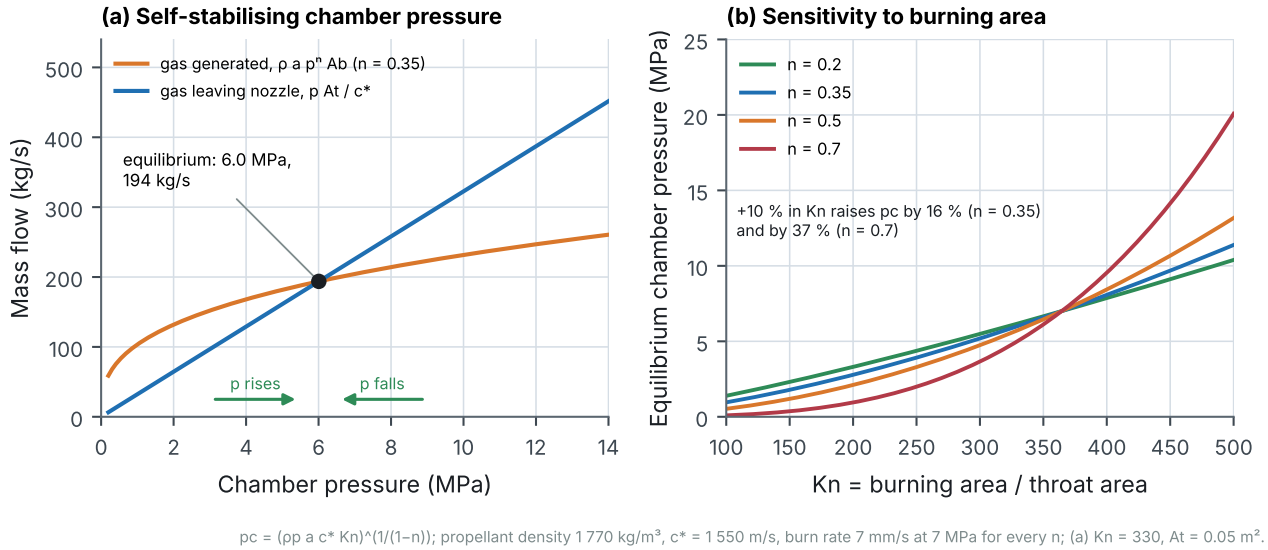
Large motors use **ammonium perchlorate composite propellant** (APCP): ammonium perchlorate oxidiser and aluminium powder in a rubbery binder. The Shuttle boosters used 69.8 % ammonium perchlorate, 16 % aluminium, 12 % PBAN binder, 2 % epoxy curing agent and 0.2 % iron oxide catalyst; newer motors use an HTPB binder. Solids are dense, store for years and give enormous thrust from a simple structure. But they have lower Isp, exhaust containing hydrogen chloride and no shut-down once lit. There is also the danger that a crack or void over-pressurises the case.

35.6.2 Burn Rate and Chamber Pressure

The burning surface recedes at a rate that depends mainly on pressure, following **Saint-Robert's law** $r = a\,p_c^{\,n}$, where a depends on the propellant and its temperature and the exponent n is typically 0.3–0.5. Gas is produced at $\rho_p r A_b$ (ρ_p the propellant density, A_b the burning area) and leaves through the nozzle at $p_c A_t / c^*$. Equating the two gives the chamber pressure:

$$p_c = \left(a\,\rho_p\,c^*\,K_n\right)^{1/(1-n)}, \qquad K_n = A_b / A_t$$

The equilibrium is stable only if $n < 1$ (Figure 35.12): a small rise in pressure then increases the nozzle outflow more than the gas production, so the pressure falls back. The exponent $1/(1-n)$ also amplifies any change in burning area. With n = 0.35, a 10 % increase in K_n — from a crack, a debond or the grain geometry — raises the pressure by 16 %; with n = 0.7, by 37 %. Designers therefore choose low-exponent propellants and inspect grains for voids by X-ray tomography. Burn rate also rises with grain temperature.



pc = (ρp a c* Kn)^(1/(1−n)); propellant density 1 770 kg/m³, c* = 1 550 m/s, burn rate 7 mm/s at 7 MPa for every n; (a) Kn = 330, At = 0.05 m².

FIGURE 35.12 Solid-motor ballistics. (a) Gas generated by the grain and gas discharged through the nozzle balance at the operating point, which is stable because n < 1. (b) Equilibrium pressure versus the ratio of burning area to throat area; a high pressure exponent makes the motor sensitive to small changes in burning area.

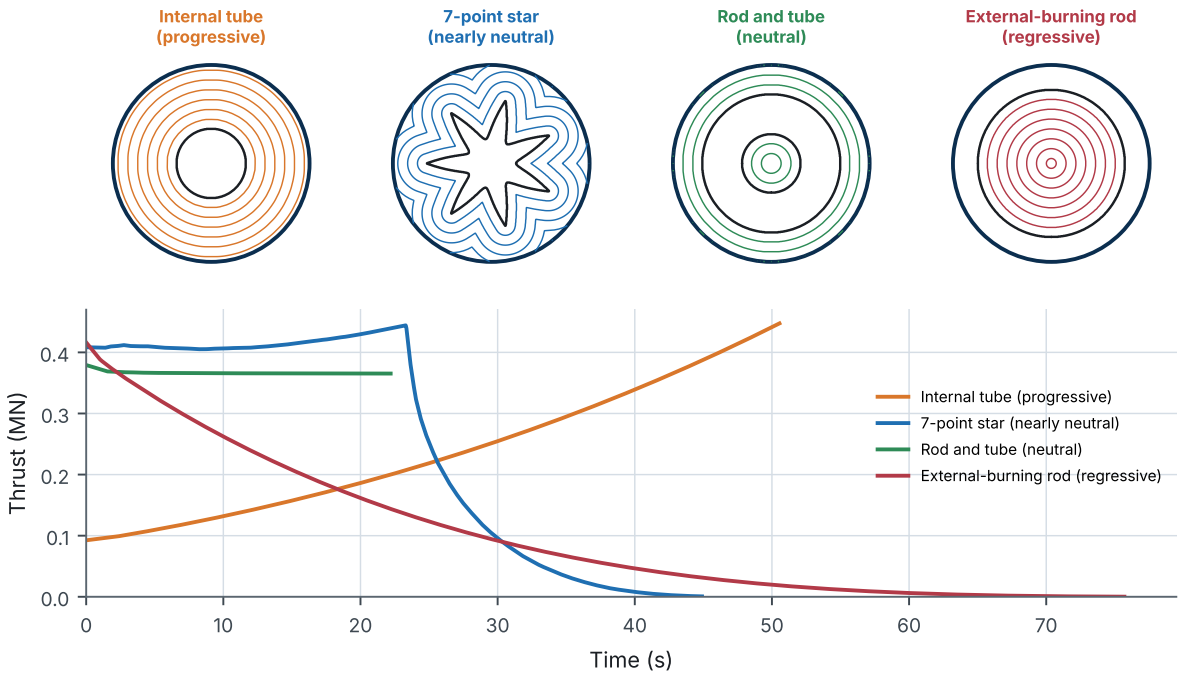
EXAMPLE: EQUILIBRIUM PRESSURE AND THRUST OF A SOLID MOTOR

A motor has A_t = 0.10 m² and A_b = 32 m² (K_n = 320). The propellant has ρ_p = 1 770 kg/m³, c^* = 1 550 m/s and n = 0.35, and burns at 7 mm/s at 7 MPa, so $a = 0.007/(7\times10^{6})^{0.35}$ = 2.81 × 10⁻⁵ in SI units.

1. **Pressure.** $a\rho_p c^* K_n$ = 2.81 × 10⁻⁵ × 1 770 × 1 550 × 320 = 24 700; raised to the power 1/(1 − 0.35) = 1.538, this gives p_c = **5.73 MPa**.
2. **Burn rate and mass flow.** $r = a p_c^n$ = 6.53 mm/s; $\dot m = \rho_p r A_b$ = 370 kg/s, equal to $p_c A_t / c^*$, as it must be.
3. **Thrust.** With a sea-level C_F of 1.55, $F = C_F p_c A_t$ = **0.89 MN** and I_{sp} = 245 s.
4. **Burn time and margin.** A neutral grain with a 0.40 m web burns for 0.40/0.006 53 ≈ **61 s**; a crack adding 10 % burning area would raise p_c to 6.63 MPa.

35.6.3 Grain Geometry

A solid motor cannot be throttled, so its thrust history is designed into the grain. The burning surface advances at the same rate everywhere; if its area grows, thrust rises (**progressive**), if constant it is **neutral**, if it shrinks **regressive** (Figure 35.13). An internal tube burns progressively; stars, wagon wheels and slotted grains are nearly neutral; combined shapes give high lift-off thrust followed by a lower sustain level. The Shuttle boosters' forward segment had an 11-point star that burned away early, cutting thrust by about a third through the period of maximum aerodynamic pressure.



Grain 1 m in diameter, 4 m long, ends inhibited; burning surface traced by a level-set (distance-field) burn-back, contours every 5 cm of web; r = a p^0.35 (7 mm/s at 7 MPa), throat sized for 6 MPa at mean area, CF = 1.6.

FIGURE 35.13 Grain geometry sets the thrust history. Top: cross-sections with burn-back contours every 5 cm of web, computed by tracing the advancing burning surface. Bottom: thrust versus time for each grain with the same size and propellant. The internal tube rises, the external-burning rod falls, and the star leaves slivers that burn out slowly.

35.6.4 Segmented and Monolithic Boosters

The largest motors are built from **segments** joined at the launch site by **field joints** sealed with O-rings. On 28 January 1986, at about 2 °C, cold O-rings in the aft field joint of Challenger's right booster failed to seal. Hot gas escaped and breached the external tank. The vehicle broke up 73 s after lift-off, killing all seven crew. The redesigned joint added a capture feature, a third O-ring and heaters.

NASA's Space Launch System uses five-segment boosters, each producing about 16 MN — together more than 75 % of lift-off thrust — for 126 s. The first composite-cased Booster Obsolescence and Life Extension (BOLE) motor ran for more than two minutes on 26 June 2025, but its nozzle suffered an anomaly near the end of the burn (Figure 35.14). Europe uses **monolithic** carbon-fibre motors: the P120C serves Ariane 6 and Vega-C, and the larger P160C first flew on Ariane 6 in June 2026. Vulcan's GEM 63XL boosters lost a nozzle in October 2024 and suffered an apparent burn-through in February 2026. After that, the US Space Force paused national-security launches on Vulcan. An improved motor was static-fired in April 2026, and Vulcan had not flown again by mid-October 2026 (Table 35.4). A large solid motor cannot be switched off; range safety destroys it by splitting the case.

TABLE 35.4 Large solid rocket motors. Values are rounded; "—" = not published.

Motor (vehicle)	Propellant (t)	Max thrust (kN)	I_{sp} vac (s)	Burn (s)	Case
SLS five-segment booster	≈ 726	≈ 16 000	—	126	Steel, 5 segments
Shuttle SRB (1981–2011)	≈ 500	≈ 14 700	268	123	D6AC steel, 4 segments
S200 (LVM3)	≈ 205	5 150	274.5	≈ 130	Maraging steel, 3 segments

Motor (vehicle)	Propellant (t)	Max thrust (kN)	I_{sp} vac (s)	Burn (s)	Case
P160C (Ariane 6, 2026)	156	≈ 4 780	—	—	Carbon fibre, monolithic
P120C (Ariane 6, Vega-C)	≈ 142	≈ 4 500	—	≈ 135	Carbon fibre, monolithic
GEM 63XL (Vulcan)	47.9	2 061	280	87	Carbon fibre, monolithic

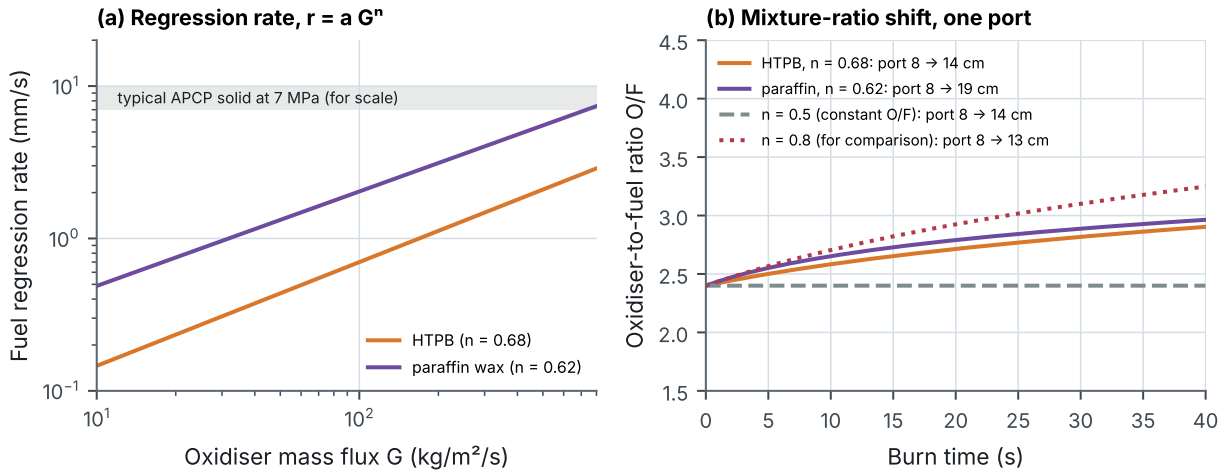


FIGURE 35.14 Northrop Grumman and NASA fire the BOLE five-segment demonstration motor horizontally at Promontory, Utah, on 26 June 2025. Image: NASA/MSFC (Northrop Grumman/Dylan Baker).

35.7 Hybrid Rockets

A **hybrid rocket** stores a liquid or gaseous oxidiser — nitrous oxide, oxygen or hydrogen peroxide — in a tank. Its solid fuel, usually a rubber such as HTPB or a wax, sits in the chamber. The oxidiser flows down a port in the fuel grain; fuel vaporises from the surface and burns in a flame within the boundary layer. With oxidiser and fuel kept apart, hybrids are safe to store, can be throttled and restarted with the oxidiser valve, and tolerate cracks in the grain.

Because the flame is fed by heat transferred through the boundary layer, the fuel **regression rate** depends not on pressure but on the oxidiser mass flux $G = \dot{m}_{ox}/A_{port}$: $r = a\,G^{\,n}$, with $n \approx 0.5$–0.8. HTPB regresses at only about 0.5–2 mm/s, several times slower than a solid propellant burns (Figure 35.15), so hybrids need long grains or several ports, which waste volume and leave unburnt slivers. Paraffin-based fuels, which form a thin liquid layer that sheds droplets into the flow, regress about three times faster. As the port widens the oxidiser flux falls, so even with constant oxidiser flow the **mixture ratio shifts**. O/F varies as $r_{port}^{\,2n-1}$, drifting upwards by about 20 % in a 40 s burn for typical exponents.



Laboratory correlations for gaseous oxygen. (b) Constant oxidiser flow 10 kg/s, single circular port of 8 cm initial radius, length set for O/F = 2.4 at ignition; O/F varies as r^(2n−1).

FIGURE 35.15 Hybrid rocket physics. (a) Regression rate versus oxidiser mass flux for HTPB and paraffin fuels, compared with a typical solid propellant. (b) Mixture ratio during a 40 s burn with constant oxidiser flow through one circular port; only for n = 0.5 would it stay constant.

Hybrids have flown on SpaceShipOne and SpaceShipTwo, whose RocketMotorTwo burned nitrous oxide with a rubber and later a polyamide fuel (Chapter 9). They have also flown on Nammo's Nucleus sounding rocket (hydrogen peroxide and HTPB), which reached about 107 km in 2018. Australia's Gilmour Space built its Eris orbital launcher around Sirius hybrids burning hydrogen peroxide with a 3-D printed fuel grain, four of about 115 kN each on the first stage. Eris's first flight failed about 14 s after lift-off on 30 July 2025; a second is expected around the end of 2026.



FIGURE 35.16 A hybrid rocket motor developed by Sierra Nevada Corporation for the Dream Chaser spaceplane, firing near San Diego in 2010. It was fired three times in one day to demonstrate restarts. Image: NASA/KSC.

35.8 Testing and Qualification

35.8.1 Test Stands and Campaigns

Engines are developed by testing: combustion stability, turbopump dynamics and chamber life cannot yet be predicted well enough to skip hot fire. A programme moves from component tests — injector elements, preburners, a turbopump on a "powerpack" rig — to complete engines, stages and often the whole vehicle on the pad. Development engines are run beyond their limits.

NASA's **Stennis Space Center** in Mississippi tested every Space Shuttle main engine and tests the RS-25s for SLS. Its Fred Haise (A-1) stand certified the new production RS-25 in 2023, running it at up to 113 % power. A January 2026 hot fire cleared engine 2063 for Artemis IV, and the B-2 stand fired the complete SLS core stage in 2021 (shown under the chapter title). SpaceX's **McGregor** site in Texas has about a dozen stands on which Merlin and Raptor engines are fired before delivery. At the German Aerospace Center's **Lampoldshausen** site the P5 stand tests Vulcain 2.1, P4.1 tests Vinci under simulated altitude conditions, and P5.2 fires the complete Ariane 6 upper stage. Upper-stage engines with large nozzles are fired in **altitude-simulation** facilities, vacuum chambers in which the exhaust, passing through a diffuser, helps hold the low pressure (Figure 35.17, Figure 35.18).



FIGURE 35.17 A Merlin engine mounted on a test stand at SpaceX's McGregor facility in Texas, where engines are hot-fired before they fly. Image: NASA/KSC (SpaceX).

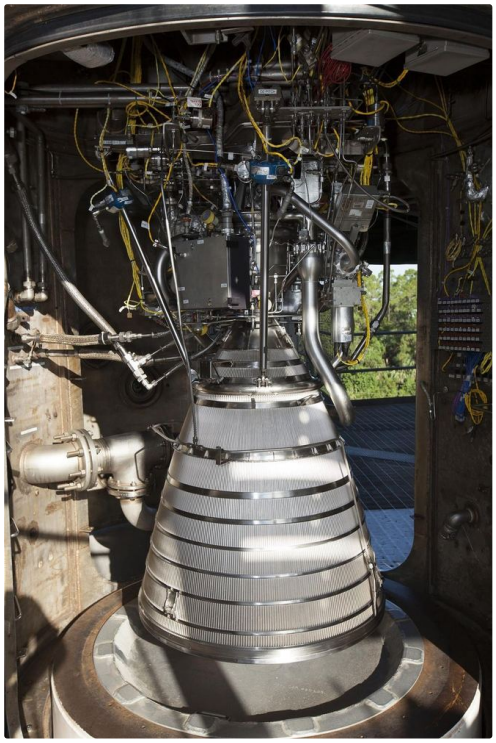


FIGURE 35.18 An RL10 in the vacuum test chamber at Aerojet Rocketdyne's West Palm Beach facility, Florida, in 2016. It was one of two engines built for the Centaur upper stage of the Atlas V that would launch Starliner's crewed flight test. Image: NASA/KSC (NASA/Dimitri Gerondidakis).

Qualification shows that a design meets its requirements with margin. Engines are run for several times their mission duration, at the corners of the allowed ranges of thrust, mixture ratio and inlet conditions, and through many starts. The RS-25 has accumulated more than 1.1 million seconds of hot fire. Every flight engine then passes an **acceptance test** to catch defects.

35.8.2 Engine-Out and Failure Modes

Clustering allows engine-out survival. Falcon 9 lost an engine on the CRS-1 mission in October 2012 and still delivered Dragon to orbit. On Starship Flight 12 one of 33 Raptor 3s shut down early during ascent, yet the vehicle still reached its planned trajectory. Controllers shut an engine

down when a **redline** — a limit on pressure, temperature, speed or vibration — is crossed. Many vehicles are held down for a few seconds after ignition, so that a fault can stop the launch (Table 35.5).

TABLE 35.5 Propulsion failures that changed engineering practice.

Event	What happened	Consequence or lesson
F-1 development, 1962–65	High-frequency combustion instability destroyed engines on the stand	Baffles, injector redesign, bomb tests
Challenger, 28 Jan 1986	Cold O-rings let hot gas escape from a booster field joint	Joint redesign; temperature limits; decision-making reform
STS-93, 23 Jul 1999	Injector pin punctured three nozzle coolant tubes	Hydrogen leak; main engines shut down early
Falcon 9 Starlink 9-3, Jul 2024	Oxygen leak chilled the Merlin Vacuum; hard start at relight	Satellites left in a 135 km perigee orbit and lost
Vulcan, Oct 2024 and Feb 2026	GEM 63XL nozzle lost; later an apparent burn-through	National-security flights paused; nozzle improved
New Glenn, 28 May 2026	Booster exploded during pre-launch hot fire	Traced to the main oxygen valve of one BE-4

35.9 Beyond Chemical Propulsion

Chemical rockets are limited by the energy stored in their propellants: hydrogen and oxygen release about 13 MJ per kilogram, which in practice caps vacuum Isp near 470 s. Missions that need large velocity changes over months use **electric propulsion**, which accelerates ions or plasma to 15–50 km/s at tiny thrust (Chapter 6). Nuclear thermal rockets, solar sails and other advanced concepts are covered in Chapter 42.

AI IN ROCKET PROPULSION

Combustion instability suits machine learning because it is hard to predict from first principles but leaves signatures in pressure data. Researchers at DLR and the University of Cambridge trained Bayesian neural networks on pressure signals from a hydrogen–oxygen research combustor. The networks forecast the onset of instability about half a second ahead — early enough for a controller to react. Engine teams also use machine-learning surrogates of slow simulations and anomaly detection on hot-fire data. **Computational engineering** tools generate whole engine geometries from requirements: in December 2024 the Dubai company LEAP 71 hot-fired a 5 kN kerosene–oxygen aerospike designed largely by its software and printed in copper. Chapter 52 discusses AI in design and manufacturing; certification still rests on hot fire.

KEY TAKEAWAYS

- Thrust is $\dot{m}v_e + (p_e - p_a)A_e$; specific impulse $F/(\dot{m}g_0)$ measures propellant efficiency, while c^* and C_F separate the performance of the chamber from that of the nozzle.
- A de Laval nozzle chokes at the throat, so $\dot{m} = p_c A_t / c^*$; exhaust velocity grows with $\sqrt{T_c/\mathcal{M}}$, which is why hydrogen engines reach 420–465 s.
- Nozzles are sized for altitude: sea-level engines use expansion ratios of about 15–40 to avoid flow separation, vacuum engines 60–280.
- Propellants trade Isp against density, storability and handling; dense propellants suit first stages, hydrogen upper stages, and methane has become the fuel of new reusable boosters.
- Engine cycles differ in how they drive the turbopumps: gas generators are simple, expanders benign, staged combustion efficient; full-flow Raptor 3 reaches 33 MPa.
- A solid motor settles at $p_c = (a\rho_p c^* K_n)^{1/(1-n)}$; grain shape sets its thrust history and a low exponent keeps it tolerant of defects. Hybrids are safe and throttleable but regress slowly and shift mixture ratio.
- Engines are proven by hot fire, from components to whole vehicles, with margins found by exceeding limits.

REVIEW QUESTIONS

1. Write the thrust equation and explain why an engine gives more thrust in vacuum than at sea level. By how much does the RS-25's thrust rise from sea level to vacuum?
2. A combustion gas has c^* = 1 800 m/s and the nozzle a vacuum C_F of 1.85. What is the vacuum Isp? For p_c = 10 MPa and a thrust of 1 MN, what throat diameter is needed?
3. Using the exit-velocity formula, explain why hydrogen–oxygen gives a higher Isp than kerosene–oxygen although its flame is cooler.
4. What happens to the flow in an over-expanded nozzle, and why do first-stage engines not use the large nozzles of upper-stage engines?
5. Using density impulse, explain why most first stages burn kerosene, methane or solid propellant while many upper stages burn hydrogen.
6. Compare the gas-generator, expander and staged-combustion cycles. Why is the closed expander limited to engines of a few hundred kilonewtons?
7. What is NPSH, and how do designers prevent cavitation at a pump inlet?
8. A solid motor with n = 0.4 runs at 6 MPa. A debond increases the burning area by 15 %. Estimate the new chamber pressure and explain why a high pressure exponent is dangerous.
9. Why does the mixture ratio of a hybrid drift during the burn, and how do paraffin fuels help with the low regression rate of HTPB?

36 CHAPTER Launch Vehicles of the World



Image: NASA/KSC (NASA/Frank Michaux)

In 2025 the world attempted 330 orbital launches, more than in any previous year, on some forty different families of rockets flown by providers in nine countries and Europe. This chapter surveys those vehicles. It traces the launch vehicle from the R-7 missile of 1957 through the Saturn V, the Space Shuttle, Energia, Proton, Ariane, Long March, H-II and PSLV, then describes the fleet of October 2026 country by country — from Falcon 9 and Starship to Ariane 6, Soyuz, the Long March family, China's commercial start-ups, H3, LVM3 and the newest small launchers. Launch data from 1957 to 2026 show who launches, how much mass reaches orbit, how reliable each class of rocket is and how fast cadence has grown. The chapter ends with the trends reshaping launch: reusability, rideshare, cadence, the heavy-lift race, the small-launcher shake-out and the demand for sovereign access to space.

IN THIS CHAPTER

36.1 A Short History of the Launch Vehicle
36.2 The World Fleet in 2026
36.3 Launch by the Numbers
36.4 Trends Reshaping Launch

Between October 1957 and October 2026 the world made about 7 400 orbital launch attempts. For most of that time the annual total stayed between 50 and 140; then, after 2018, it took off. In 2025 there were 330 attempts, of which 318 succeeded, and by 13 October 2026 another 244 had been made. Half of all launches in 2025 used a single type of rocket, SpaceX's Falcon 9, and most of those carried the company's own Starlink satellites. At the same time more countries and companies than ever were building rockets: China alone flew sixteen different families of launch vehicles in 2025.

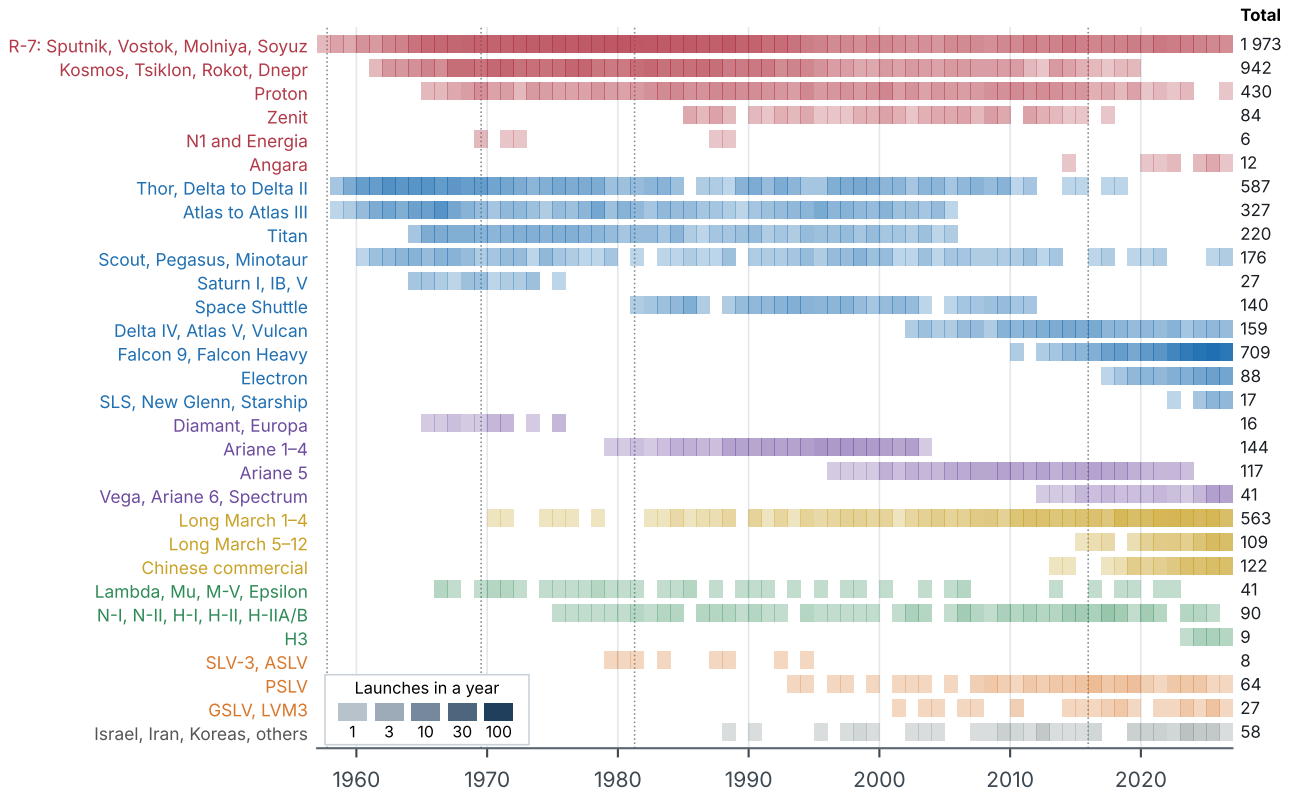
This chapter is a field guide to that fleet. It does not repeat the engineering of rockets: the rocket equation, staging, structures and guidance are the subject of Chapter 34, and engines of Chapter 35. Instead it uses that engineering to compare vehicles. It looks at how much they lift, what they burn, whether they can be reused, how often they fly and fail, and what they cost. Launch sites and ground systems are covered in Chapter 37, and suborbital rockets in Chapter 9. The launch statistics here come from Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) as of 13 October 2026. Appendix E lists every launch of 2025 and 2026 and the all-time totals by country, vehicle family and launch site.

36.1 A Short History of the Launch Vehicle

36.1.1 Missiles Become Launchers

The first launch vehicles were ballistic missiles with a satellite in place of a warhead. Sergei Korolev's **R-7** was built to throw a nuclear warhead of about 5 t across continents. It had a central core and four strap-on boosters, all burning kerosene and liquid oxygen and all ignited on the ground. A modified R-7 launched Sputnik on 4 October 1957, and a version with an upper stage carried Yuri Gagarin in 1961. Its descendants — Vostok, Voskhod, Molniya and Soyuz — have flown about 1 970 times, more than any other rocket family (Figure 36.1). Crews still fly to the International Space Station on a Soyuz-2, a modernised R-7 with digital avionics.

The United States followed a different route. The civilian Vanguard exploded on its first orbital attempt in December 1957, and the Army's Juno I, derived from the Redstone missile, put Explorer 1 into orbit on 31 January 1958. Three military missiles then became the American workhorses for half a century. The **Thor** intermediate-range missile grew, stage by stage, into the **Delta** family; the last Delta II flew in 2018. The **Atlas** missile, with its balloon tanks held rigid by internal pressure, launched the orbital Mercury flights. Topped by the hydrogen-fuelled **Centaur** stage (first successful flight in 1963), it also launched the Surveyor Moon landers and many planetary probes. The **Titan II** missile carried the Gemini crews, and with large solid boosters became the Titan III and IV, which launched heavy military satellites until 2005.



Each cell is one year; shading shows the number of orbital launch attempts (log scale). Dotted lines: Sputnik 1957, Apollo 11 1969, first Shuttle 1981, first Falcon 9 landing 2015. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

FIGURE 36.1 Orbital launch activity of the main launch-vehicle families, 1957–2026. Each cell is one year, shaded by the number of launch attempts; the totals at the right count all attempts since the first flight. Families born as missiles in the 1950s dominated until the 1990s; since 2015 the activity has shifted to Falcon 9, the new Long March rockets and Chinese commercial vehicles.

36.1.2 The Moon Race and the Super-Heavies

The race to the Moon produced the largest rockets of the century. NASA's **Saturn V** stood 110.6 m tall and weighed almost 3 000 t at lift-off. Its five F-1 engines (Chapter 35) gave 34 MN of thrust. The three stages could place about 140 t in low Earth orbit (LEO), including the third stage and the Apollo spacecraft bound for the Moon. All thirteen Saturn Vs launched between 1967 and 1973 reached space (Figure 36.2); the last lifted the Skylab station. The Soviet answer, the **N1**, had thirty engines in its first stage. Its first stage was never fired on a test stand as a complete stage. All four launches between 1969 and 1972 failed in that stage — through engine fires, a turbopump explosion and control problems — before the programme was cancelled. The next Soviet super-heavy, **Energia**, flew only twice, in 1987 and 1988, the second time carrying the uncrewed Buran shuttle; its four strap-on boosters were essentially first stages of the Zenit rocket.

36.1.3 Shuttles and Workhorses

The **Space Shuttle** (Chapter 10) promised cheap, routine access by reusing its orbiter and solid boosters. It flew 135 missions between 1981 and 2011 and carried most of the International Space Station into orbit (Figure 36.3). But refurbishment was slow and costly, and two orbiters were lost with their crews, *Challenger* in 1986 and *Columbia* in 2003. After *Challenger* the United States stopped launching commercial satellites on the Shuttle; the business moved back to expendable rockets, revived the Delta, Atlas and Titan lines and gave Europe's new Ariane its opening.

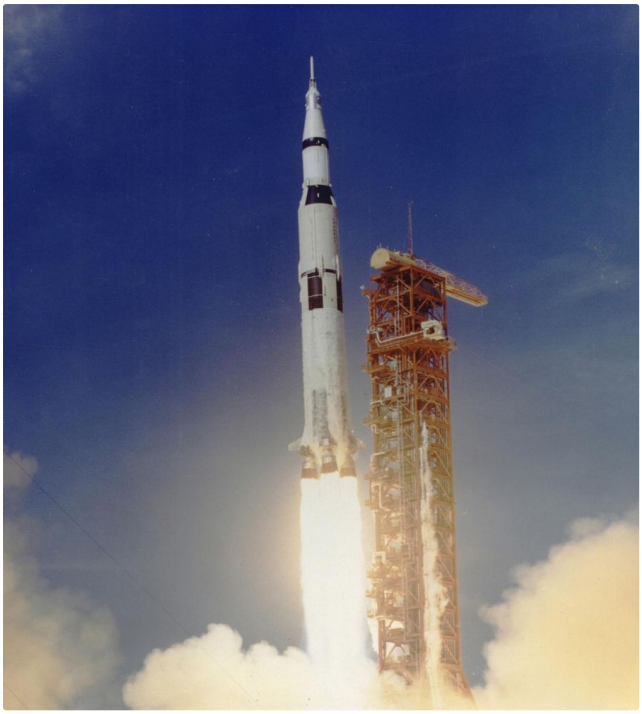


FIGURE 36.2 A Saturn V lifting off from Kennedy Space Center with Apollo 11 on 16 July 1969. Each of its five F-1 engines burned about 2.6 t of kerosene and oxygen per second. Image: NASA/MSFC.



FIGURE 36.3 The exhaust plume of Space Shuttle Atlantis climbing away from Pad 39A on STS-135, the last Shuttle mission, on 8 July 2011, seen from NASA's Shuttle Training Aircraft. Image: NASA (NASA/Dick Clark).

In the Soviet Union, two families did most of the work. The **Proton**, first launched in 1965, burns storable nitrogen tetroxide and unsymmetrical dimethylhydrazine (UDMH) — toxic but able to sit in the tanks for weeks. It lifted the Salyut and Mir stations, the Russian modules of the ISS and many geostationary satellites. About 430 have flown. The **Soyuz-U** flew 787 times between 1973 and 2017 (counting its PVB variant), at up to 47 launches a year. In the early 1980s the USSR launched about 100 rockets a year — 108 in 1982 — mostly carrying short-lived reconnaissance satellites that returned their film to Earth.

36.1.4 New Launching Nations

France became the third nation to launch its own satellite, with the Diamant rocket in 1965. Japan followed in February 1970 with the small solid Lambda-4S, China in April 1970 with the Long March 1, the United Kingdom in 1971 with Black Arrow and India in 1980 with the SLV-3. Israel (1988), Iran (2009), North Korea (2012) and South Korea (2013, with a Russian first stage; 2022 with its own Nuri) joined later. Table 36.1 lists the landmark vehicles.

After the failure of the multinational Europa rocket, Europe built **Ariane 1**, first launched on 24 December 1979. Its successor Ariane 4 (1988–2003) took about half of the commercial geostationary market. **Ariane 5** (1996–2023) flew 117 times, carrying two large satellites at once to geostationary transfer orbit (GTO). In its most famous flight it carried the James Webb Space Telescope (Figure 36.4). China's Long March family began with the Long March 1. Next came the

Long March 2, which returned film capsules and, as the 2F, has launched every Shenzhou spacecraft since 1999 and every crew since 2003. Then came the hydrogen-fuelled Long March 3 for geostationary missions. Japan developed both a solid-rocket line for science (Mu, M-V and Epsilon) and liquid rockets. Its liquid rockets were first built under licence from the Delta, followed by the all-Japanese **H-II** (1994) and **H-IIA**. The H-IIA flew 50 times from 2001 to 2025 with a single failure (Figure 36.5). India's four-stage **PSLV** (1993) launched Chandrayaan-1 and the Mars Orbiter Mission, and its LVM3 (originally GSLV Mk III) carried Chandrayaan-2 and -3.



FIGURE 36.4 An Ariane 5 lifts off from Kourou with the James Webb Space Telescope on 25 December 2021. The two solid boosters provided about 90 % of the lift-off thrust. Image: NASA (NASA/Bill Ingalls).



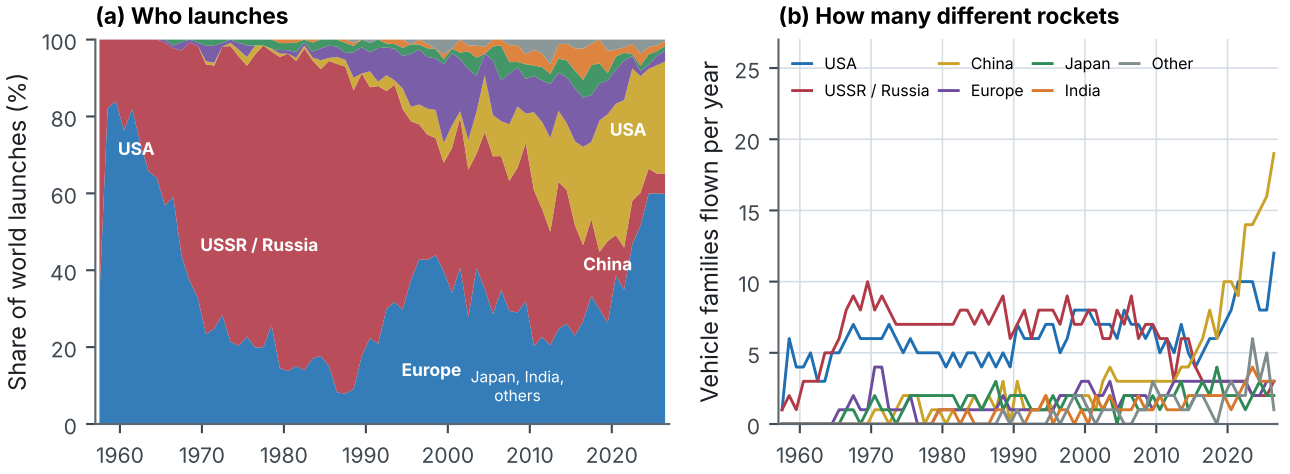
FIGURE 36.5 An H-IIA launches the NASA–JAXA Global Precipitation Measurement Core Observatory from Tanegashima on 28 February 2014. The H-IIA retired in June 2025 after 50 flights. Image: NASA (NASA/Bill Ingalls).

TABLE 36.1 Landmark launch vehicles. Flight counts are orbital launch attempts in GCAT to October 2026; payloads are approximate maxima to low orbit.

Vehicle	Country	First flight	Max. LEO payload	Flights (failures)	Why it matters
R-7 / Soyuz family	USSR/Russia	1957	1.3–8 t	≈ 1 970	First satellite and first human; still flying
Thor–Delta	USA	1958	to 6 t (Delta II)	≈ 590	Commercial workhorse; ended 2018
Saturn V	USA	1967	≈ 140 t	13 (0)	Apollo Moon landings; Skylab
Proton	USSR/Russia	1965	20–23 t	≈ 430	Space stations; storable propellants
N1	USSR	1969	95 t (design)	4 (4)	Lunar rocket that never worked
Space Shuttle	USA	1981	27.5 t	135 missions (1)	Partly reusable; built the ISS
Energia	USSR	1987	100 t	2 (1)	Launched Buran; boosters became Zenit

Vehicle	Country	First flight	Max. LEO payload	Flights (failures)	Why it matters
Ariane 4 / 5	Europe	1988 / 1996	7.6 / 21 t	116 / 117	Commercial GTO market; Webb
Long March 2–4	China	1974–88	3.5–11.5 t	≈ 530	Basis of China's programme, crews since 2003
H-II / H-IIA	Japan	1994 / 2001	10 t	7 / 50	Independent Japanese hydrogen rockets
PSLV	India	1993	3.8 t	64 (4)	Low-cost rideshare; Moon and Mars probes

The balance of power shifted twice (Figure 36.6). Until the mid-1960s the United States made most launches. From 1970 to the end of the 1980s the Soviet Union made 70–85 % of them. After its collapse the totals fell, and Europe, China, Japan and India took a growing share. In 2025 the United States and China made nearly nine launches in ten. Diversity has grown too: in 2026 Chinese providers flew 19 different vehicle families and American providers about a dozen, against two or three in Russia, Europe and Japan.

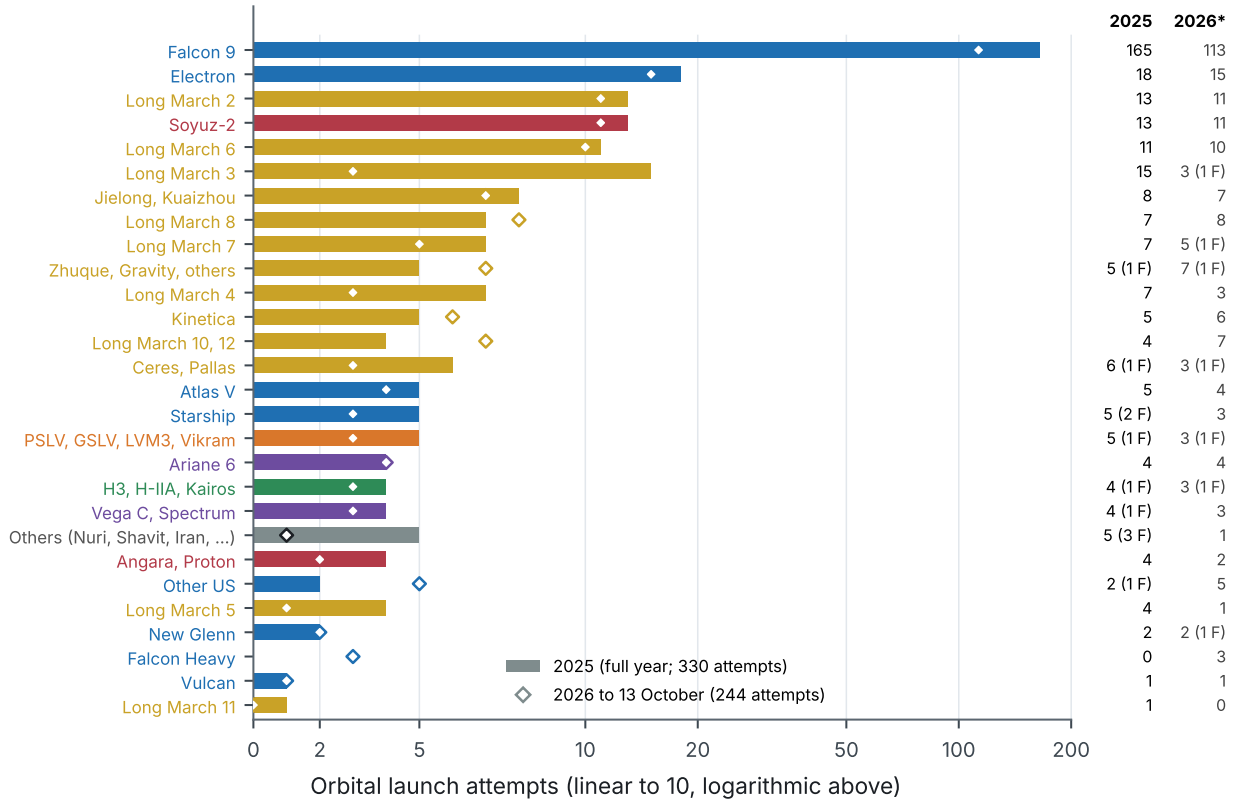


Country of the launch provider (Rocket Lab counted as USA). A family groups the variants of one vehicle (Long March split by number). 2026 to 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

FIGURE 36.6 Left: share of world orbital launch attempts by the country of the launch provider, 1957–2026. Right: the number of different vehicle families each country flew per year. China's commercial boom since 2020 has produced the most varied fleet any nation has operated.

36.2 The World Fleet in 2026

In 2025, 330 orbital launch attempts were made: 198 by US providers (including Rocket Lab's launches from New Zealand), 93 Chinese, 17 Russian, 8 European, 5 Indian, 4 Japanese and 5 by other countries. Eleven failed. Figure 36.7 breaks these down by vehicle family, together with the 244 attempts made in 2026 up to 13 October; Table 36.2 compares the main medium and heavy vehicles, and Figure 36.8 and Figure 36.9 place them against the rockets of the past.



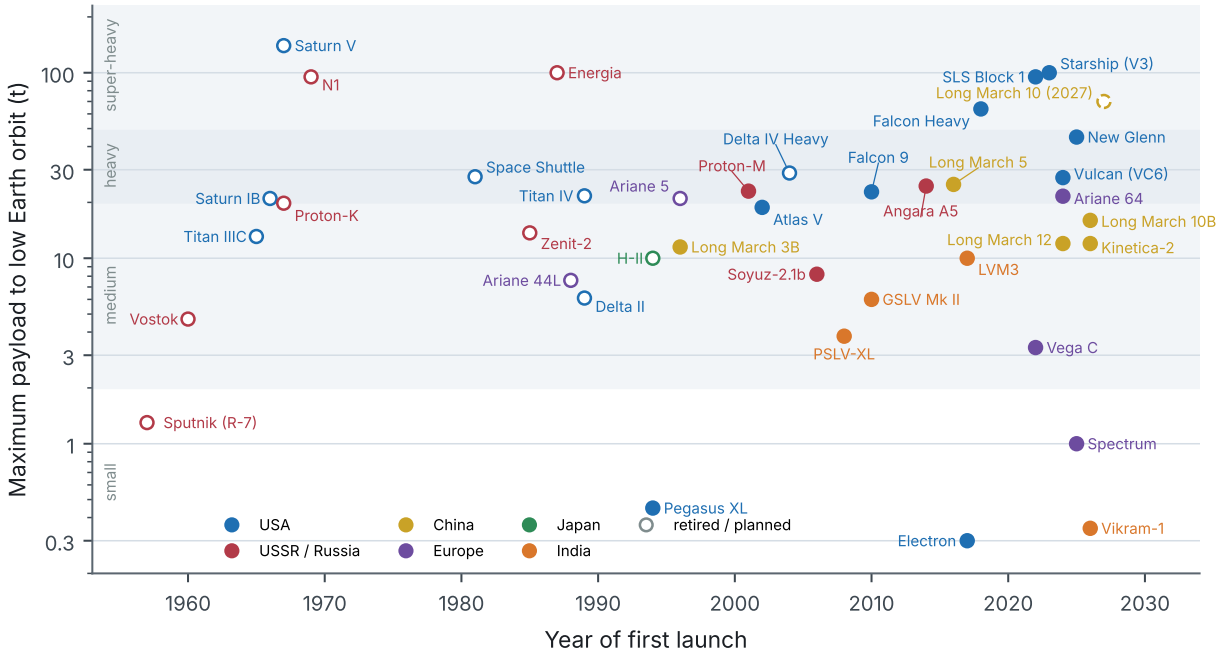
Colour = country of the launch provider (blue USA, red Russia, gold China, violet Europe, green Japan, orange India). F = failures. *To 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

FIGURE 36.7 Orbital launch attempts in 2025 (bars) and in 2026 up to 13 October (diamonds) by vehicle family, with failures (F). Falcon 9 made half of all launches in 2025; the Long March rockets, split here by series, made 69.

TABLE 36.2 Main medium and heavy launch vehicles in 2026. Payloads are approximate published maxima (LEO at low inclination; GTO with 1 800 m/s to geostationary orbit); counts are orbital attempts from GCAT (*2026 to 13 October); prices are published list prices or estimates where available.

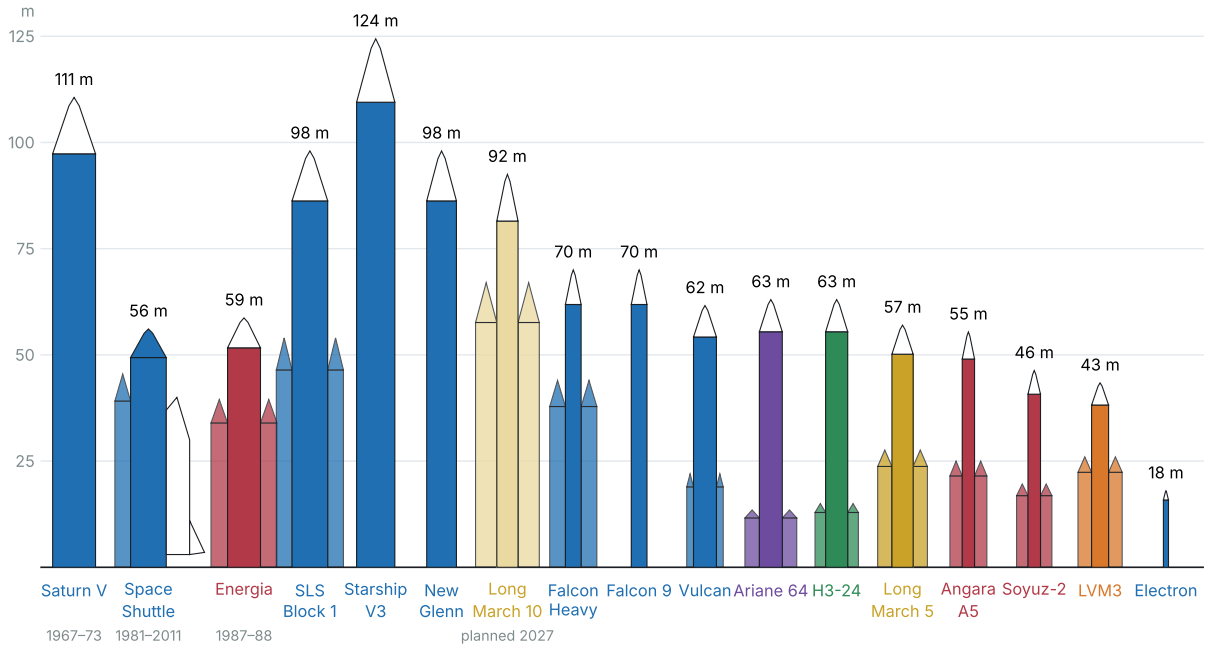
Vehicle (first flight)	LEO / GTO (t)	Height	Stages and propellants	Reuse	Launches 2025 / 2026*	Price (public)
Falcon 9 (2010), USA	22.8 / 8.3 (17.5 / 5.5 reused)	70 m	2; LOX/kerosene	Booster, fairing	165 / 113	$74 M
Falcon Heavy (2018), USA	63.8 / 26.7	70 m	2 + 2 side cores; LOX/kerosene	Boosters	0 / 3	≈ $97 M (reported)
Starship (2023; V3 2026), USA	≈ 100 (reusable)	124 m	2; LOX/methane	Full (planned)	5 / 3	—
SLS Block 1 (2022), USA	95 (27 to the Moon)	98 m	Core + 2 solids + upper stage; LOX/LH_2	None	0 / 1	≈ $2.2 billion (NASA OIG est.)
Vulcan Centaur (2024), USA	27.2 / 14.4 (VC6)	62 m	2 + 0–6 solids; LOX/methane, LOX/LH_2	None	1 / 1	—
New Glenn (2025), USA	45 / 13	98 m	2; LOX/methane, LOX/LH_2	Booster	2 / 1	—
Atlas V (2002), USA	18.8 / 8.9 (551)	58–62 m	2 + 0–5 solids; LOX/kerosene, LOX/LH_2	None	5 / 4	Retiring
Ariane 62 / 64 (2024), Europe	10.4 / 4.5; 21.7 / 11.5	63 m	2 + 2 or 4 solids; LOX/LH_2	None	4 / 4	€100 M / €115 M (est.)
Vega C (2022), Europe	2.3 to SSO	35 m	3 solid + 1 storable	None	3 / 2	—
Soyuz-2 (2006), Russia	7–8.7 / 3.1	46 m	4 boosters + core + 3rd stage (+ Fregat); LOX/kerosene	None	13 / 11	—
Angara A5 (2014), Russia	24.5 / 5.4	55 m	5 modules + 2nd + upper stage; LOX/kerosene	None	1 / 0	—

Vehicle (first flight)	LEO / GTO (t)	Height	Stages and propellants	Reuse	Launches 2025 / 2026*	Price (public)
Proton-M (2001), Russia	23 / 6.4	58 m	3 + upper; N_2O_4/UDMH	None	0 / 1	Retiring
Long March 2C/2D (1982/92), China	3.5–3.9	41–42 m	2; N_2O_4/UDMH	None	10 / 9	—
Long March 3B (1996), China	11.5 / 5.5	56 m	3 + 4 boosters; storable, LOX/LH_2	None	13 / 3	—
Long March 5 / 5B (2016), China	25 / 14	57 m	2 + 4 boosters; LOX/LH_2, LOX/kerosene	None	4 / 1	—
Long March 6A / 7A / 8A / 12 (2020–24), China	8–12 / up to 7	50–60 m	2–3 (+ boosters); LOX/kerosene (+ LH_2)	None (12A/B: tests)	25 / 22	—
Long March 10B (2026), China	16	70 m	2; LOX/kerosene, LOX/methane	Booster (caught at sea)	— / 1	—
H3 (2023), Japan	6.5 GTO (H3-24); 4 SSO (H3-30)	57–63 m	2 + 0–4 solids; LOX/LH_2	None	3 / 2	≈ ¥5 billion target (H3-30)
LVM3 (2017), India	10 / 4.2	43 m	2 solids + 2; storable, LOX/LH_2	None	2 / 0	—
PSLV / GSLV Mk II (1993/2001), India	3.8 / 1.3; 6 / 2.3	44 / 52 m	4 / 3 stages; solid, storable, LOX/LH_2	None	3 / 2	—



Approximate published maximum LEO payloads (expendable configuration, low inclination; definitions vary between sources). Filled: flying in 2026.

FIGURE 36.8 Maximum payload to low Earth orbit of major launch vehicles against the year of their first launch. The super-heavies of the 1960s–80s were retired; the 2020s have brought a second generation — SLS, Starship and, from 2027, China's Long March 10 — alongside reusable heavies such as Falcon Heavy and New Glenn.



Heights and diameters to a common scale (simplified outlines; side boosters shown in pairs). Grey dates: retired or planned vehicles.

FIGURE 36.9 Launch vehicles of the past and present drawn to the same scale. Starship V3, at 124 m, is the tallest rocket ever flown; Electron, at 18 m, is the smallest in regular service.

36.2.1 United States

SpaceX dominates the world's launch market. Falcon 9 made 165 of the 330 attempts in 2025 and 113 more by mid-October 2026. Its first stage lands on a droneship or a landing zone after almost every flight (Chapter 34): the fleet leader flew for the 37th time in August 2026. SpaceX lists a Falcon 9 at $74 million in 2026. **Falcon Heavy**, three Falcon 9 cores strapped together, did not fly in 2025 but launched three times in 2026, including NASA's Nancy Grace Roman Space Telescope on 30 August. **Starship**, designed for full reuse, is the largest rocket ever built. The Version 3 vehicle stands 124.4 m tall and weighs about 5 000 t. It has 33 Raptor engines on its Super Heavy booster and six on the ship. After three ships were lost in the first half of 2025, the last two flights of Version 2 succeeded. On Flight 14, on 28 September 2026, a ship reached orbit for the first time and deployed 26 Starlink V3 satellites. It returned to splash down in the Pacific three hours later. Catching the ship on the launch tower, the step needed for full reuse, was the next goal, expected on Flight 15. NASA's Artemis lunar landings depend on refuelling Starships in orbit, which has yet to be demonstrated.

NASA's **Space Launch System** (SLS) is the government's own super-heavy rocket. It uses Space Shuttle engines and five-segment solid boosters and is thrown away on every flight. After the uncrewed Artemis I in 2022 it launched the four astronauts of Artemis II around the Moon on 1 April 2026 (Figure 36.10). In February 2026 NASA dropped plans for the larger Exploration Upper Stage, and in March it chose a Centaur V upper stage for later SLS rockets. NASA's Inspector General has estimated that each Artemis launch costs more than $4 billion, including the Orion spacecraft and ground systems.



FIGURE 36.10 The SLS rocket lifts off from Pad 39B with the four Artemis II astronauts aboard Orion on 1 April 2026. The two solid boosters provide more than three-quarters of the lift-off thrust. Image: NASA/MSFC (NASA/Michael DeMocker).

United Launch Alliance (ULA), owned equally by Boeing and Lockheed Martin, is replacing its Atlas V and the retired Delta IV with **Vulcan Centaur**. Vulcan's first stage burns liquefied natural gas in two Blue Origin BE-4 engines. Atlas V flew 110 times from 2002 without losing a payload; its last Amazon Leo launch, on 2 July 2026, left six rockets reserved for Boeing's Starliner capsule. Vulcan has had a difficult start. On its second flight in October 2024 the nozzle fell off one of its solid boosters. When a booster anomaly recurred on the USSF-87 mission in February 2026, the Space Force paused national-security launches on Vulcan. ULA had planned 16–18 Vulcan launches in 2026; by mid-October it had made one, and return to flight was reported for late October.

Blue Origin's New Glenn is a 98 m, 7 m diameter rocket with a reusable first stage powered by seven BE-4 engines. It reached orbit on its first flight in January 2025. It landed its booster on a ship on its second flight, which launched NASA's ESCAPADE Mars probes in November 2025 (Figure 36.12). It re-flew that booster in April 2026 — but the upper stage under-performed and left AST SpaceMobile's BlueBird 7 satellite in a useless orbit. On 28 May 2026 a New Glenn exploded during a hot-fire test before its fourth flight, wrecking the transporter-erector at Launch Complex 36. Blue Origin traced the cause to an oxygen valve on one BE-4 and aimed to fly again before the end of 2026.



FIGURE 36.11 Vulcan Centaur lifts off from Cape Canaveral on its first flight, carrying Astrobotic's Peregrine lunar lander, on 8 January 2024. Image: NASA/KSC (NASA/Isaac Watson).



FIGURE 36.12 New Glenn lifts off from Launch Complex 36 with NASA's twin ESCAPADE Mars probes on 13 November 2025. Its first stage then made Blue Origin's first booster landing on a ship at sea. Image: NASA/KSC (Blue Origin).

Smaller American rockets serve satellites below about a tonne. **Rocket Lab's Electron** is an 18 m carbon-composite rocket with electric-pump-fed engines (Figure 36.13). It has made 88 orbital launches with four failures and flew 18 times in 2025, more than any other single rocket type except Falcon 9. Its average revenue per launch was $9.1 million in mid-2026. Rocket Lab's reusable medium-lift **Neutron** slipped to 2027 after a tank failed in testing. **Firefly Alpha** returned to flight in March 2026 after a failure in April 2025 and the loss of a booster in a ground test; Firefly and Northrop Grumman are developing the larger Eclipse for 2027. Northrop Grumman's solid **Minotaur IV** flew once in each of 2025 and 2026. The air-launched **Pegasus XL**, dropped from an L-1011 airliner, launched a robotic servicer in July 2026. The servicer is intended to raise the orbit of NASA's ageing Swift observatory. Relativity's Terran R and Stoke Space's Nova had not flown by October 2026.



FIGURE 36.13 A Rocket Lab Electron lifts off from Māhia, New Zealand, on 26 May 2023 with two of NASA's TROPICS storm-observing CubeSats. Image: NASA/KSC (Rocket Lab).

36.2.2 Europe

Ariane 6 replaced Ariane 5 in July 2024. It keeps a hydrogen-fuelled core stage with a Vulcain 2.1 engine, adds a restartable Vinci upper stage (Chapter 35) and carries either two (Ariane 62) or four (Ariane 64) P120C solid boosters. By October 2026 it had made nine flights: the first in 2024,

four Ariane 62s in 2025, then the first Ariane 64 on 12 February 2026, which placed 32 Amazon Leo satellites in orbit. In June 2026 the first Ariane 64 with the larger P160C boosters carried 36 of them, and Amazon raised its order from 18 to 24 Ariane 64 launches. **Vega C** is a four-stage rocket whose solid first stage is the same P120C motor used as Ariane 6's booster. It returned from a 2022 failure in December 2024 and is now marketed by its builder Avio. It launched the European–Chinese SMILE mission in May 2026.

Europe's dependence on so few rockets became painful in 2023–24. Ariane 5 retired, Vega C was grounded and Soyuz launches from Kourou ended after Russia's invasion of Ukraine. ESA had to buy Falcon 9 launches for Euclid, Galileo satellites and Hera. One response is the **European Launcher Challenge**, to which ESA's member states committed €902 million at their November 2025 ministerial meeting. In August 2026 ESA signed conditional contracts with Isar Aerospace (€197.8 million), Rocket Factory Augsburg (€186.9 million) and PLD Space (€158.9 million). Each is required to reach orbit by the end of 2027. Isar's **Spectrum** failed 30 s into its first flight in March 2025 but reached orbit from Andøya, Norway, on 5 September 2026 — the first satellite delivery by a European commercial launch company. RFA One and PLD's Miura 5 were preparing for first flights, MaiaSpace was developing a partly reusable rocket, and Orbex had closed down.

36.2.3 Russia

Russia launched only 17 rockets in 2025 and 13 by mid-October 2026. The workhorse is **Soyuz-2** (Figure 36.14), which launches crews and Progress cargo ships to the ISS from Baikonur in Kazakhstan, military satellites from Plesetsk and other missions from Vostochny in the Far East. When the Soyuz MS-28 crew launched in November 2025, a service platform fell into the flame trench of Site 31, Russia's only pad for crewed flights; it was repaired by March 2026. The modular **Angara** family, built around a common kerosene core, is meant to replace Proton and give Russia heavy launches from its own territory, but has flown only a dozen times since 2014. **Proton-M** made one launch in 2026. The new **Soyuz-5** medium rocket, with a single RD-171MV engine on its first stage, made a suborbital test flight from Baikonur on 30 April 2026.



FIGURE 36.14 A Soyuz-2.1a launches the Soyuz MS-26 crew to the ISS from Baikonur on 11 September 2024. Its four strap-on boosters, which separate in the classic "Korolev cross", trace back to the R-7 of 1957. Image: NASA (NASA/Victor Zelentsov).

36.2.4 China

China launched 93 rockets in 2025 and 71 by mid-October 2026, more than half of them Long Marches built by the state-owned China Aerospace Science and Technology Corporation (CASC). The older Long March 2, 3 and 4 burn storable N_2O_4/UDMH. A new generation burns kerosene and oxygen (Long March 6, 7, 8 and 12) or hydrogen (the Long March 5 heavy-lift rocket and upper stages). Launches increasingly carry China's two broadband mega-constellations, Guowang and Qianfan, from the new commercial launch site next to Wenchang on Hainan Island. For crewed lunar missions CASC is developing the **Long March 10**: a 92.5 m, three-core rocket with 21 kerosene engines, able to send 27 t towards the Moon, due to fly in 2027. A single-core version, the Long March 10A, made a suborbital test with an abort of the Mengzhou crew capsule in February 2026. The commercial **Long March 10B** reached orbit on its first flight on 10 July 2026. Its first stage was caught at sea by hooks snagging cables stretched over a ship — China's first recovery of an orbital booster.

Since 2015 Chinese policy has opened launch to private and semi-private companies, and their rockets made about a quarter of China's launches in 2025. LandSpace's **Zhuque-2** became the first methane-fuelled rocket to reach orbit in 2023. The company's stainless-steel **Zhuque-3** reached orbit in December 2025. In August 2026 it landed its first stage on legs at a desert site in Gansu province, although a fire then broke out at its base. Galactic Energy's solid **Ceres-1** has flown 23 times, but its liquid Ceres-2 failed on its debut in January 2026 before the company's larger **Pallas-1** succeeded in September. CAS Space flies the solid **Kinetica-1** and debuted the 12 t **Kinetica-2** in March 2026. Orienspace launches the solid **Gravity-1** from a ship at sea. Space Pioneer's **Tianlong-3**, designed as a Falcon 9 rival, failed half a minute after lift-off on its first flight in April 2026. CASC's own reusable test vehicle, the Long March 12A, reached orbit in December 2025 but its booster crashed near the landing pad.

36.2.5 Japan, India and Other Nations

Japan relies on **H3**, a two-stage hydrogen rocket with the expander-bleed LE-9 engine. After a failed first flight in 2023 it succeeded five times, but on its seventh launch (flight F8) in December 2025 the QZS-5 navigation satellite was lost. JAXA's investigation concluded that the most likely cause was bonded joints in the honeycomb satellite adapter that had come apart during manufacture: the adapter buckled and broke piping on the second-stage hydrogen tank. H3 returned to flight in June 2026 with the first H3-30, which has no solid boosters. It launched the QZS-7 satellite in August and was scheduled to launch the Martian Moons eXploration (MMX) mission on 20 October 2026. The H-IIA retired in June 2025. The small solid Epsilon S has been delayed by two second-stage motor explosions in ground tests and is now planned with the older Epsilon second stage by March 2027. The private **Kairos** of Space One failed on all three of its attempts, most recently in March 2026.

India suffered two consecutive failures of the PSLV, in May 2025 and January 2026, both in its solid third stage, and had not flown again by October 2026. The **GSLV Mk II** launched the NASA–ISRO NISAR radar satellite in July 2025. The **LVM3** — renamed HLVM3 in its human-rated form — is being prepared for the first uncrewed Gaganyaan flight. That flight is expected at the end of 2026 or in 2027. The small SSLV has flown three times. On 18 July 2026 Skyroot Aerospace's **Vikram-1**, a three-stage solid rocket with a liquid kick stage, reached orbit on its first attempt, the first private Indian rocket to do so.

South Korea's three-stage kerosene **Nuri** made its fifth flight on 7 October 2026, delivering five satellites and nine CubeSats — its fourth success in five attempts. By contrast, the private Innospace's Hanbit-Nano failed shortly after lift-off from Brazil in December 2025. **Israel** launches its Ofeq reconnaissance satellites westward over the Mediterranean on the solid **Shavit-2**, accepting a retrograde orbit to keep spent stages away from populated land; the latest flew in September 2025. **Iran** launched satellites on the Simorgh and the solid Qaem-100 in 2024, but its Zoljanah failed in September 2025, and **North Korea** has not attempted an orbital launch since a failure in May 2024. **Australia's** first orbital rocket, Gilmour Space's hybrid-propellant Eris, failed 14 s after lift-off in July 2025; a second attempt was expected at the end of 2026. Table 36.3 summarises the small launchers.

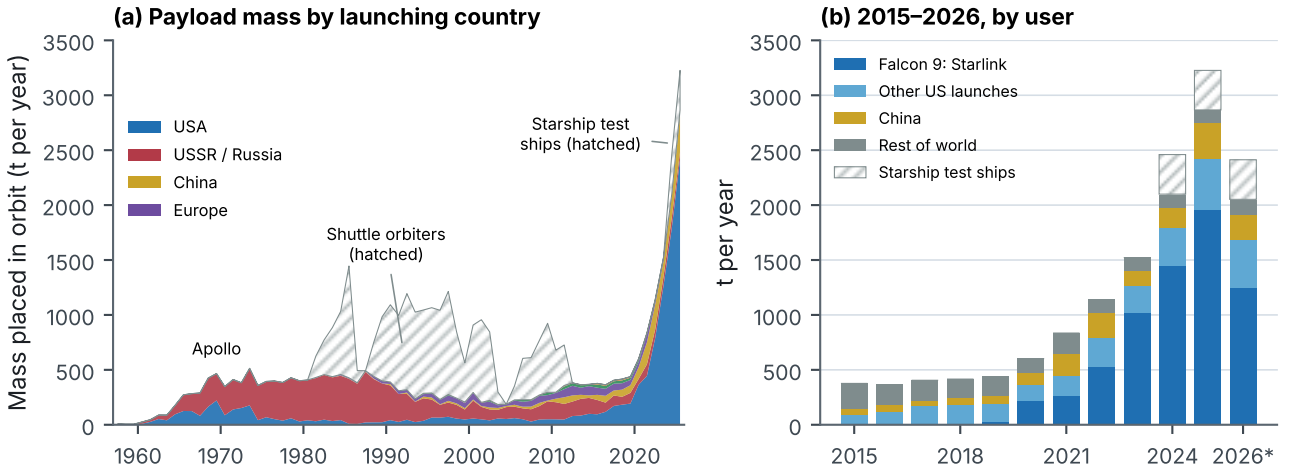
TABLE 36.3 Small launchers (under about 2 t to LEO) of the 2020s. Attempts from GCAT to 13 October 2026 (a Kuaizhou-1A pad explosion in 2025 not counted); payloads are approximate published values.

Vehicle	Country	LEO payload	Attempts (failures)	Status, October 2026
Electron	USA/New Zealand	0.3 t	88 (4)	15 launches in 2026 to date
Kuaizhou-1A / -11	China	0.3 / 1.5 t	30 (2) / 7 (1)	Active (solid)
Ceres-1	China	0.4 t	23 (2)	Active; land and sea launch
Kinetica-1	China	2 t	16 (1)	Active (solid)
Jielong-3	China	1.6 t to SSO	12 (0)	Active; sea launch
Firefly Alpha	USA	1 t	7 (3)	Returned to flight March 2026
Spectrum	Germany	1 t	2 (1)	Reached orbit September 2026
SSLV	India	0.5 t	3 (1)	No launch since 2024
Vikram-1	India	0.35 t	1 (0)	Reached orbit July 2026
Kairos	Japan	0.25 t	3 (3)	Three failures
Eris	Australia	0.3 t	1 (1)	Second flight expected late 2026
Astra Rocket 3, LauncherOne	USA	0.05–0.5 t	7 (5), 6 (2)	Retired in 2022–23

36.3 Launch by the Numbers

36.3.1 Mass to Orbit

Launch counts say little about what is lifted. Figure 36.15 sums the payload mass of every successful launch. The Moon race lifted the world total to almost 500 t a year. From 1981 to 2011 the Space Shuttle added several hundred tonnes a year. However, GCAT counts each orbiter — about 100 t that came back to Earth — as payload. Real growth began after 2018. In 2025 launches delivered about 2 870 t of satellites, cargo and crew vehicles, plus 360 t of Starship test ships. Including payloads lost in failures, the GCAT total is about 3 500 t, nearly nine times that of 2015. Falcon 9 carried 2 330 t of the 2 870 t — 81 % — and Starlink satellites alone 1 960 t. China lifted 330 t, about 11 %.

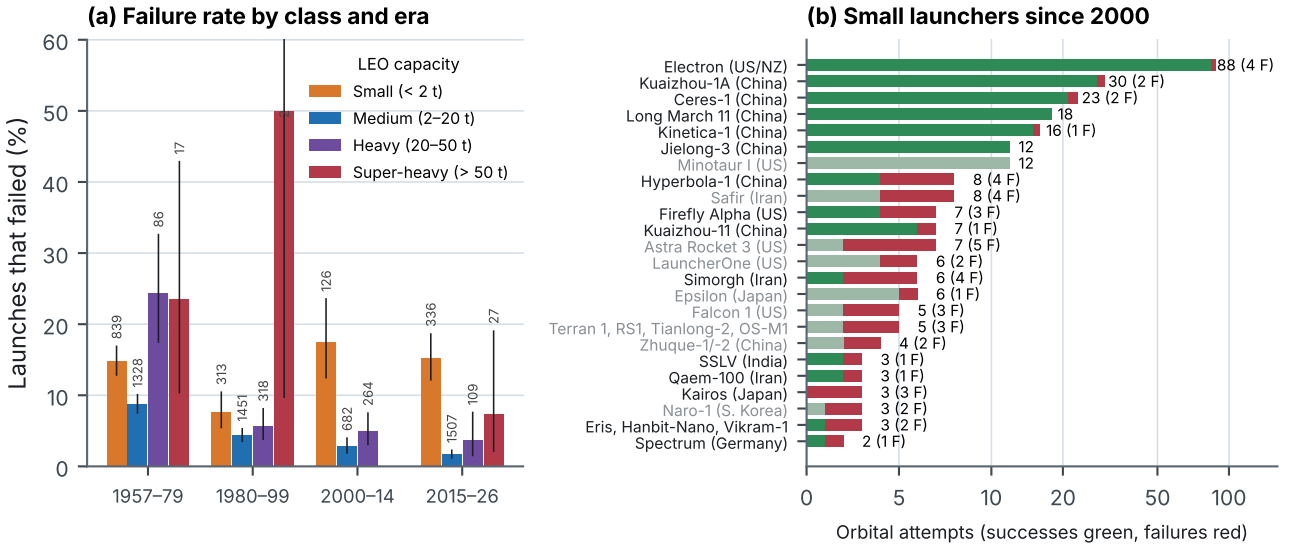


Sum of GCAT payload masses on successful launches. GCAT counts reusable orbiters (Shuttle, Buran, about 100 t each) and Starship test ships (120 t each) as payload: shown hatched. *2026 to 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

FIGURE 36.15 Left: payload mass placed in orbit each year by the country of the launch provider, 1957–2025. Right: 2015–2026 by user. GCAT counts reusable orbiters and Starship test ships (hatched) as payload; without them, 2025 saw about 2 870 t of satellites and spacecraft delivered, two-thirds of it Starlink.

36.3.2 Reliability by Class

Figure 36.16 adds two views to the learning curve of Chapter 34. Grouped by size, small launchers have always failed most often: 15 % of the 336 small-launcher attempts since 2015 failed, against 1.7 % of 1 507 medium launches and 3.7 % of 109 heavy launches. Part of the reason is that many small rockets are first or early flights of new designs by new companies, and few of those companies survive long enough to climb the learning curve. The right-hand panel shows the shake-out. Electron, the Chinese solid rockets and Jielong-3 have built long records. Astra and Virgin Orbit stopped flying after a handful of launches, and Kairos has yet to reach orbit.

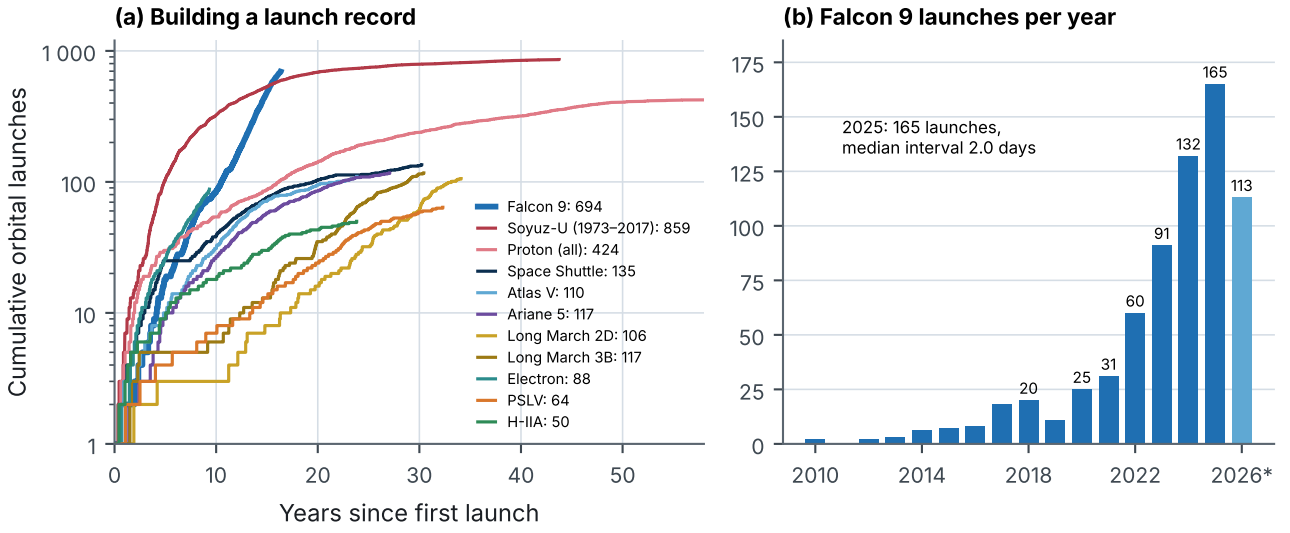


(a) Classes by approximate maximum LEO payload (Falcon 9 as medium); bars: failed share; whiskers: 90 % Jeffreys interval; numbers: attempts; pad aborts and explosions excluded. (b) Grey labels: retired or company closed. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

FIGURE 36.16 (a) Share of orbital launch attempts that failed, by vehicle size class and era. (b) Successes and failures of small launchers first flown since 2000.

36.3.3 Cadence

The most striking change of the 2020s is cadence — how often a rocket flies (Figure 36.17). Rockets used to need weeks or months between flights from one pad, because each was assembled and checked for weeks before launch. Falcon 9 flew 165 times in 2025, a median of 2.0 days between launches. The Soviet Soyuz-U had flown almost as often in the 1970s — it reached 500 launches in 14 years, Falcon 9 in 15 — but only by building a new rocket every time. Falcon 9 reached its 100th launch in late 2020, ten years after its first, and its 600th early in 2026, with most flights on re-used boosters.



Pad aborts excluded. *2026 to 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

FIGURE 36.17 (a) Cumulative launches against years since each vehicle's first flight (pad aborts excluded). (b) Falcon 9 launches per year; the fleet averaged one launch every 2.2 days in 2025.

Table 36.4 lists the first flights of new launch vehicles and major new versions in 2025–26. There were 19 debuts in 21 months, of which 7 failed (37 %). This is in line with the 38 % maiden-flight failure rate found in Chapter 34.

TABLE 36.4 First orbital flights of new vehicles and major new versions, 2025–26 (dates UTC or local as reported).

Date	Vehicle	Country	Outcome
16 Jan 2025	New Glenn	USA	Orbit; booster lost before landing
16 Jan 2025	Starship V2	USA	Ship lost during ascent
11 Feb 2025	Long March 8A	China	Success
30 Mar 2025	Spectrum	Germany	Lost control at about 30 s
30 Jul 2025	Eris	Australia	Failed at about 14 s
18 Sep 2025	Zoljanah	Iran	Failed to reach orbit
3 Dec 2025	Zhuque-3	China	Orbit; landing attempt failed
22 Dec 2025	Hanbit-Nano	South Korea	Failed shortly after lift-off
23 Dec 2025	Long March 12A	China	Orbit; landing attempt failed
17 Jan 2026	Ceres-2	China	Failed to reach orbit
12 Feb 2026	Ariane 64	Europe	Success, 32 Amazon Leo satellites
30 Mar 2026	Kinetica-2	China	Success
3 Apr 2026	Tianlong-3	China	Failed at about 33 s

Date	Vehicle	Country	Outcome
22 May 2026	Starship V3	USA	Ship splashed down as planned; booster lost
1 Jun 2026	Long March 12B	China	Success
12 Jun 2026	H3-30	Japan	Success (no solid boosters)
10 Jul 2026	Long March 10B	China	Success; first stage caught at sea
18 Jul 2026	Vikram-1	India	Success
1 Sep 2026	Pallas-1	China	Success

36.4 Trends Reshaping Launch

36.4.1 Reusability

Before December 2015 no orbital booster had ever landed. By 2026 Falcon boosters had landed more than 650 times. Three other organisations had recovered orbital first stages: Blue Origin (on a ship), CASC (caught by cables at sea) and LandSpace (on legs, though a fire broke out after landing). Others are following — Rocket Lab's Neutron, Europe's Themis demonstrator, Japan's and India's experimental vehicles — and SpaceX aims to recover both stages of Starship. The economics are discussed in Chapter 34: reuse pays only at a high flight rate, which is why the leaders are operators with their own constellations to launch.

36.4.2 Rideshare

A **rideshare** mission carries dozens of small satellites from many customers to a common orbit. India's PSLV made it famous by launching 104 satellites at once in 2017. SpaceX's **Transporter** missions to sun-synchronous orbit (Transporter-1 carried 143 in January 2021; Transporter-18 flew on 1 October 2026) and **Bandwagon** missions to mid-inclination orbits now dominate. The published price is $350 000 for 50 kg. Rideshare undercuts dedicated small launchers on price; its drawbacks are that the satellite goes where and when the main mission goes, which **orbital transfer vehicles** (Chapter 11) partly overcome.

36.4.3 The Heavy-Lift Race

Between Energia's last flight in 1988 and Falcon Heavy's first in 2018, no rocket able to place more than 30 t in orbit flew. Now four are competing (Figure 36.8). Falcon Heavy (64 t expendable) and New Glenn (45 t with a reusable booster) are in service. SLS (95 t) is expensive and has flown only twice, in 2022 and 2026. Starship aims at 100 t or more with full reuse and orbital refuelling. China's Long March 10 (70 t to LEO, 27 t towards the Moon) is due in 2027 to support crewed lunar landings planned by 2030, and Russia has proposed a heavy Angara A5V. Heavy-lift rockets matter for crewed exploration, space stations and large telescopes, but much of the demand that pays for them is the launch of constellation satellites in large batches.

36.4.4 The Small-Launcher Shake-Out

Around 2018 more than a hundred small-launcher projects were under way worldwide. Few survived: rideshare took much of their market, and the cost of a dedicated small launch per kilogram is high (see the example below). Survivors offer what rideshare cannot — a specific orbit

at a specific time, for constellation replenishment, defence "responsive launch" or sovereign needs. Many are growing into medium rockets: Rocket Lab is moving to Neutron, Galactic Energy from Ceres-1 to Pallas-1 and Firefly to Eclipse.

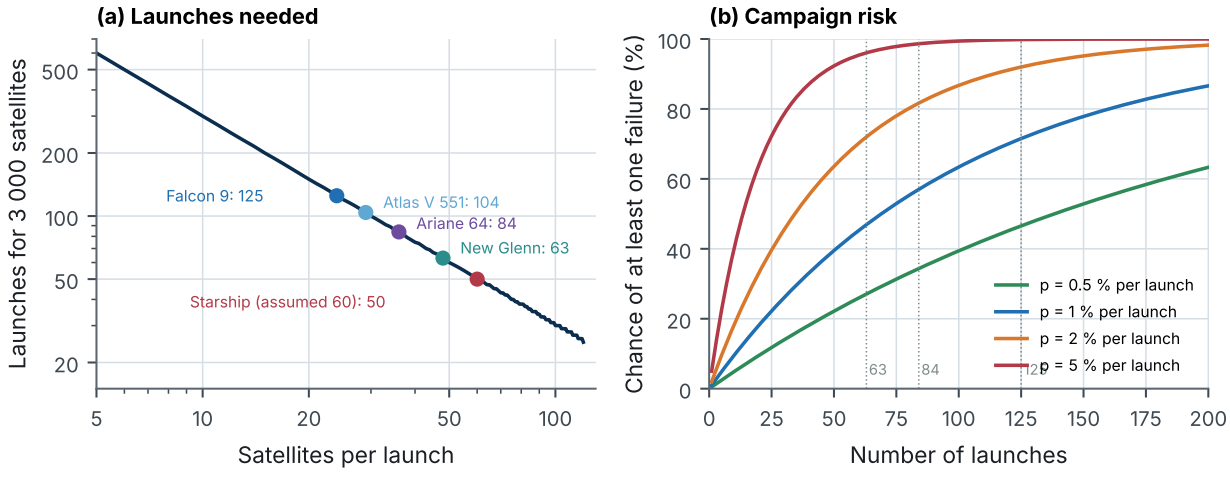
36.4.5 Sovereign Access

Governments value independent access to space as a matter of security. Europe's crisis of 2023–24, Russia's loss of Western customers after 2022, India's PSLV failures and Japan's H3 failure all show how fragile a fleet of one or two rockets is. The policy response is to fund several competing launchers. Examples are the European Launcher Challenge, Japan's support for private rockets, India's opening to Skyroot and others, and the US Space Force's practice of buying launches from several providers. The response also includes building new spaceports in Norway, Sweden, the United Kingdom and Australia (Chapter 37).

EXAMPLE: PRICE PER KILOGRAM AND PAYLOAD FRACTION

Compare three vehicles at their published prices, assuming each flies full to LEO. **Payload fraction** is payload mass divided by lift-off mass.

1. **Falcon 9** with booster landing: \$74 M for 17.5 t → $74 \times 10^6 / 17\,500 \approx$ **\$4 230/kg**. Lift-off mass 549 t → payload fraction 17.5/549 = **3.2 %**. Expending the booster raises the payload to 22.8 t (4.2 %), but the booster is then lost.
2. **Ariane 64**: estimated €115 M for 21.65 t → **€5 310/kg**, about \$6 100/kg at an assumed \$1.15 per euro. Lift-off mass 860 t → **2.5 %**. The hydrogen core gives high specific impulse, but the four solid boosters add mass.
3. **Electron**: \$9.1 M (average revenue per launch, 2026) for 0.3 t → **\$30 300/kg**; lift-off mass 13 t → **2.3 %**. Small rockets cost more per kilogram because avionics, range services and launch crews cost almost as much for a small rocket as for a big one.
4. Real missions are rarely full. A 150 kg satellite flying alone on Electron pays about \$60 000/kg, while large batches of constellation satellites fill a rocket almost to capacity. The comparison explains why customers who can accept a shared ride choose rideshare at \$7 000/kg, and why a dedicated small launch is bought for schedule and orbit, not price.



(a) Launches = ceil(3 000 / satellites per launch); per-launch counts from Amazon Leo missions (Starship value assumed). (b) 1 − (1 − p)^n for independent launches.

FIGURE 36.18 (a) Launches needed to deploy 3 000 satellites as a function of how many each launch carries, with per-launch counts from Amazon Leo missions. (b) The chance that at least one launch fails during a campaign of n launches, for four per-launch failure probabilities.

EXAMPLE: LAUNCHING A 3 000-SATELLITE CONSTELLATION

A broadband operator must deploy 3 000 satellites of about 600 kg — 1 800 t in all. How many launches are needed, and what is the risk (Figure 36.18)?

1. Satellites per launch are limited by the fairing volume and dispensers as much as by mass. Amazon Leo's missions carried 24 satellites on Falcon 9, 27–29 on Atlas V and 32–36 on Ariane 64; New Glenn was to carry 48.
2. Launches needed = $\lceil 3\,000 / k \rceil$: **125** on Falcon 9, **104** on Atlas V, **84** on Ariane 64, **63** on New Glenn, and about **50** on a Starship carrying 60.
3. At four launches a month, the Falcon 9 campaign takes 31 months, the New Glenn campaign 16 months. Real campaigns use several vehicles at once, because no single provider has enough capacity.
4. Risk: with independent launches each failing with probability p, the chance of at least one failure is $1 - (1 - p)^n$. For $p = 1$ % and 125 launches it is $1 - 0.99^{125} =$ **72 %**. For 63 launches on a new rocket with $p = 2$ %, it is $1 - 0.98^{63} =$ **72 %** as well.
5. The expected number of satellites lost is $N\,p$ — 30 at $p = 1$ %, whichever rocket is used. Bigger rockets reduce the number of launches but put more satellites at risk on each, so operators spread a constellation across several vehicles and keep spares in orbit.

AI IN LAUNCH OPERATIONS AND RANGE SAFETY

Machine-learning models are being developed to forecast the weather that most often delays launches — lightning, upper-level winds and anvil clouds. They are also meant to help evaluate launch-commit criteria. Other models scan thousands of telemetry channels during countdowns for anomalies that a human console operator might miss. Range safety, once a human officer with a destruct button, is now largely automated: autonomous flight-safety systems on Falcon 9, Electron and many new rockets decide on board whether to terminate a flight. With hundreds of launches a year, optimisation algorithms schedule pads, ranges, airspace closures and recovery ships, and estimate the risk that falling stages or debris pose to people and aircraft. Because these decisions affect public safety, the algorithms on board stay deterministic and verifiable, while learning-based tools support the humans who certify them; Chapter 52 discusses AI across launch and its assurance.

KEY TAKEAWAYS

- About 7 400 orbital launches have been attempted since 1957; the record year, 2025, saw 330 attempts and 11 failures, 60 % by US and 28 % by Chinese providers.
- Most early launchers were converted missiles: the R-7 (Soyuz) family has flown about 1 970 times and still carries crews. Saturn V, N1, Energia and the Shuttle were the super-heavies of the twentieth century.
- Falcon 9 dominates: half of all launches and 81 % of the satellite mass delivered in 2025, with a median of two days between flights and routinely re-used boosters.
- China flies the most varied fleet — 16–19 vehicle families a year — and in 2026 recovered orbital boosters for the first time; its Long March 10 is due in 2027.
- Europe (Ariane 6, Vega C, Spectrum), Japan (H3), India (LVM3, Vikram-1) and Russia (Soyuz-2, Angara) maintain sovereign access, but setbacks in 2025–26 show how fragile small fleets are.
- Small launchers fail far more often than larger ones (15 % since 2015) and many have disappeared; dedicated small launches cost several times more per kilogram than rideshare.
- Reusability, rideshare, high cadence and a new generation of super-heavies (Starship, SLS, Long March 10) are the trends shaping launch in the late 2020s.

REVIEW QUESTIONS

1. Why were most of the first launch vehicles converted ballistic missiles? Give two examples from the USSR and two from the United States, and name what each became.
2. Compare the N1 and the Saturn V. Why did the N1 fail, and what lesson does the comparison hold for vehicles with many engines?
3. From Figure 36.6, describe how the shares of world launches changed between 1965, 1985, 2005 and 2025, and explain the main causes.
4. Using Table 36.2, compare Falcon 9, Ariane 64 and Long March 5 in payload to LEO and GTO, propellants and reusability. Why do hydrogen-fuelled vehicles often still need solid boosters?
5. GCAT reports about 3 500 t launched in 2025, yet only about 2 870 t of satellites and spacecraft were delivered. Explain the difference.
6. In 2015–2026, 15 % of small-launcher attempts failed against 1.7 % of medium launches. Give three reasons for this difference.
7. Falcon 9 and Soyuz-U both reached 500 launches in about 15 years. What made Falcon 9's cadence fundamentally different?
8. A rideshare costs $7 000/kg and a dedicated Electron launch about $30 000/kg when full. Under what circumstances would a satellite operator still choose the dedicated launch?
9. Estimate the number of launches needed to deploy 1 500 satellites if each launch carries 40, and the probability of at least one failure if each launch has a 1.5 % chance of failing.
10. What is "sovereign access to space"? Using examples from 2023–2026, explain why governments fund more than one launch vehicle.

CHAPTER

37 Spaceports and Ground Infrastructure



Image: NASA/KSC

Every orbital rocket leaves from a few hectares of concrete surrounded by thousands of square kilometres of cleared sky and sea. This chapter explains how launch sites are chosen — latitude, reachable inclinations, safe corridors over the ocean, weather and logistics — and surveys the world's spaceports, from Florida, which hosted a third of all launches in 2025, to new sites in Norway, Australia and India. It then looks inside a launch complex: flame trenches, water deluge, umbilical towers, propellant farms and the three ways of assembling a rocket. It follows a launch campaign through the range, the countdown and the weather rules, and ends with noise, environmental licensing, and launching from aircraft and ships.

IN THIS CHAPTER

37.1 Choosing a Launch Site
37.2 The World's Spaceports
37.3 Inside a Launch Complex
37.4 The Range: Tracking, Safety and Recovery
37.5 The Launch Campaign and Countdown
37.6 Environment, Community and Regulation
37.7 Launching from the Air and the Sea

A rocket spends weeks at its launch site and minutes flying from it. A modern **spaceport** combines an assembly hall and a chemical plant storing hundreds of tonnes of liquids at −253 °C to −183 °C. It has a deluge system that pours more than a million litres of water in half a minute, and a tracking network watching thousands of kilometres of ocean. It also includes a marine and aviation operation that clears the rocket's path and recovers what comes back.

In 2025 the world made 330 orbital launch attempts. More than half — 175 — left from just two places in the United States: Florida's Space Coast (109) and Vandenberg in California (66). China flew 93 times from four centres and from ships; Rocket Lab's private site on New Zealand's Mahia Peninsula launched 17 times, more than Russia's Baikonur and Plesetsk together. This chapter explains why the sites are where they are and what happens on them; the vehicles are described in Chapters 34 and 36.

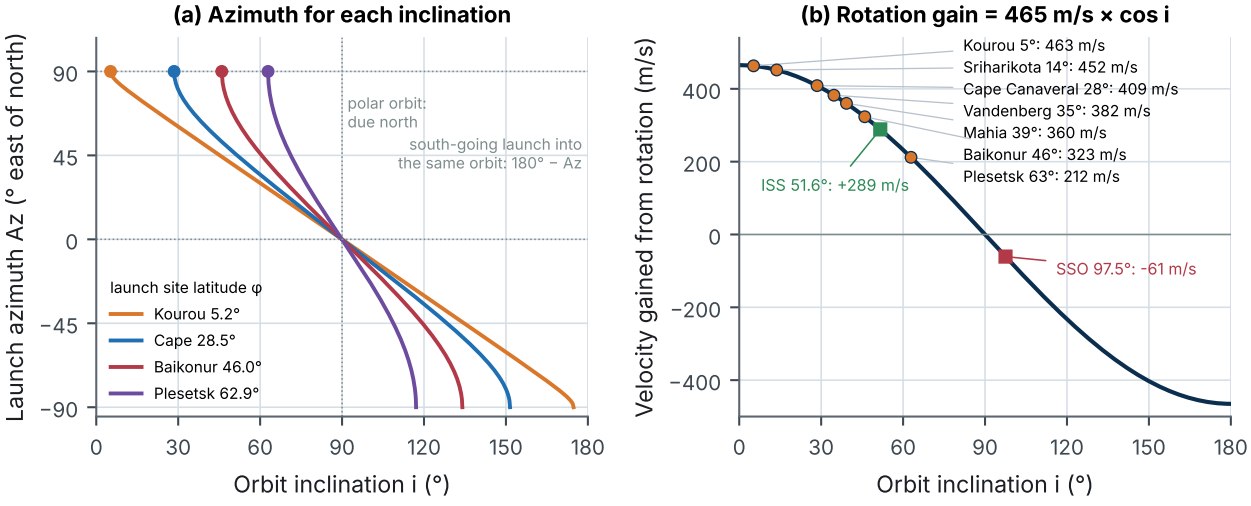
37.1 Choosing a Launch Site

37.1.1 Latitude, Azimuth and Inclination

Chapter 34 showed that a rocket launched from latitude ϕ on **azimuth** Az (the compass direction of flight, measured clockwise from north) enters an orbit of inclination i given by

$$\sin Az = \frac{\cos i}{\cos \varphi}$$

Because $|\sin Az|$ cannot exceed 1, a direct launch can reach only inclinations between ϕ and $180° - \phi$. Kourou (5.2° N) can reach almost any orbit. Plesetsk (62.9° N) can reach nothing below 62.9° without an expensive turn. Each allowed inclination can be reached flying north-east or south-east (north-west or south-west for a retrograde orbit), because the equation has two solutions, Az and $180° - Az$ (Figure 37.1). A launch due east gives the lowest inclination, $i = \phi$; a polar orbit needs a launch due north or south; a sun-synchronous orbit (97–98°) a launch slightly west of north or south.



sin Az = cos i / cos φ; surface speed 465.1 m/s × cos φ. Dots in (b): the most each site can gain, launching due east into i = φ. Range-safety limits on azimuth ignored.

FIGURE 37.1 (a) Launch azimuth needed to reach each orbit inclination from four sites (north-going solution; the south-going launch into the same orbit flies at 180° − Az). (b) The velocity a rocket gains from Earth's rotation depends only on the inclination of the orbit, 465 m/s × cos i; the dots mark the largest gain each site can obtain, by launching due east.

Earth's rotation carries the pad east at 465 m/s × cos ϕ: 463 m/s at Kourou, 409 m/s at Cape Canaveral, 212 m/s at Plesetsk. Only the component along the orbit helps, and it equals 465 m/s × cos i — the same for every site that can reach that inclination; the component across the orbit

must be cancelled. For the International Space Station's 51.6° orbit latitude therefore matters little. It matters greatly for orbits inclined *less* than the site's latitude, above all the geostationary orbit, because turning the orbital plane after launch is expensive.

EXAMPLE: AN ISS ORBIT AND A GEOSTATIONARY TRANSFER FROM KOUROU AND FROM BAIKONUR

Kourou lies at 5.24° N; Baikonur at 45.96° N. Assume a 200 km parking orbit (7 784 m/s).

1. **ISS orbit (51.64°).** From Kourou, $\sin Az = \cos 51.64° / \cos 5.24° = 0.6206 / 0.9958 = 0.6232$, so $Az = 38.6°$. From Baikonur, $\sin Az = 0.6206 / 0.6952 = 0.8927$, so $Az = 63.2°$.
2. **Rotation gain.** Kourou's surface moves at 463 m/s, Baikonur's at 323 m/s. The speed the rocket must add is the vector difference between orbital velocity and ground velocity, $\sqrt{v^2 - 2vv_r \sin Az + v_r^2}$. This gives a gain of 280 m/s at Kourou and 287 m/s at Baikonur: the equatorial site's extra rotation points across the orbit and is wasted.
3. **Geostationary transfer.** Launching due east, Kourou reaches a transfer orbit inclined about 5°. Baikonur's drop zones keep its launches at 51.5° or more, so a satellite would have to remove 51.6° of inclination at apogee, where it moves at 1.60 km/s and must reach 3.07 km/s. The combined burn is $\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1 v_2 \cos \Delta i}$ = 1.49 km/s from Kourou against 2.43 km/s from Baikonur.
4. **Propellant.** With a 320 s apogee engine (v_e = 3.14 km/s), the propellant share of the satellite's mass is $1 - e^{-\Delta v / v_e}$: 38 % from Kourou and 54 % from Baikonur. This is why Proton's Breeze-M upper stage performs most of the plane change itself in a long, multi-burn ascent, and why Europe built its spaceport near the equator.

37.1.2 What the Rotation Is Worth

Of the roughly 9.3 km/s of ideal Δv needed to reach low orbit (Chapter 34), typically 0.2–0.45 km/s comes free from the spin of the Earth. The next example puts a price on it.

EXAMPLE: THE PAYLOAD VALUE OF EARTH'S ROTATION

Take an expendable two-stage kerosene rocket similar to Falcon 9: first stage 411 t of propellant, 25.6 t dry, average specific impulse 296 s; second stage 107.5 t of propellant, 4.0 t dry, 348 s.

1. Solving the rocket equation stage by stage for an ideal Δv of 9 300 m/s gives a payload of **21.9 t**.
2. Repeating for 9 400 m/s gives 21.1 t: each extra 100 m/s costs about 0.82 t, or 3.8 % of the payload.
3. A due-east launch from Kourou (463 m/s) rather than Cape Canaveral (409 m/s) gains 54 m/s, about 0.45 t; rather than Plesetsk (212 m/s), 251 m/s, or **2.2 t** — 10 % of the payload.
4. A sun-synchronous orbit (97.5°) is slightly retrograde: the rocket loses 61 m/s to rotation wherever it launches, about 0.5 t.

37.1.3 Safety Corridors and Overflight

Geometry says which orbits a site *could* reach; safety says which it *may*. Rockets drop spent stages and fairings along their path, and a failing vehicle can scatter debris wherever its **instantaneous impact point** — where it would land if the engines stopped — happens to be. Ranges therefore allow only azimuths whose early flight and drop zones lie over the sea or empty land (Figure 37.2). Cape Canaveral launches between north-east, along the US coast to inclinations of about

57°, and east-south-east over the Atlantic. Vandenberg launches south over the Pacific, and Kourou has open ocean to the north and east. Starship leaves Starbase over the Gulf of Mexico and passes between Florida and Cuba.

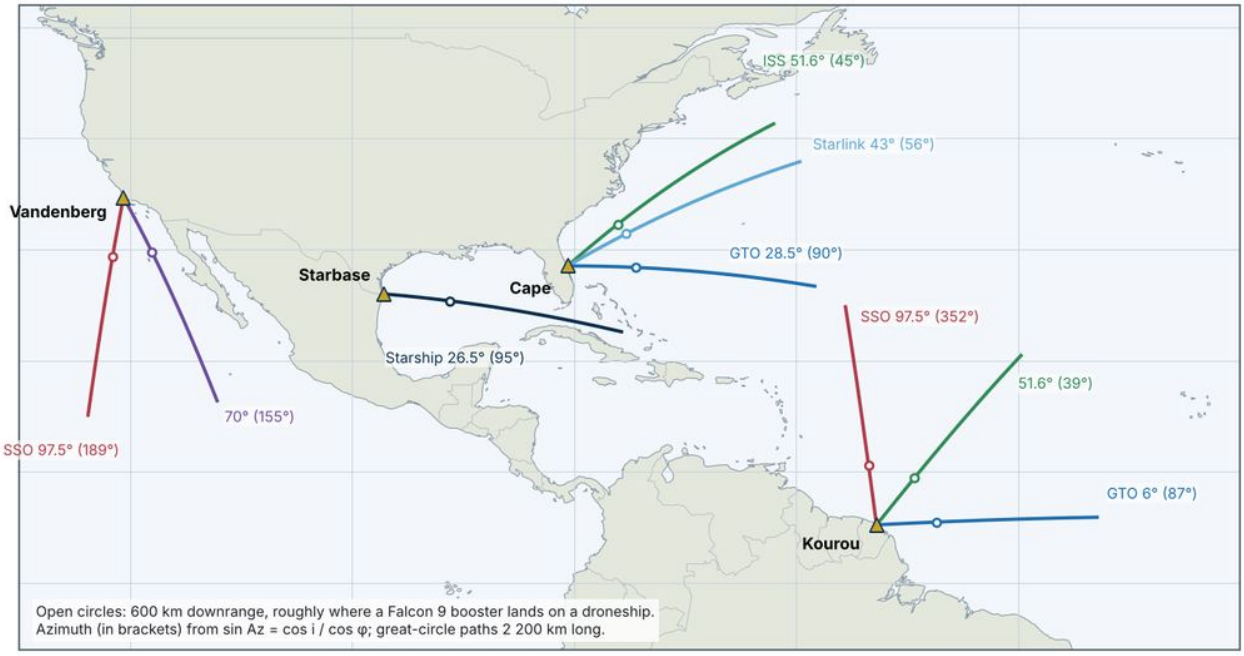
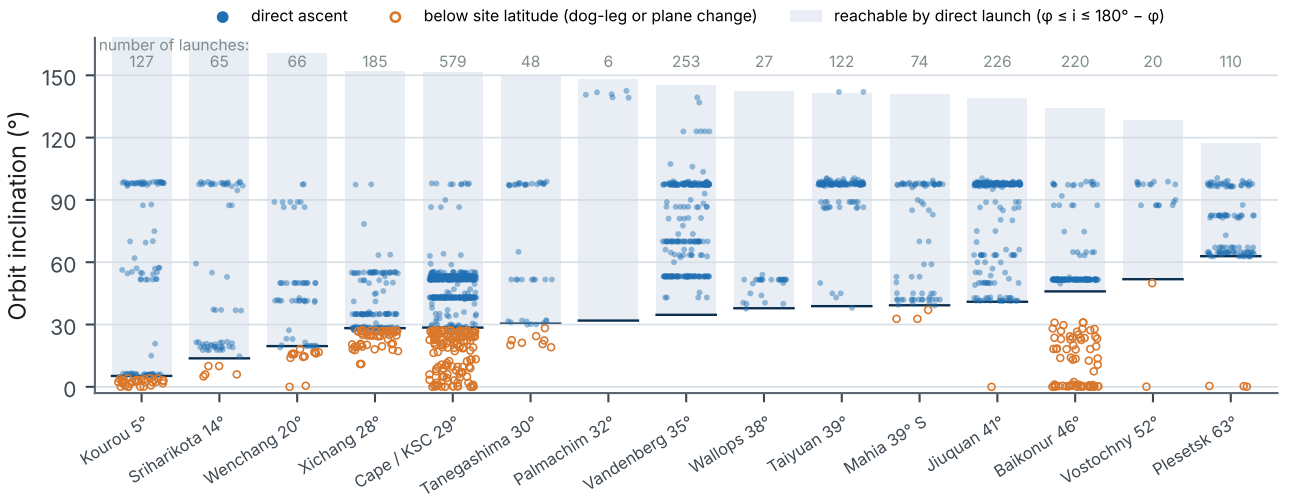


FIGURE 37.2 Initial launch corridors from four spaceports in the Americas: great-circle paths for the first 2 200 km on the azimuth needed for each inclination.

The corridors leave a fingerprint in the orbits each site actually uses (Figure 37.3). Baikonur, deep inside Kazakhstan, drops its stages in designated zones on the steppe; partly to keep them out of China, it launches no lower than about 51.5°, which is why the ISS orbits at 51.6°. Palmachim, on Israel's coast, launches westward over the Mediterranean to avoid overflying its neighbours, so every satellite launched from Israel flies a retrograde orbit inclined 139–143°. Sriharikota's polar launches must turn ("dog-leg") to avoid Sri Lanka, one reason India is building a second spaceport farther south. Florida had not launched into a polar orbit since February 1969 until a Falcon 9 flew south along the coast on 30 August 2020. This "polar corridor" has since carried the PACE satellite and, in 2025, the first crewed polar flight, Fram2.



Successful orbital launches 2010 – 13 Oct 2026 (GCAT, J. McDowell). Inclination of the final rocket stage (or first payload). Sites ordered by latitude.

FIGURE 37.3 Inclinations reached by successful orbital launches from 15 spaceports, 2010–2026, against the band each site's latitude allows (shaded). Clusters reveal the corridors: 51.6° at Baikonur, 43° and 53° (Starlink) at the Cape, 97–98° at Vandenberg, Taiyuan and Jiuquan, 139–143° at Palmachim. Open circles, below the site's latitude, are mostly geostationary missions whose upper stages removed inclination in orbit.

37.1.4 Weather, Logistics and Politics

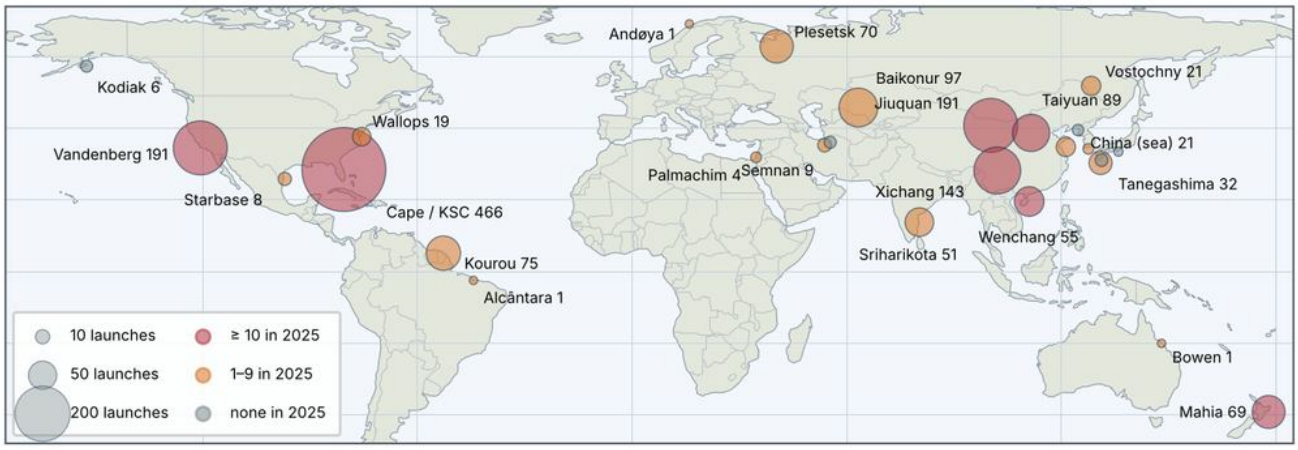
A launch site also needs tolerable weather (Florida's thunderstorms are discussed below), ports and roads for stages up to 9 m in diameter, power, fresh water, a workforce and an airport. Large stages are often built far away: the SLS core stage reaches Florida by barge from New Orleans, Ariane 6 stages cross the Atlantic by ship and Soyuz rockets reach Baikonur by rail. SpaceX builds Starship beside its pads at Starbase, a company town incorporated as a city in May 2025.

Politics shapes the map as much as physics. Russia leases Baikonur from Kazakhstan for $115 million a year until 2050 and built Vostochny, in its own Far East, to depend less on it. China placed Jiuquan, Taiyuan and Xichang far inland for security, which is why their spent stages fall on land. And many countries want a "sovereign" launch site even if it flies only a few times a year.

37.2 The World's Spaceports

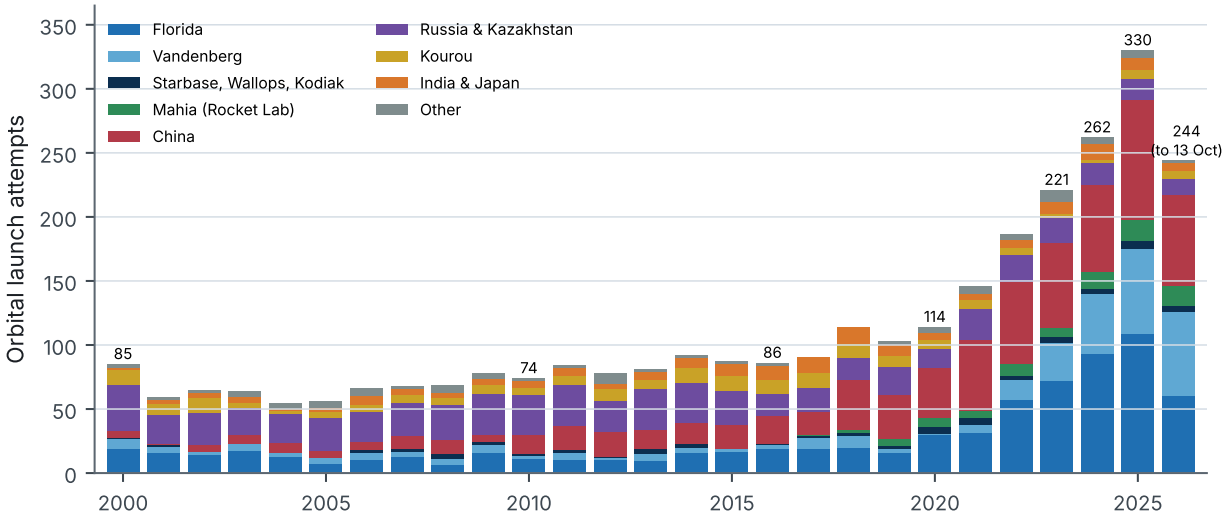
37.2.1 Where Rockets Fly From

About 40 sites, counting airfields used for air launch, have sent rockets towards orbit; 22 did so in 2025 (Figure 37.4). Over the whole space age the busiest has been Plesetsk, with 1 688 launches, mostly Soviet military satellites, followed by Baikonur (1 560) and Cape Canaveral with Kennedy (1 401). Since 2016 the world launch rate has nearly quadrupled, and the growth came almost entirely from Florida, Vandenberg and China (Figure 37.5).



Orbital launch attempts 2016–2025 by spaceport (GCAT, J. McDowell; Wenchang includes the Hainan commercial site; Chinese sea launches plotted at Haiyang).

FIGURE 37.4 Orbital launch attempts from each spaceport, 2016–2025 (bubble area), coloured by activity in 2025. Florida, Vandenberg and China's four centres dominate; Rocket Lab's Mahia site in New Zealand outflew every Russian site.



GCAT (J. McDowell), orbital and deep-space launch attempts including failures and Starship test flights; 2026 to 13 October.

FIGURE 37.5 Orbital launch attempts per year by spaceport, 2000–2026. Florida and Vandenberg together hosted 175 of the 330 attempts in 2025; China's launch rate has risen sixfold since 2010.

Table 37.1 summarises the busiest spaceports and Table 37.2 the smaller and newer ones.

TABLE 37.1 The world's busiest spaceports. "2025" is the number of orbital launch attempts that year (GCAT, J. McDowell).

Spaceport	Latitude	Main pads and vehicles (2026)	2025	Notes
Cape Canaveral SFS & Kennedy Space Center (USA)	28.5° N	SLC-40, LC-39A (Falcon 9, Falcon Heavy); SLC-41 (Atlas V, Vulcan); LC-36 (New Glenn); LC-39B (SLS)	109	Busiest site; Starship pads at LC-39A and SLC-37 under construction
Vandenberg SFB (USA)	34.7° N	SLC-4E (Falcon 9); SLC-6 (Falcon, being rebuilt); SLC-2W (Firefly Alpha); SLC-8 (Minotaur)	66	Polar and sun-synchronous launches south over the Pacific
Starbase, Texas (USA)	26.0° N	Pads 1 and 2 (Starship)	5	Up to 25 launches a year approved in 2025
Jiuquan (China)	41.0° N	LC-43/91 (Long March 2F, crews); LC-43/94 (Long March 2C/2D, 4B/4C); commercial pads (Kuaizhou, Ceres-1, Zhuque, Long March 12B)	34	China's crewed launch site, in the Gobi Desert

Spaceport	Latitude	Main pads and vehicles (2026)	2025	Notes
Wenchang (China)	19.6° N	LC-101 (Long March 5), LC-201 (Long March 7/7A), LC-301 (Long March 10); commercial Pads 1–2 (Long March 8, 12, 10B)	21	Coastal; heavy lift, station cargo and lunar missions
Xichang (China)	28.2° N	LC-2, LC-3 (Long March 3A/3B/3C)	19	Geostationary and BeiDou launches; stages fall inland
Mahia, LC-1 (New Zealand)	39.3° S	LC-1A, LC-1B (Electron)	17	Private; up to 120 launch opportunities a year
Taiyuan (China)	38.8° N	LC-9, LC-9A, LC-16 (Long March 2, 4, 6, 6A)	12	Sun-synchronous and polar orbits
Plesetsk (Russia)	62.9° N	Site 43 (Soyuz-2), Site 35 (Angara)	9	Military; high inclinations only
Kourou, Guiana Space Centre (France/ESA)	5.2° N	ELA-4 (Ariane 6), ZLV (Vega-C); ELM-Diamant (micro-launchers, planned)	7	Near-equatorial; ocean to the north and east
Chinese sea launches	≈ 35° N	Ships from Haiyang, Shandong (Long March 11, Jielong-3, Ceres-1S, Gravity-1)	7	28 launches since 2019, all successful
Baikonur (Kazakhstan, leased by Russia)	46.0° N	Site 31 (Soyuz crew and cargo); Site 81 (Proton); Site 45 (Soyuz-5)	6	Site 31 damaged in November 2025, repaired by March 2026
Sriharikota, SDSC-SHAR (India)	13.7° N	First and Second Launch Pads (PSLV, GSLV, LVM3, SSLV)	5	Third Launch Pad approved January 2025
Tanegashima (Japan)	30.4° N	Yoshinobu LP-1, LP-2 (H3)	4	H-IIA retired June 2025

37.2.2 The Americas

Florida is two neighbouring sites run as one range. NASA's Kennedy Space Center owns Launch Complex 39, built for the Saturn V. Pad 39A is leased to SpaceX for Falcon 9 and Falcon Heavy and is being equipped for Starship. Pad 39B was rebuilt for the Space Launch System, which flew Artemis II from it on 1 April 2026. Cape Canaveral Space Force Station hosts SpaceX's busiest pad, SLC-40, which has also launched crews since 2024, and United Launch Alliance's SLC-41 for Atlas V and Vulcan. It also hosts Blue Origin's LC-36 for New Glenn, where a booster exploded during a hot-fire test in May 2026. Other pads are being rebuilt for newcomers such as Relativity and Stoke. In late 2025 the Air Force approved SLC-37 for up to 76 Starship launches a year.

Vandenberg, on the Californian coast, is America's polar spaceport; its 66 launches in 2025 were almost all Falcon 9s from SLC-4E. In October 2025 the Air Force approved up to 100 Falcon launches a year from SLC-4 and a rebuilt SLC-6. The California Coastal Commission, worried about sonic booms and wildlife, objected, and the dispute went to court. **Wallops**, in Virginia, hosts Antares and Minotaur pads and Rocket Lab's LC-2 for Electron and LC-3 for Neutron, whose debut has slipped to 2027. At **Starbase**, in southern Texas, Starship's second pad, with a water-cooled flame trench, flew all three Starship flights of 2026, including the first orbital flight on 28 September.

Europe's spaceport is the **Guiana Space Centre** at Kourou, run by the French space agency CNES with ESA. Ariane 6 flies from ELA-4 and Vega-C from the Vega pad; the Soyuz pad fell idle when Russia withdrew in 2022, and the old Diamant site is being rebuilt for European micro-launchers. Brazil's **Alcântara**, at 2.3° S, hosted its first orbital attempt in more than 20 years in December 2025, when Innospace's HANBIT-Nano failed about 30 s after lift-off.

37.2.3 Russia and Kazakhstan

Baikonur, on the Kazakh steppe, launched Sputnik and Gagarin and is still Russia's only site for crews and ISS cargo. Soyuz rockets are assembled horizontally, rolled to the pad by train and raised by a hydraulic erector (Figure 37.6). On 27 November 2025, during the launch of Soyuz MS-28, the mobile service cabin under the rocket at Site 31, not properly secured after retraction, fell into the flame trench. With Gagarin's original pad retired in 2019, Russia briefly had no pad for crewed flights. More than 150 workers rebuilt the cabin, installing 19 m structural members of about 17 t. The pad was cleared for a Progress launch in March 2026. Site 45 flew the first Soyuz-5 on a suborbital test on 30 April 2026.



FIGURE 37.6 A Soyuz rocket rolled out by train to Site 31 at Baikonur, September 2023. Assembled horizontally, it travels on the transporter-erector that will raise it at the pad. Image: NASA (NASA/Bill Ingalls).



FIGURE 37.7 The erector raising a Soyuz into the launch table at Site 31, September 2024. The four support arms of the pad will hold the rocket by its waist until lift-off. Image: NASA (NASA/Bill Ingalls).

Plesetsk, 800 km north of Moscow, launches military Soyuz-2 and Angara rockets; **Vostochny**, in the Far East, has launched Soyuz-2 since 2016 and Angara A5 since 2024, but only twice in 2025.

37.2.4 China

China has four state launch centres and a growing commercial sector. **Jiuquan**, in the Gobi Desert, launches Shenzhou crews and, increasingly, commercial rockets from new pads nearby — among them LandSpace's Zhuque-3, whose booster touched down on its legs in August 2026, although a fire followed. **Taiyuan** specialises in sun-synchronous orbits and **Xichang**, in Sichuan, in geostationary and BeiDou missions. Both lie inland: spent stages carrying toxic hypergolic residue fall in mountain valleys, and residents downrange are warned to shelter or leave.

Wenchang, on Hainan Island, is the coastal site for Long March 5 and 7 and the new crew-capable Long March 10. The new rocket's first stage flew a suborbital test from the new pad LC-301 in February 2026. Beside it, the **Hainan commercial launch site**, China's first commercial spaceport, opened its two pads in 2024–25. It flew Long March 10B's maiden flight on 10 July 2026 and was building Pads 3 and 4 for more than 60 launches a year. China also launches solid rockets from **ships** sailing from Haiyang in Shandong (see the last section).

37.2.5 India, Japan, Korea and the Middle East

India's **Satish Dhawan Space Centre** on Sriharikota Island has two pads; in January 2025 the government approved a third (₹3 985 crore, about four years to build) for the Next Generation Launch Vehicle. **Kulasekarapattinam**, at 8.4° N in Tamil Nadu, is being built for SSLV and private rockets, with a first launch planned for 2027.

Japan launches H3 from the Yoshinobu complex at **Tanegashima**, where rockets are stacked in an assembly building and moved to the pad on a mobile launcher (Figure 37.8). The H-IIA made its final flight in June 2025; H3 failed in December 2025 and returned to flight in mid-2026. The solid Epsilon S, delayed by two motor explosions in ground tests, is due to fly from **Uchinoura** by March 2027. South Korea's **Naro Space Center** launched Nuri successfully in November 2025, the first time with Hanwha Aerospace building the rocket, and again in October 2026.



FIGURE 37.8 An H-IIA rocket carrying the Global Precipitation Measurement Core Observatory rolls out on its mobile launcher to Launch Pad 1 at Tanegashima in February 2014. The red-and-white masts are lightning towers. Image: NASA (NASA/Bill Ingalls).

Israel launches Shavit rockets from **Palmachim** air base; Iran uses the **Imam Khomeini Spaceport** near Semnan and the military site at **Shahroud**; North Korea's **Sohae** last launched, unsuccessfully, in May 2024.

37.2.6 New and Small Spaceports

Small launchers have prompted a wave of new sites, many at high latitudes where polar orbits can be reached over empty sea (Table 37.2). Isar Aerospace's Spectrum failed 30 s into its first flight from **Andøya**, Norway, in March 2025 but reached orbit on 5 September 2026 — the first orbital launch from continental Western Europe.

TABLE 37.2 Selected new and smaller spaceports.

Spaceport	Latitude	Operator and vehicle	Status (October 2026)

Spaceport	Latitude	Operator and vehicle	Status (October 2026)
Andøya (Norway)	69.3° N	Isar Aerospace Spectrum	Failed March 2025; reached orbit 5 September 2026
SaxaVord, Shetland (UK)	60.8° N	RFA One	Awaiting first launch; rocket de-stacked July 2026
Esrange (Sweden)	67.9° N	Firefly Alpha (LC-3C)	First orbital launch targeted 2028
Kulasekarapattinam (India)	8.4° N	ISRO SSLV, private rockets	Under construction; first launch 2027
Bowen (Australia)	20.0° S	Gilmour Space Eris	First flight failed after 14 s (July 2025); second planned late 2026
Alcântara (Brazil)	2.3° S	Innospace HANBIT-Nano	First flight failed (December 2025)
Spaceport Kii (Japan)	33.5° N	Space One Kairos	Three failures (2024–26)
Spaceport America (USA)	33.0° N	Virgin Galactic (suborbital)	Delta-class spaceplane being prepared for flight tests

37.3 Inside a Launch Complex

37.3.1 The Pad: Mounts, Trenches and Deflectors

At the heart of every pad is the **launch mount**, a steel table that carries the rocket through **hold-down clamps** (Figure 37.9). Liquid-fuelled rockets ignite a few seconds before release while computers confirm that every engine has reached full thrust; if one misbehaves, everything shuts down with the vehicle still clamped — a **hold-down abort**.

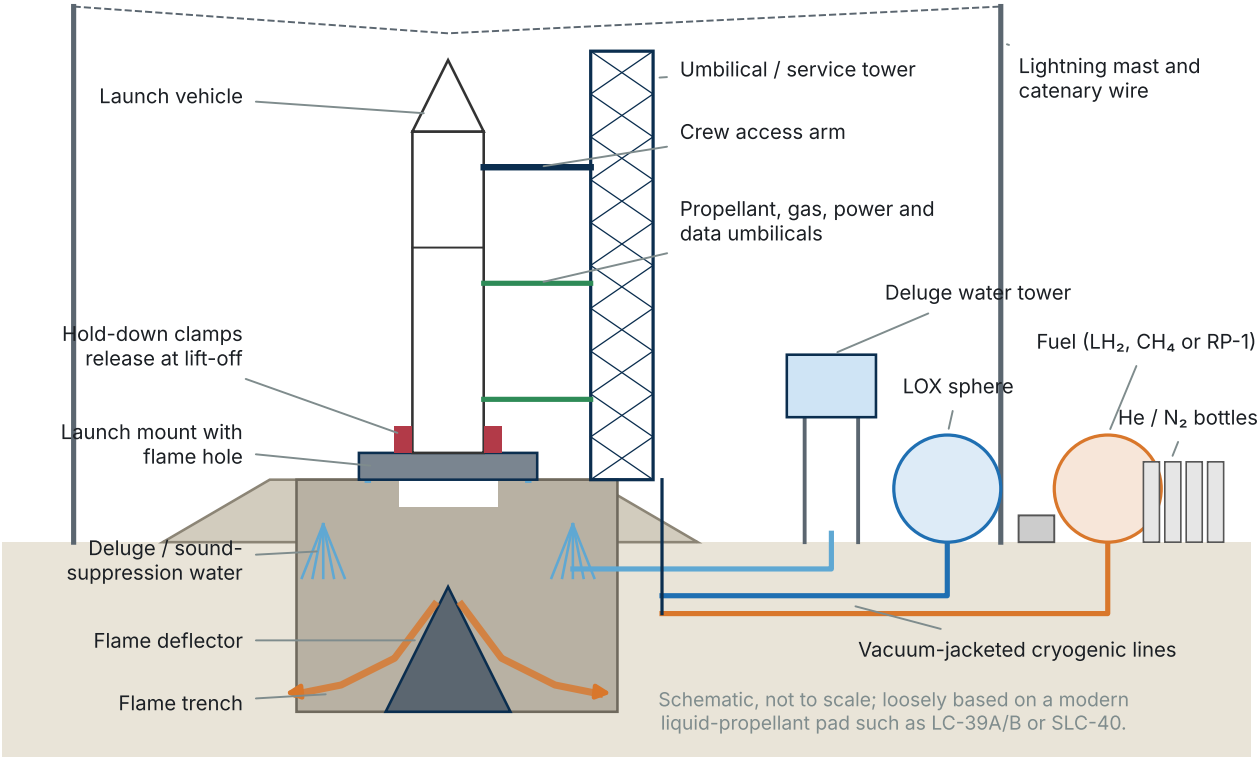


FIGURE 37.9 Main elements of a liquid-propellant launch pad. The exhaust passes through the flame hole into a trench, where a deflector turns it sideways; water floods the mount and trench at ignition; umbilicals deliver propellant, gas, power and data until lift-off; lightning masts and propellant storage stand at a safe distance.

The exhaust, at 2–4 km/s and up to about 3 000 °C, passes through a hole in the mount into a **flame trench**, a concrete channel where a wedge-shaped **flame deflector** turns it sideways. At Pad 39B the trench is more than 130 m long and lined with about 96 000 heat-resistant bricks. The deflector, completed in 2018, is built from some 150 steel plates of up to 1.8 t each (Figure 37.10). Without a trench the jet strikes the ground directly: on Starship's first flight in April 2023, Super Heavy's 33 engines dug a crater under the mount and threw concrete across the site. SpaceX then fitted a water-cooled steel plate under the mount, and its second Starbase pad has a full trench.



FIGURE 37.10 The new main flame deflector in the flame trench at Launch Complex 39B in May 2018, built to divert the exhaust of the Space Launch System sideways along the trench. Image: NASA/KSC (NASA/Kim Shiflett).

37.3.2 Water: Deluge and Sound Suppression

Ignition creates a sharp pressure pulse, the **ignition overpressure**, which reflects from the trench onto the rocket. On the first Space Shuttle flight in 1981 it caused most of the damage to the orbiter's heat shield, on which 16 tiles were lost and 148 damaged. Lift-off also radiates about 200 dB of sound power, which shakes the payload (Chapter 34). The cure is water sprayed into the exhaust and over the mount, which absorbs energy as it is shattered into droplets and turned to steam. At Pad 39B a 400 000-gallon (1.5 million litre) tower empties in under 30 s, at a peak of about 70 m^3/s (Figure 37.11). The Shuttle's 300 000 gallons, released just before main-engine ignition, emptied in 41 s. Ariane 6 receives about a quarter of an Olympic pool in 20 s. Most of the white cloud seen at lift-off is this water turned to steam.



FIGURE 37.11 A test of the Pad 39B sound-suppression system in September 2019: water pours over the SLS mobile launcher and through its flame hole into the trench. Image: NASA/KSC (NASA/Kim Shiflett).

37.3.3 Umbilicals, Towers and Catch Arms

Until lift-off the rocket is fed through **umbilicals** — hoses and cables for propellant top-up, pressurant gas, conditioned air, power and data. They are attached by quick-disconnect plates that separate in the last seconds or as the vehicle rises. Upper-stage umbilicals hang from swing arms on a **service tower**, which for crewed launches also carries a **crew access arm** and an escape system; tail umbilicals at the base retract into protective housings at lift-off. These interfaces are a common cause of scrubs: Artemis I was delayed from August to November 2022 partly by hydrogen leaks at a quick-disconnect on its tail service mast.

SpaceX's Starship towers add a new function. Their two "chopstick" arms stack the booster and ship on the mount and, since Flight 5 in October 2024, catch the returning booster in mid-air. This makes the tower a crane, umbilical tower and landing pad in one.

37.3.4 Propellants and Gases

A pad is also a cryogenic plant (Table 37.3). NASA's newest hydrogen sphere at Pad 39B, finished in 2022, holds about 4 700 m^3 of usable liquid hydrogen — the largest such tank in the world. Its glass-bubble insulation cuts boil-off by 40–50 % compared with the perlite of its 1960s neighbour. Transfer lines are **vacuum-jacketed** to limit heat leaks.

TABLE 37.3 Commodities stored and handled at a typical launch site.

Commodity	Storage	Density (kg/m^3)	Used for	Main hazard
Liquid oxygen (LOX)	−183 °C; Falcon 9 subcools to about −207 °C	1 141 (more when subcooled)	Oxidiser for most liquid rockets	Intense fire with any fuel; cold burns
Liquid hydrogen (LH_2)	−253 °C	71	Fuel (SLS, Ariane 6, H3, Centaur)	Invisible flame; leaks; embrittlement
Liquid methane (LCH_4)	−162 °C	422	Fuel (Starship, Vulcan, New Glenn, Zhuque)	Flammable vapour cloud
RP-1 kerosene	Ambient; Falcon 9 chills to about −7 °C	≈ 810	Fuel (Falcon, Soyuz, Electron, Long March 6/7)	Fire

Commodity	Storage	Density (kg/m³)	Used for	Main hazard
Hydrazine, MMH, N_2O_4	Ambient, sealed	870–1 450	Spacecraft thrusters; some upper stages	Toxic, carcinogenic; ignite on contact
Helium, nitrogen	Gas at 20–40 MPa	—	Pressurant; purges	Asphyxiation; stored energy

Cold propellant is denser: Falcon 9 chills its oxygen to about −207 °C, 24 °C below its boiling point, and its kerosene to about −7 °C. Because the propellant then warms in the tanks, it is loaded only in the last 35 minutes, with the crew already aboard on Dragon flights — a "load and go" procedure NASA approved in 2018. Spacecraft propellants are **hypergolic** (they ignite on contact) and toxic. Technicians in protective suits load them days or weeks earlier in a payload processing facility (Figure 37.12). This is done before the spacecraft is sealed in its fairing (Chapter 34).



FIGURE 37.12 Fuelling technicians in self-contained atmospheric protective ensemble (SCAPE) suits take a sample of monomethylhydrazine before loading it into NASA's Solar Dynamics Observatory at a payload processing facility near Kennedy Space Center in January 2010. Image: NASA/KSC.

37.3.5 Assembling the Rocket

Rockets are assembled in one of three ways (Figure 37.13). **Vertical assembly and transfer** stacks the vehicle on a mobile launcher inside a tall building, such as NASA's 160 m Vehicle Assembly Building, and carries it to the pad. The crawler-transporter weighs about 3 000 t, can carry 8 200 t and creeps the 6.8 km to Pad 39B at 1.6 km/h. It burns some 390 litres of diesel per kilometre, while hydraulic jacks keep its load level on the ramp (Figure 37.14). A second, taller mobile launcher for SLS Block 1B was abandoned in 2026 after its cost grew from $383 million to about $1.8 billion. Japan, China and United Launch Alliance use the same principle with platforms on rails or tow vehicles.

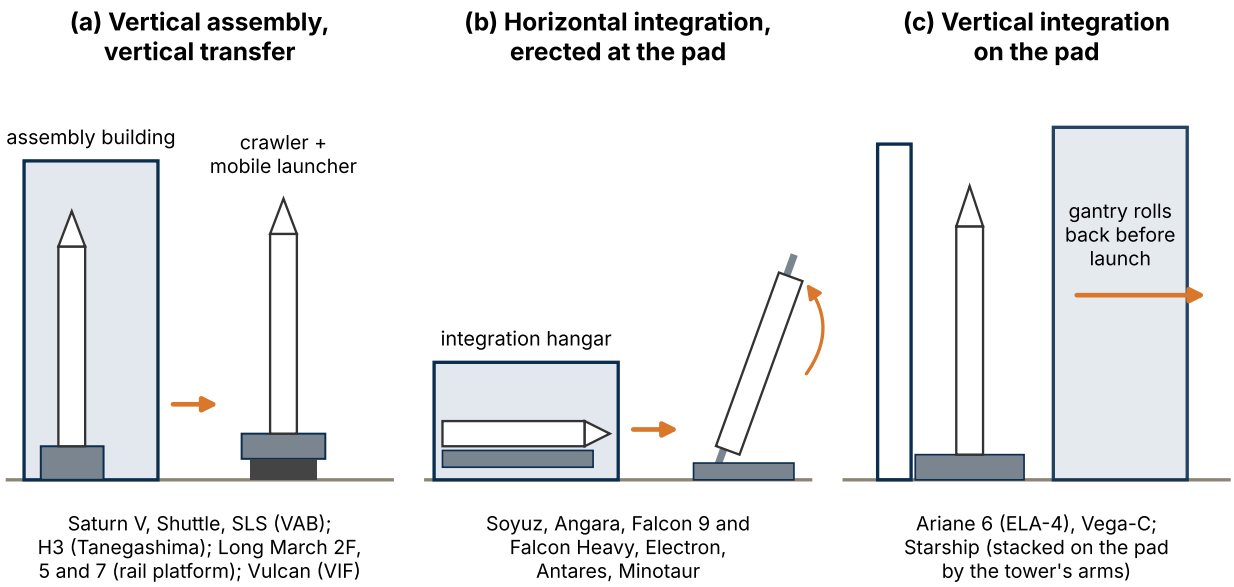


FIGURE 37.13 Three ways to assemble a launch vehicle: (a) stacked vertically in an assembly building and carried to the pad on a mobile launcher; (b) integrated horizontally in a hangar and raised at the pad by a transporter-erector; (c) assembled vertically on the pad inside a mobile gantry that rolls away before launch.

Horizontal integration, the Soviet tradition followed by SpaceX and Rocket Lab, joins the stages lying down in a simple hangar; a transporter-erector then raises the rocket at the pad (Figure 37.6). The payload is usually mated horizontally too, which some national-security satellites cannot tolerate, so in 2020 SpaceX proposed a mobile service tower at Pad 39A for vertical integration. **Vertical integration on the pad** builds the rocket in place inside a mobile gantry: Ariane 6's, at ELA-4, is 90 m tall, weighs about 8 200 t and rolls 120 m away before launch.

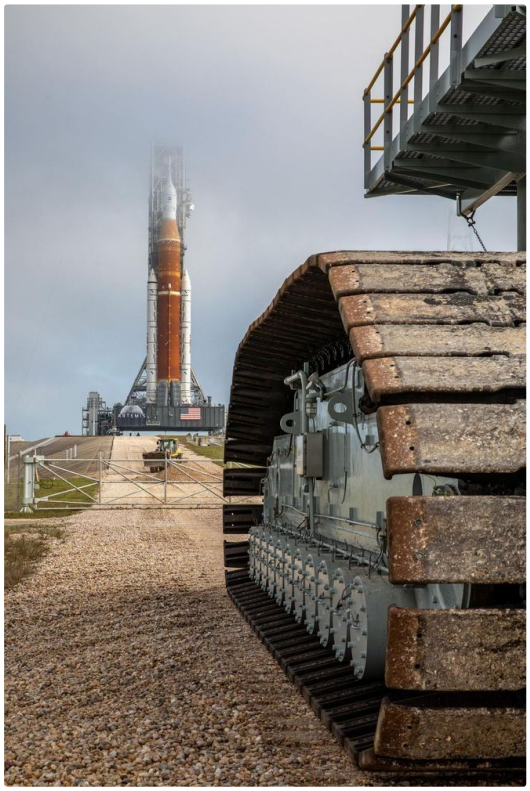


FIGURE 37.14 The crawler-transporter that carried NASA's Space Launch System to Launch Pad 39B in March 2022, with the rocket standing on its mobile launcher at the pad in the background. The crawler rides on four double-tracked trucks, each driven by four 375 hp electric motors. Image: NASA/KSC (NASA/Kim Shiflett).

37.4 The Range: Tracking, Safety and Recovery

37.4.1 Instrumentation and Telemetry

The **range** is everything beyond the pad that keeps a launch safe and observed: radars, telemetry antennas, optical trackers, weather instruments and a control centre. The US Space Force runs the Eastern Range from Florida (Space Launch Delta 45) and the Western Range from Vandenberg (Space Launch Delta 30); NASA runs Wallops and CNES Kourou, each with downrange tracking stations. During ascent the range receives telemetry and compares the instantaneous impact point with pre-set limits. Since 2017 most American launches have carried an **autonomous flight safety system** that makes any destruct decision on board (Chapter 34), so ranges depend less on tracking radars and can support launches in quick succession.

37.4.2 Hazard Areas in the Sky and at Sea

Before each launch the range defines **hazard areas** where falling stages, fairings or failure debris could land. Aircraft are kept out by notices issued through the Federal Aviation Administration (FAA), ships by navigational warnings, and a single boat inside the zone can halt the count — the range is "fouled". If a vehicle breaks up unexpectedly, the FAA can activate a **debris response area**. When Starship Flight 7 broke up over the Caribbean on 16 January 2025, dozens of airliners were held or diverted. Debris was reported on the Turks and Caicos Islands.

37.4.3 Recovery Fleets and Landing Zones

Reusable rockets have turned ranges into recovery operations. SpaceX has autonomous **droneships**, one in the Atlantic and one in the Pacific since a third was reassigned to Starship in April 2026. They wait about 600 km downrange for Falcon boosters (Figure 37.2), and other ships collect fairing halves. Boosters that fly back land on concrete **landing zones**. At Cape Canaveral, Landing Zone 1 was retired when its lease ended in July 2025. LZ-40 beside SLC-40 took its first booster on 13 February 2026. Dragon capsules have splashed down off California since 2025, after pieces of their jettisoned trunks reached land in Australia and North Carolina. Blue Origin lands New Glenn boosters on the ship *Jacklyn*, and in July 2026 China caught a Long March 10B first stage in a net of cables on the ship *Linghangzhe*.

37.5 The Launch Campaign and Countdown

37.5.1 From Arrival to Lift-off

A campaign begins weeks before the countdown. A large satellite typically arrives four to eight weeks before launch, is tested, fuelled, mated to its adapter and encapsulated. The rocket's stages are checked, joined and often test-fired on the pad — a **static fire** — or put through a **wet dress rehearsal**, a countdown with propellant loading that stops before ignition. Falcon 9 is usually rolled out within a day of launch, Soyuz two days before, while SLS spends weeks at the pad.

37.5.2 The Countdown

The countdown is a scripted sequence (Table 37.4). The **launch director** polls each team at set points ("go/no-go"), and built-in **holds** give time to catch up. In the last minutes an automatic sequencer takes over, ready to stop if any measurement drifts out of limits.

TABLE 37.4 Falcon 9 countdown milestones for a Dragon cargo mission from LC-39A (CRS-29, NASA, November 2023; times approximate).

Time (min:s)	Event
−38:00	Launch director verifies "go" for propellant load
−35:00	RP-1 loading begins; first-stage LOX loading begins
−16:00	Second-stage LOX loading begins
−07:00	Pre-launch engine chill begins
−05:00	Dragon transfers to internal power
−01:00	Flight computer begins final checks; tanks pressurised for flight
−00:45	Launch director verifies "go" for launch
−00:03	Engine controller commands ignition sequence
00:00	Lift-off

The **launch window** may last hours for a sun-synchronous satellite, or be **instantaneous** for a mission that must meet the ISS, leaving no time to resolve a problem.

37.5.3 Launch Commit Criteria

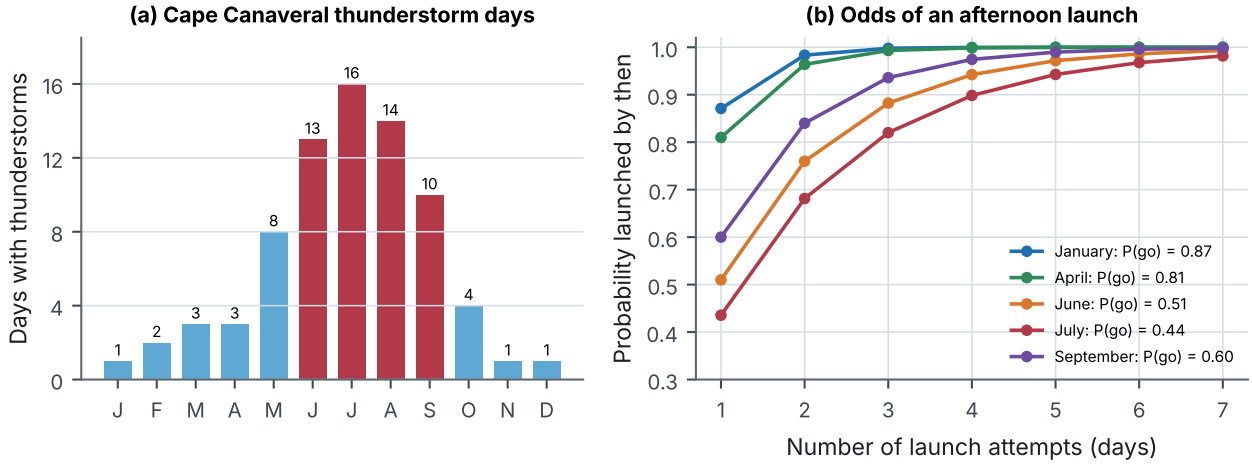
Weather causes about half of all scrubs at Cape Canaveral: in 2020 the 45th Weather Squadron reported that 49 % of scrubs were due to weather, which affected 24 % of all countdowns. The rules are written as **launch commit criteria** (Table 37.5). Wind limits protect the vehicle on the pad and in the jet stream, and cloud and lightning rules protect it from **triggered lightning**. A rocket and its long, ionised plume can trigger a strike in a field too weak to produce lightning on its own. This happened to Apollo 12 in 1969 and to Atlas-Centaur 67, destroyed in 1987. Electric-field mills around the pads measure the field continuously.

TABLE 37.5 Weather launch commit criteria for a crewed Falcon 9 from Florida (NASA fact sheet, abridged).

Parameter	Do not launch if…
Ground wind	Sustained wind at 49 m (162 ft) on the pad exceeds 13.4 m/s (30 mph)
Upper-level winds	Wind shear aloft could cause control problems
Lightning	Lightning within 18.5 km (10 nautical miles) of the pad or flight path in the last 30 min, unless specific conditions are met
Anvil clouds	Within 18.5 km of an attached anvil (with exceptions) or of any detached anvil
Cumulus clouds	Within 18.5 km of cumulus with tops colder than freezing, unless distance rules are met
Thick clouds	Flight through a layer more than 1.4 km thick extending into freezing temperatures
Electric field	Field mills within 9.3 km of the pad read more than ±1 500 V/m (or ±1 000 V/m under some conditions) in the last 15 min
Abort weather	Conditions at any of 50+ downrange sites would exceed Dragon's abort splashdown limits

Florida is America's thunderstorm capital. In 1950–80 the Cape averaged 76 thunderstorm days a year, 80 % of them from May to September, peaking at 16 in July (Figure 37.15). Summer storms build in the afternoon, when sea breezes from both coasts collide. A 2024 study looked at 18 years

of field-mill records. It found that 88 % of successful launches in 2005–2020 lifted off in hours when the climatological chance of violating the electric-field rule was 15 % or less.



(a) NASA TM-103498 (1950–52, 1957–80). (b) Assumes an afternoon window is lost on any thunderstorm day, a 10 % chance of a non-weather scrub, and independent days.

FIGURE 37.15 (a) Mean number of days with thunderstorms per month at Cape Canaveral, 1950–52 and 1957–80. (b) The probability that an afternoon launch has flown after a given number of daily attempts, if any thunderstorm day forbids launch and there is a 10 % chance of a non-weather scrub each day.



FIGURE 37.16 Lightning near Launch Complex 39B on 10 June 2022, with the Artemis I rocket on the pad. The pad's three 180 m masts and their wires carry strikes to ground away from the vehicle. Image: NASA/KSC (NASA).

EXAMPLE: THE ODDS OF A SUMMER SCRUB

A satellite must launch in a two-hour window starting at 14:00 local time from Cape Canaveral. Use the thunderstorm climatology of Figure 37.15 as the chance that lightning rules forbid an afternoon window, and assume a 10 % chance that a technical problem scrubs any attempt, independent of the weather.

1. **July:** 16 thunderstorm days in 31, so the chance of good weather is $1 - 0.516 = 0.484$. The chance of launching on a given attempt is $0.484 \times 0.90 =$ **0.44**.
2. The number of attempts follows a geometric distribution, with mean $1/0.44 =$ **2.3 attempts**. The chance of having launched within three days is $1 - (1 - 0.44)^3 = 0.82$, so about one campaign in five will still be waiting after three days.
3. **January:** 1 thunderstorm day in 31 gives $0.97 \times 0.90 = 0.87$ per attempt, a mean of 1.15 attempts, and a 99.8 % chance of launching within three days. But winter cold fronts bring strong upper-level winds instead.
4. Over the year, 76 thunderstorm days in 365 is a 21 % chance, close to the 24 % of countdowns that weather affects. A morning window, when storms are rare, is the best cure.

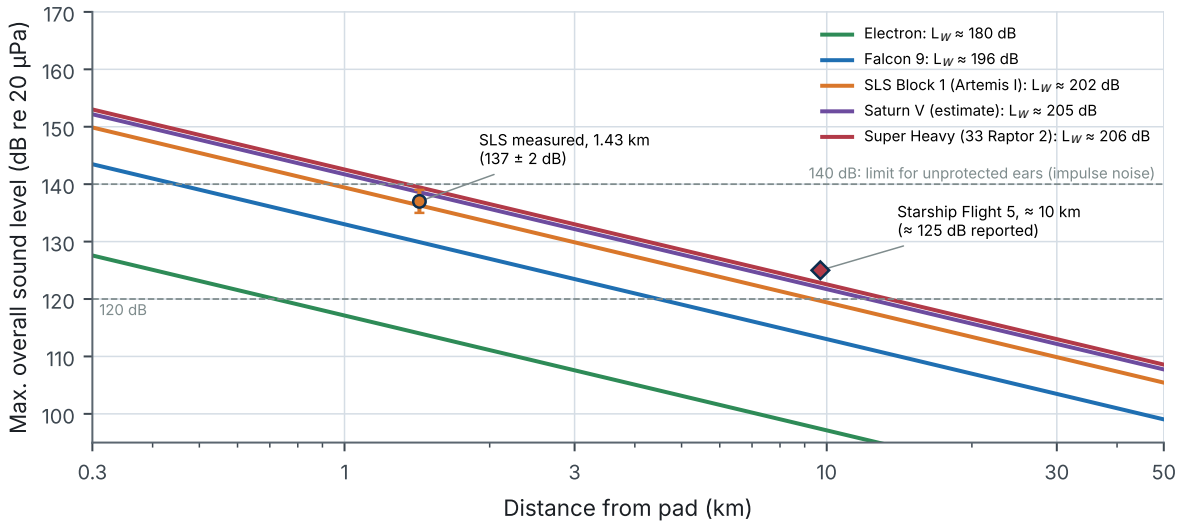
37.5.4 Scrubs, Holds and Recycles

When a criterion is violated the count **holds**; if the problem cannot be cleared within the window, the launch is **scrubbed**. The **turnaround** to the next attempt depends on draining and replacing propellant and fixing the fault. Falcon 9 often tries again 24 hours later, and a hydrogen-fuelled vehicle may need two days or more. A mission tied to a planetary window may wait weeks.

37.6 Environment, Community and Regulation

37.6.1 Noise and Sonic Booms

A rocket's **sound power** is a small fraction η, the **acoustic efficiency**, of the mechanical power of its jet, $\tfrac{1}{2} \times$ thrust $\times$ exhaust speed. Measurements 1.4–1.8 km from the Artemis I launch gave SLS a sound power level of 202.4 dB (re 1 pW). The measured efficiency was 0.33 %, below the 0.5 % long assumed. Falcon 9 radiates about 196 dB and the Saturn V about 205 dB. Treating the rocket as a source above hard ground gives the levels of Figure 37.17. The model reproduces SLS's measured 137 dB at 1.43 km and predicts 123 dB at 10 km from Super Heavy. This is close to the 125 dB recorded by Brigham Young University researchers about 10 km from Starbase on Starship Flight 5. The 120 dB contour lies about 4.5 km from a Falcon 9, 9 km from SLS and 13 km from Super Heavy.



$L_p = L_W + 5$ dB (directivity) − 10 $\log_{10}(2\pi r^2)$: hemispherical spreading over hard ground, no atmospheric absorption. $L_W = 10 \log_{10}(\eta \cdot ½ \cdot F \cdot v_e$ / 1 pW) with η = 0.3 %, or measured/estimated values (Kellison & Gee 2023, 2024). Super Heavy: 33 × 2.26 MN, $v_e \approx 3.2$ km/s.

FIGURE 37.17 Maximum overall sound level against distance for five launch vehicles, from their sound power assuming hemispherical spreading over hard ground and a 5 dB directivity gain. Points are measurements at the Artemis I launch and during Starship Flight 5.

The model ignores absorption by the air, small at the low frequencies (10–100 Hz) where rocket noise is strongest, and weather, which can focus sound far from the pad. Returning boosters add **sonic booms**: at Starbase the boom of the descending Super Heavy was the strongest pressure event of the flight, and booms lie at the centre of California's objections to more Vandenberg launches.

37.6.2 Debris, Wildlife and Water

Spaceports often share land with wildlife because their safety zones keep people out. Kennedy Space Center shares its land with the Merritt Island National Wildlife Refuge, and Starbase lies among protected tidal flats. Vandenberg's coast hosts nesting shorebirds and seals. Operators control lighting that disorients nesting sea turtles, monitor deluge water (which picks up hydrochloric acid from solid motors) and survey debris thrown out at lift-off. Downrange, falling stages have contaminated drop zones in Kazakhstan and China with the toxic fuel UDMH.

37.6.3 Licensing and Environmental Review

In the United States the FAA licenses launches, re-entries and spaceport operators. Its performance-based rule, **14 CFR Part 450**, adopted in 2021, became the only licensing route in March 2026, when the older licences expired. It caps the expected casualties from each launch at 1×10^{-4} (Chapter 34). Federal decisions also need review under the **National Environmental Policy Act** — an environmental assessment where impacts are expected to be modest, or a full environmental impact statement that can take years. The May 2025 approval of 25 Starship launches a year from Starbase followed an assessment; SLC-37 and the Vandenberg increase needed full statements. Executive Order 14335 of August 2025 told the FAA to revise Part 450 and agencies to use simpler "categorical exclusions" for routine launch actions.

Elsewhere, the UK Civil Aviation Authority licenses spaceports and launches under the Space Industry Act 2018. France's 2008 Space Operations Act gives CNES responsibility for safety at Kourou. Australia, New Zealand and Japan passed launch laws in 2016–18, and India authorises

private launches through its regulator IN-SPACe. All implement the Outer Space Treaty principle that a state is liable for damage caused by objects it launches (Chapter 1).

37.7 Launching from the Air and the Sea

A rocket launched from an aircraft or a ship can choose its latitude and azimuth, avoid overflying land and escape a fixed site's weather. Air launch also saves a little Δv by starting above the densest air (Table 37.6). Northrop Grumman's **Pegasus** has flown 46 times since 1990, latterly from the L-1011 *Stargazer* at about 12 km (Figure 37.18). Its latest flight, on 3 July 2026 from Kwajalein, carried a servicer to raise the orbit of NASA's Swift observatory. Virgin Orbit's **LauncherOne**, dropped from a Boeing 747, reached orbit four times in six attempts. After a failure from Spaceport Cornwall on 9 January 2023 — the first orbital attempt from the UK — the company closed.



FIGURE 37.18 A Pegasus XL rocket carrying NASA's NuSTAR telescope, attached beneath the L-1011 carrier aircraft Stargazer in June 2012. The aircraft carries the rocket to about 12 km and releases it over the ocean. Image: NASA/JPL (NASA).

Sea Launch, an American–Russian–Ukrainian–Norwegian venture, sailed a converted oil platform from Long Beach to the equator and launched Zenit-3SL rockets 36 times from 1999 to 2014; in 2007 one exploded on the platform. China is now the leading sea launcher. A Long March 11 flew from a barge in the Yellow Sea in June 2019. Since then, solid rockets such as Jielong-3, Ceres-1S and Gravity-1 have flown 28 times from ships based at Haiyang and Yangjiang. Solid rockets need little ground support; cryogenic propellants are far harder to handle on a moving deck.

TABLE 37.6 Air and sea launch systems (flights from GCAT, J. McDowell).

System	Platform	Flights	Period	Status (2026)
Pegasus (Northrop Grumman)	L-1011 *Stargazer* (earlier NB-52)	46, of which 3 failed and 2 partly failed	1990–2026	Flown occasionally
LauncherOne (Virgin Orbit)	Boeing 747 *Cosmic Girl*	6, of which 4 succeeded	2020–23	Company closed 2023
Sea Launch (Zenit-3SL)	*Odyssey* platform at the equator, 154° W	36, of which 3 failed	1999–2014	Ended; platform moved to Russia

System	Platform	Flights	Period	Status (2026)
Chinese sea launches	Barges and ships in the Yellow and South China Seas	28, all successful	2019–26	Expanding
Russian submarine launches	Shtil' and Volna from submarines in the Barents Sea	3, of which 1 failed	1998–2006	Ended

AI ON THE LAUNCH RANGE

As launch rates climb, ranges are turning to software (Chapter 52). Scheduling several launches a day is an optimisation problem — windows, hazard areas, airspace closures and shared tracking and recovery assets — for which algorithms can propose conflict-free timetables. Weather is a second target: the 45th Weather Squadron's lightning forecasts still rely on statistical tools. But the Air Force Institute of Technology has trained machine-learning models on a decade of weather-balloon data to automate them. A model from NASA Marshall and the University of Alabama in Huntsville detected 84 % of lightning 15–30 minutes ahead with a 22 % false-alarm rate (Chapter 47). Autonomous flight safety systems remain deliberately simple, verifiable rules; machine learning for debris-risk estimates or faster airspace release must meet the same safety standards.

KEY TAKEAWAYS

- A site at latitude ϕ can launch directly into inclinations from ϕ to $180° - \phi$, with $\sin Az = \cos i / \cos \phi$. Rotation adds 465 m/s × $\cos i$; latitude matters most for geostationary missions.
- Safety corridors decide which azimuths are allowed, explaining Baikonur's 51.6°, Palmachim's retrograde orbits and Florida's polar corridor of 2020.
- In 2025 Florida (109) and Vandenberg (66) hosted more than half of the world's 330 orbital attempts; China flew 93 times, and Rocket Lab's Mahia site alone matched all 17 Russian launches.
- A pad combines a flame trench and deflector, a deluge of up to 70 m^3/s, umbilical towers, lightning masts and a cryogenic farm. Rockets are stacked vertically and carried out, integrated horizontally and erected, or built on the pad.
- The range tracks the flight, clears hazard areas in the sky and at sea, and recovers boosters, fairings and capsules.
- Weather causes about half of Florida's scrubs; on a July afternoon the chance of launching on a given day can fall below one in two.
- Noise, sonic booms, debris and wildlife make licensing and environmental review a pacing item for new launch sites.

REVIEW QUESTIONS

1. Derive the range of inclinations reachable by a direct launch from latitude ϕ. Which inclinations can Vostochny (51.8° N) reach, and in which two directions can it launch into a 70° orbit?
2. Why does a launch into the ISS orbit gain almost the same velocity from Earth's rotation at Kourou as at Baikonur, while a geostationary mission gains far more at Kourou?
3. Using the worked example's sensitivity (0.82 t per 100 m/s), estimate how much payload a sun-synchronous mission loses to Earth's rotation, and how much a due-east launch from Kourou gains over one from Cape Canaveral.
4. Explain the reasons for Palmachim's westward launches, Sriharikota's dog-leg and Baikonur's minimum inclination of 51.6°.

5. List the main elements of a launch pad and the purpose of each. Why do pads need water at ignition, and what happened when Starship first launched without a flame trench or deluge?
6. Compare vertical assembly and transfer, horizontal integration and vertical integration on the pad. Give one advantage and one drawback of each, with an example vehicle.
7. What is triggered lightning, and which launch commit criteria are designed to prevent it?
8. A launch window opens at 15:00 in August at Cape Canaveral. Using Figure 37.15 and a 10 % technical-scrub probability, estimate the expected number of attempts and the chance of launching within two days.
9. Using the noise model, estimate the distance at which a Falcon Heavy (three Falcon 9 cores) would produce 120 dB. Why might real levels differ?
10. What are the advantages and limitations of air and sea launch? Why are most sea launches today made by solid-fuelled rockets?